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OECD ECONOMIC SURVEYS

1998

SPECIAL FEATURES
Financial deregulation and Other structural policies

ICELAND

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1997-1998

ICELAND

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF ICELAND

THE LAND

Area (1 000 sq. km)	103
Productive area (1 000 sq.km)	21
of which:	
Cultivated area	1.1
Rough grazings	20
Unproductive area (1 000 sq. km)	82
of which:	
Glaciers	12
Other area devoid of vegetation	70

THE PEOPLE

Population, 1st December 1997	272 064
Net increase 1987-97, annual average (per cent)	1.0
Occupational distribution, 1996 (per cent):	
Agriculture	4.5
Fishing and fish processing	10.9
Other manufacturing	11.1
Construction, total	6.5
Commerce	13.7
Communications	7.1
Services and other	46.2
	100.0

GOVERNMENT AND PARLIAMENT

	1991	1995
Parliament, number of seats:		
Independence Party (Lib. Cons.)	26	25
Progressive Party (Agrarians)	13	15
Peoples' Alliance (Socialists, Communists)	9	9
Social Democratic Party	10	7
Citizen's Party	–	–
Women's Alliance	5	3
Other	1	4
	63	63

Last general election: April 1995

PRODUCTION AND CAPITAL FORMATION

Gross national product in 1997:	
IKr million	515 420
Per head, US dollars	26 694
Gross fixed capital formation in 1997:	
IKr million	97 694
Per cent of GNP	19.0

FOREIGN TRADE

Exports of goods and services in 1997, per cent of GNP	36.9
Main exports in 1997 (per cent of merchandise exports):	
Fish products	71.3
Aluminium	11.8
Other manufacturing products	10.3
Agricultural products	1.6
Miscellaneous	5.1
Imports of goods and services in 1997, per cent of GNP	36.2
Imports in 1996, by use (per cent of merchandise imports)	
Consumer goods	31.1
Investment goods	31.1
Intermediate goods (excl. fuels)	30.1
Fuels and lubricants	7.7

THE CURRENCY

Monetary unit: Krona

Currency units per US dollar, averages of daily figures:	
Year 1997	70.97
March 1998	72.53

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Iceland by the Economic and Development Review Committee on 23 March 1998.

•

After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 6 April 1998.

•

The previous Survey of Iceland was issued in May 1997.

Assessment and recommendations

The economy is growing fast...

When the OECD reviewed the Icelandic economy a year ago, there were clear signs that the economic recovery was proceeding rapidly. In the event, GDP rose even more than expected, growing on average by about 5 per cent, which is well above the growth of potential as estimated by the OECD Secretariat. Buoyant domestic demand fuelled this strong performance. Business investment has picked up, as foreign companies continued to finance large-scale projects designed to take advantage of Iceland's relatively inexpensive electricity. At the same time, with households' real income rising, personal consumption has accelerated rapidly. Reflecting these developments, the labour market improved significantly, with a marked fall in youth unemployment, bringing the overall unemployment rate down to below 4 per cent – which is similar to its currently estimated structural level, *i.e.* a level consistent with stable inflation.

... and appears to be over stretched

With the economy thus apparently running close to full capacity, there are some indications that both external and domestic imbalances may worsen. Strong investment growth has led to a surge in imports, mainly of capital goods, which, with exports growing less rapidly due to an unfavourable mix of the fish catch, has been reflected in a widening of the current account deficit. Moreover, the 1997 wage agreements have set the stage for rising labour costs. Indeed, these settlements are proving to be expensive for

the business sector, in part because of the incorporation of certain bonuses into payments for non-daylight hours, which are particularly important in Iceland. Also, given the already tight labour market conditions, buoyant demand for labour and recent strikes in the key fishing industry seem likely to add to the substantial basic wage increases by boosting wage drift.

This could generate inflation risks...

So far, annual price increases have remained limited to 2 per cent, a better performance than expected one year ago. The rise in the exchange rate in the first half of 1997 helped to keep inflation low, especially when coupled with falling oil prices and low costs among Iceland's trading partners. Domestic deregulation also contributed to disinflation, with the liberalisation of air transport pushing down prices by a half on certain routes. However, inflation pressures have tended to arise more recently. Given the tightness of the labour market, there is a risk that such pressures may intensify, the more so as productivity growth is expected to slacken somewhat. Although weak import prices may offset some of these problems in the immediate future, rising utilisation rates could push inflation to above 3 per cent.

... as the upturn is expected to continue

This risk of accelerating prices is heightened by the prospect of continuing slightly above-potential economic growth this year and next. The traditional source of export earnings – fish – should be buoyant in the near future, although sales to Asia may be hit by weaker demand in that area. Conservation policies have led to a recovery in Icelandic cod stocks and increased catch limits for this species, offsetting lower landings of other varieties. In addition, some of the foreign investment projects should come on stream, boosting exports. At the same time, gains in employment and real wages should ensure continued strength in private consumption. On the other hand, the

growth of capital spending may slacken, slowing the increase of imports. Nonetheless, the current account deficit may remain high in comparison to the recent past. Over the medium term, though, a fall in the external deficit could occur as the investment boom draws to an end and the new foreign-owned plants start to produce at full capacity for export markets.

Tighter monetary policy would thus seem necessary...

The strength of the economic upswing and the associated risk of a resurgence of inflation suggest that monetary policy might have to be tightened further. In line with the recommendation made in the previous OECD *Survey*, the Central Bank has raised interest rates twice in the past eighteen months, with the latest move occurring in November 1997. However, the increases were relatively small, totalling 70 basis points. As a result, the demand for credit has remained high, with loans growing 9¼ per cent and the money supply increasing by a similar amount during 1997. Additional increases in interest rates would thus seem to be called for in order to prevent these developments being translated into higher inflation expectations.

... which might imply further appreciation of the currency

As current exchange rate arrangements allow fluctuation margins of 6 per cent on either side of the central rate of the krona, the required monetary tightening might imply that the exchange rate should be permitted to drift upwards within this band. This would seem to be sufficient to maintain low inflation while ensuring broad exchange rate stability. However, should the upward pressure on the currency become excessive, a wider band or a shift in the central point of the exchange rate might be needed to avoid large-scale intervention in the exchange market that would undermine the effectiveness of monetary control. In either case, the resulting appreciation of the currency would mean that there was symmetry in the current policy framework and that the devaluations which occurred in recessions

(such as 1992 and 1993) could be reversed at the top of the cycle.

Additional fiscal consolidation would also help stabilise the economy...

A tighter fiscal stance, could, in the short run, take some of the pressure off monetary policy in keeping inflation under control. Already the government has pursued a successful policy of fiscal consolidation since 1990. During this period, public spending has been reduced by 5 per cent of GDP, and the general government deficit is now close to balance, as, in 1997, the central government achieved a budget surplus. At the same time, gross public debt has declined to 52 per cent, with net debt being significantly lower. In 1998, the central government projects a budget surplus similar to that of last year, despite a large increase in social security benefits and income tax cuts contained in the second stage of a three-year income tax reform programme. This continued surplus, however, mainly stems from profits from the privatisation of state-owned companies (see below) and a cyclical upswing in tax receipts. As a result, the Secretariat estimates that the cyclically-adjusted general government balance for 1998 will still display a deficit, and one that is widening. Given the need for fiscal policy to support monetary management in restraining demand and to contribute to reducing the external deficit, the government should aim for larger savings in the period ahead. This would improve the mix of macroeconomic policy instruments in stabilising the economy.

... which would require more spending restraint

Achieving this would call for some further reductions in outlays on subsidies and capital transfers, which still represent 4 per cent of GDP. In addition, a deceleration in real public consumption would be desirable given the pay increases embodied in the current three-year wage settlement. Also, after the strong growth of social transfers in the previous two budgets, additional increases in this area should be limited to the rate of inflation.

The increased transparency of government accounts should aid in this context

The case for such supplementary government savings is highlighted by the welcome introduction of accrual accounting techniques for the presentation of the budget. The adoption of this standard, for instance, has shown that conventional debt figures understate the full liabilities of the government, in particular with respect to the pension fund for state employees that, by 1995, had accumulated a deficit amounting to 15 per cent of GDP. While contributing in this way to increasing the transparency of the budget, such a new accounting framework also presents the advantage for a country like Iceland of correctly measuring the borrowing costs associated with extensive index-linked loans. It also constitutes an aid to decentralised management control, a feature of the ongoing reform of public sector administration that has been introduced by the government.

Despite significant financial liberalisation...

As part of the steps towards a more market-oriented economy, there has been a substantial liberalisation of financial markets since 1992. Controls over inward and outward capital flows have been progressively reduced, culminating in the complete abolition of restrictions on short-term flows in 1995. Two years later, the Central Bank ceased to be a market maker for foreign exchange transactions. In addition, a money market was created, first to provide short-term finance to the Treasury following the removal of the government's automatic overdraft facility with the Bank and, more recently, to satisfy commercial banks' needs for short-term liquidity. The bond market has also been improved, being both widened and deepened. Moreover, the government no longer attempts to set real long-term interest rates, as it did between 1992 and 1994. As a result of these measures, there has been a marked increase in the size of capital markets, which are now exerting growing competitive pressure on commercial banks.

... the government still plays a big role in the banking sector

By contrast, most of the banking sector still remains under government control – a situation that has persisted for nearly 70 years. Currently two of the three commercial banks are state-owned, accounting for 77 per cent of bank deposits. While 29 savings banks compete with them in certain sectors, they are small. Moreover, the ownership structure of the savings banks is opaque, with nominal control vested in founding depositors and the local government of the area where they operate. Such a fragmented nature of the banking system and its public or mutual ownership have resulted in costs and interest rate spreads that are amongst the highest in the OECD area. In addition, the government owns a newly-created investment bank, into which four of the largest investment credit funds were merged. It also possesses a number of small investment credit funds that provide long-term credit and venture capital to the business sector. Overall, publicly owned bank and non-bank institutions provided 85 per cent of outstanding credit to the business sector in 1996. As such, they represented at that time a major asset of the public sector, with their total equity capital amounting to 12 per cent of GDP.

While privatisation plans are welcome in this regard...

The government has started recently to change the status of some of these institutions. The two government-owned commercial banks have been made into joint stock companies. An offer of new equity in these banks will be opened to the public in the future, but the private sector will be limited to a 35 per cent stake. Any further sale of equity will require permission from the legislature. Later this year, the government proposes to sell a 49 per cent stake in its investment bank.

... they should be more ambitious...

Despite these reforms, the government will retain a majority stake in all these institutions. The possibility of the capital markets exerting effective control over these partially privatised institutions will, therefore, be limited, as

will the incentives for cost reduction. Thus the efficiency gains from privatisation may not occur to the extent that would be desirable. Such considerations argue for a more complete privatisation of these credit institutions in a relatively short time frame. Steps should also be taken to merge and privatise the remaining investment credit funds. Moreover, a rapid reduction through mergers in the number of mutually owned savings banks would benefit consumers by increasing efficiency and lowering interest rate spreads. Such a move would require legislation to clarify the procedures for demutualisation of these companies.

... and extended to housing finance

The government is also reviewing the role of the public sector in the provision of housing finance. Government-owned housing funds are the predominant source of finance for residential investment and are far bigger than the investment credit funds serving enterprises. The government continues to borrow for some of these funds or guarantee the bonds they issue. While such a procedure does lower costs to borrowers, it imposes a burden on the taxpayer. Given the move towards increased transparency in Icelandic public finance, a bill before Parliament proposes to replace low interest lending with the existing tax rebate system for housing. Moreover, it is planned that the housing funds be merged into a single institution. Once all this is done, there would be little reason to keep this new institution in the public sector. It could be transformed into a housing bank and sold to the private sector. Indeed, with the rapid development of capital markets and the growth of private sector pension schemes, such a bank would be in a position to finance itself through access to the capital market without having to rely on government guarantees.

At the same time, a bigger role for private pension funds should be considered

A further extension of capital markets in Iceland would require more rapid growth of pension funds. These institutions provide a second-tier income-related pension to all economically active people who are obliged to join a scheme. Their assets have grown to 56 per cent of GDP in 1996, up from 10 per cent in 1980. The recent decision by the government to establish a funded pension scheme for all new entrants to public-sector employment should speed this process and may result in the assets of all pension funds approaching 150 per cent of GDP in 50 years time. A high proportion of personal wealth is now invested in these funds, which are now gradually approaching the size of the government as a source of income for the retired. In this context, the new legislation, establishing the regulatory regime for pension plans, which was adopted in 1997, is to be welcomed, as it will ensure that the supervisor has power to compel funds to restore their solvency. In addition, it will introduce some competition in the selection of fund managers for the new supplementary retirement accounts.

Outside the financial sector, deregulation has proceeded significantly...

In parallel with the liberalisation of the financial sector, the government has embarked, over the past five years, on a policy that favours free competition in the rest of the economy. Most notably, it introduced new competition legislation in 1993 that has been enforced quite vigorously since. It is now established that competition law takes precedent over other legislation. As a result, various monopolistic practices in the economy have been changed. There has also been a move to deregulate a number of industries, such as domestic air transport. Moreover, some central and local government enterprises have been privatised. For the future, the government plans to gradually introduce more competition into the energy and telecommunication sector. A competitive electricity industry is envisaged at the latest by 2009, a time frame that could be usefully speeded up.

Steps have also been taken to improve the functioning of the labour market, in line with certain recommendations of the *OECD Jobs Study*. In particular, a national employment service has been created that is linked to the unemployment benefit administration. This will enable the grant of benefits, whose duration is now limited to five years, to be clearly connected to the job search activity of the unemployed.

... but additional progress is still needed in some areas

Additional reform initiatives would, nonetheless, be useful to strengthen the dynamism and adaptability of the economy. With respect to the *labour market*, this would imply an eventual shortening of the time for which unemployment benefits are paid. Further flexibility could be introduced by an agreement to reduce the high payments made for night time work – a move that might give employers greater scope to use part-time employees and so reduce costs. In addition, contributions to the unemployment insurance system should reflect the need to reduce the implicit subsidy given to sectors with a strong seasonal demand for labour. In the field of environment, some further measures will be required, following the Kyoto agreement, to meet the targets for the emission of greenhouse gases. Another area requiring substantial reform is *agriculture* where the high level of support should be reduced further.

To sum up

In summary, there has been an outstanding transformation of the Icelandic economy in the 1990s. Inflation has been brought under control, financial markets have been deregulated and the budget balanced. However, after two years of rapid economic growth, the Icelandic economy appears to be confronted again with the risk of rising external and internal imbalances. Although inflation has, so far, remained low, it looks set to increase this year and next as labour costs are accelerating in the wake of the 1997 wage agreement. This suggests that interest rates should be raised

further and, in case of resulting upward pressure on the currency, that the exchange rate be allowed to appreciate sufficiently to ensure the efficacy of monetary policy. By restraining demand, a tighter fiscal stance would assist monetary policy in keeping inflation under control and would contribute to reducing the external deficit over the medium term. At the same time, the emphasis on regulatory reform should be maintained to strengthen economic efficiency. This would imply notably speeding up the privatisation of the large government-owned financial sector, introducing more flexibility into the labour market, as well as reforming agriculture.

I. Recent trends and prospects

An economic recovery has been under way in Iceland since 1994 after a prolonged period of stagnation. The upswing gained momentum in 1996 with renewed foreign interest in developing low-cost energy sources to supply the metal industries and as the benefits of a successful fisheries management policy began to be felt. Real incomes began to improve, and private consumption turned up. By the start of 1997, it was clear that the economy was poised for another year of solid growth, as recognised in the last *OECD Economic Survey* (1997).

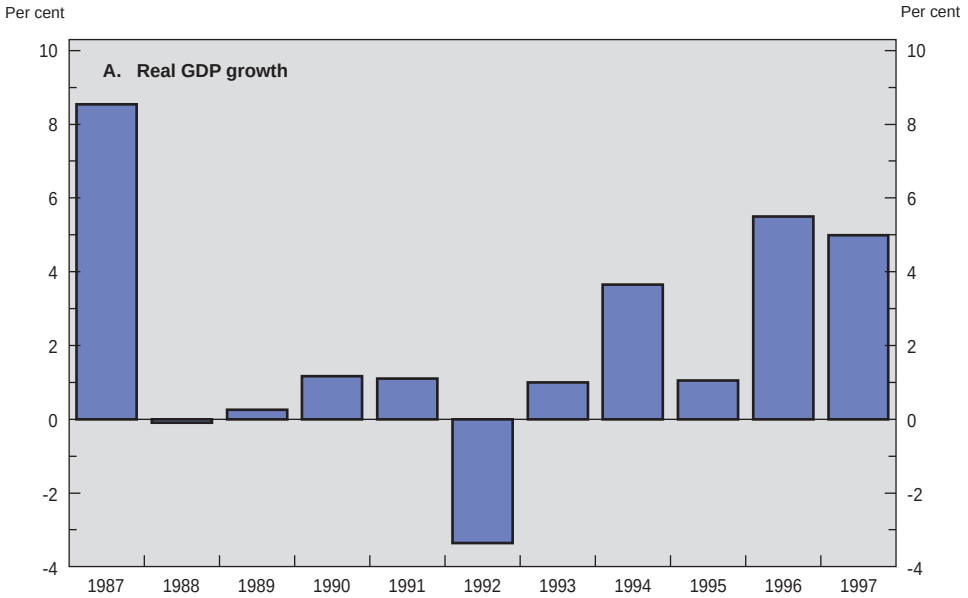
The economy is approaching capacity limits...

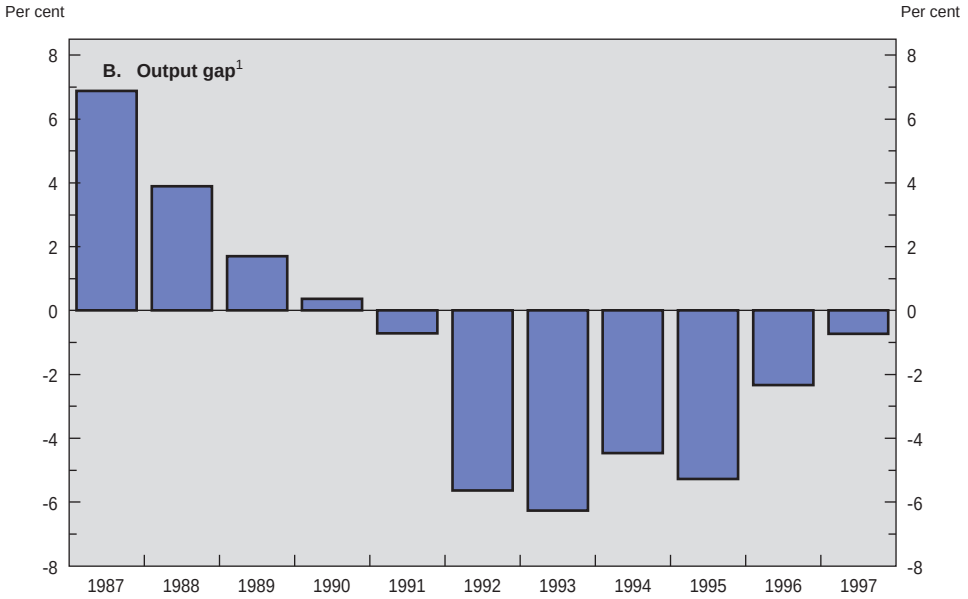
In the event, output grew even faster than expected, and the economy has become increasingly stretched. Supported by strong domestic demand, real GDP grew 5 per cent in 1997, bringing output closer to potential than at any time since 1991 (Figure 1). As a result, the labour market has tightened, with the unemployment rate falling to 3.9 per cent, despite rising participation rates. Such developments have led to a substantial acceleration in the growth of wages, but so far, price inflation has only edged up slightly. Strong domestic demand pushed the current account further into deficit in 1997. Indeed, imports grew rapidly for a second year with most of the capital equipment necessary for the investment boom coming from abroad. An unfavourable mix of the fish catch prevented exports from keeping pace.

... as business investment strengthens...

For the second year real business fixed investment surged, growing 12.9 per cent in 1997 after jumping 53 per cent the previous year (Figure 2). The major

Figure 1. **GROWTH PERFORMANCE**





1. Percentage difference between output and estimated potential output.
Source: National Economic Institute and OECD.

Table 1. **Output and demand**

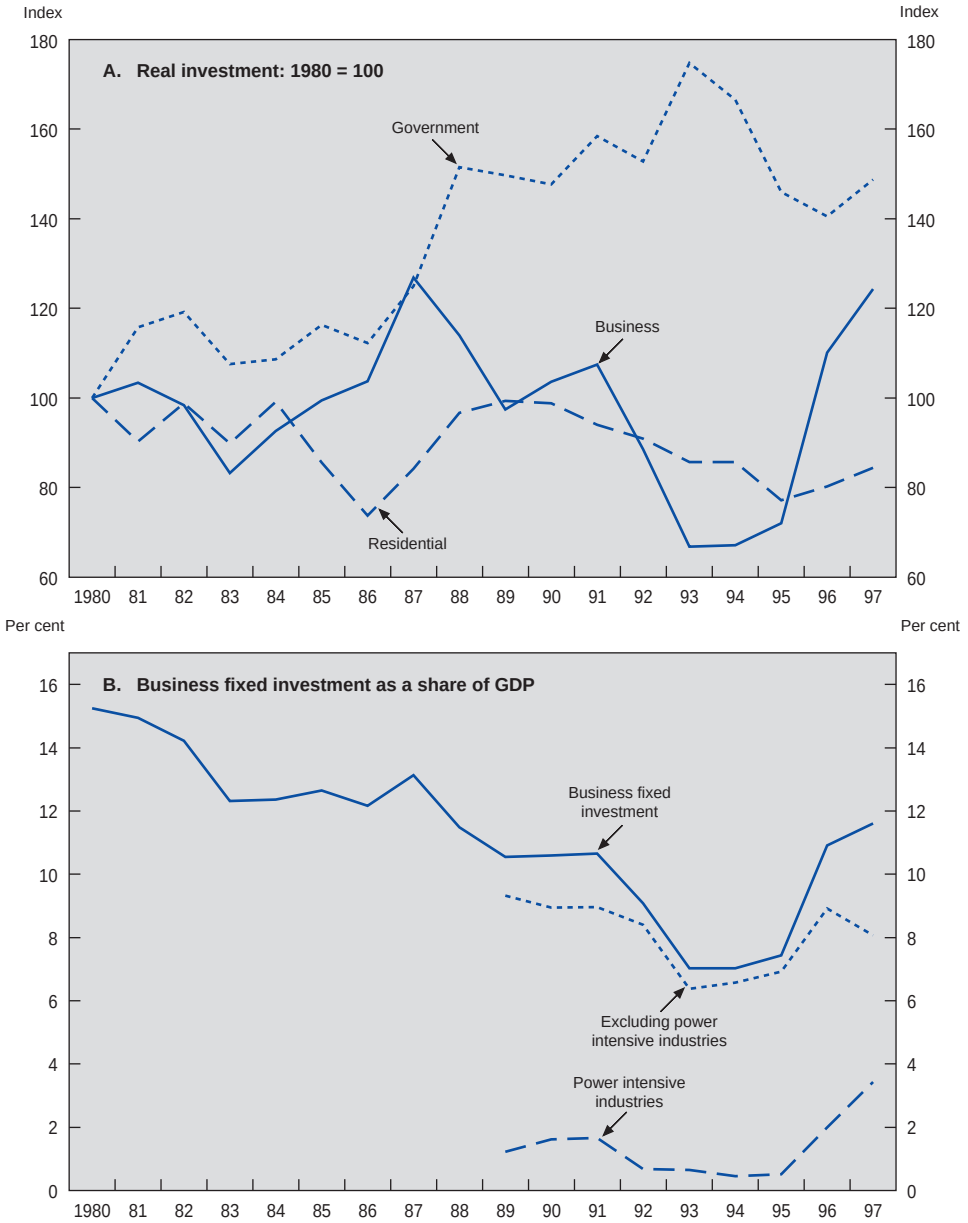
Per cent change in volume terms, 1990 prices

	1992	1993	1994	1995	1996	1997 estimate
Private consumption	–4.5	–4.5	1.9	4.2	6.4	6.0
Government consumption	–0.8	2.3	3.7	1.3	1.0	1.5
Gross fixed investment	–11.3	–11.4	–1.1	–2.8	26.5	9.9
Residential	–3.3	–5.8	0.0	–9.9	4.0	5.1
Business	–17.5	–24.6	0.5	7.3	53.0	12.9
Government	–3.6	14.4	–4.7	–12.4	–3.7	5.8
Final domestic demand	–5.0	–4.4	1.8	2.5	8.3	5.8
Stockbuilding[1]	–0.3	0.3	–0.2	0.6	–0.7	0.0
Total domestic demand	–5.3	–4.1	1.5	3.1	7.5	5.8
Exports of goods and services	–1.9	7.1	10.0	–2.1	9.6	5.6
Imports of goods and services	–8.0	–8.6	4.2	3.8	16.6	8.0
Foreign balance[1]	2.1	5.0	2.1	–1.9	–1.7	–0.6
GDP	–3.4	1.0	3.7	1.0	5.5	5.0

1. Contribution to growth; change in aggregate as a per cent of GDP of the previous year.
Source: Statistics Iceland, National Economic Institute.

part of the jump in capital expenditure represents continued foreign investment in aluminium and ferrosilicon production as the construction of a new aluminium plant started following the expansion of existing facilities. In addition, there have been increases in investment in electrical power generation facilities and related infrastructure that is needed for these power-intensive industries. Outlays in other industries have also been increasing, raising the share of such spending in GDP towards its historical norm (Figure 2). In particular, capital formation in the manufacturing, construction and communication industries posted large increases, with commercial property and office development also being strong. In contrast, after a large run up in 1996, investment in the fishing and processing industries appears to have fallen back somewhat in 1997. Elsewhere, growth in less capital-intensive industries has also been brisk. For example, a biotechnology firm signed a large contract with a European pharmaceutical company, and the software industry has continued to expand.

Figure 2. **CATEGORIES OF INVESTMENT**



Source: National Economic Institute and OECD.

... personal demand expands...

Households' demand was a second driving force behind the expansion of demand in 1997. Given the slackening of inflation, large gains in nominal incomes (Table 2) have been translated into strong growth of real incomes that have also been boosted by tax cuts. As a result, both pre and post-tax real wages have come close to recovering to the peak levels registered a decade ago (Figure 3). Buoyant incomes boosted private consumption, especially in the more cyclically-sensitive categories such as motor vehicles, furniture and household equipment, and recreation in 1996. In 1997, purchases of durable goods increased rapidly, with new car registrations jumping by more than a quarter. This continued the trend of the two previous years, when consumers rebuilt their stock

Table 2. **Household income and balance sheet**

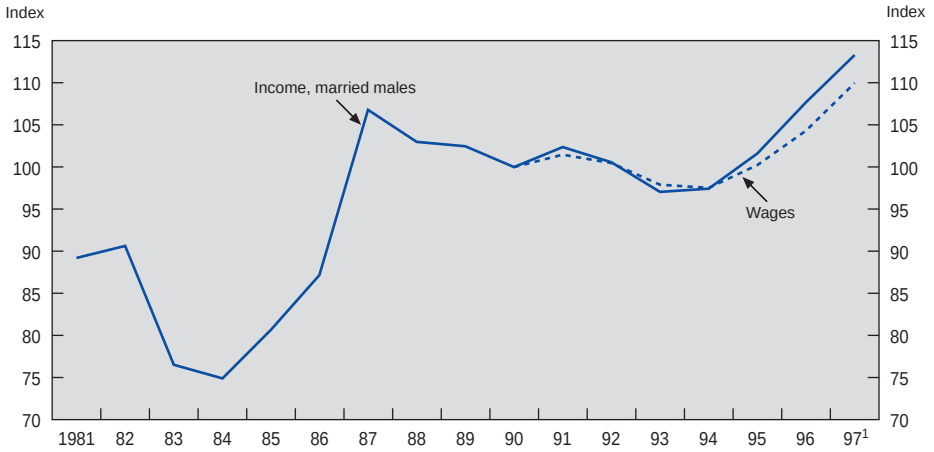
Nominal growth rates

	Share of 1992 pre-tax income	1993	1994	1995	1996	1997 estimate
Income						
Pre-tax income	100.0	–0.7	3.4	5.8	8.1	7.7
Labour income	82.5	–0.1	3.2	7.0	10.0	8.9
Net transfers	4.2	22.1	5.1	8.1	–5.7	–7.7
Other income	13.3	10.2	3.3	2.6	4.1	8.2
Direct taxes	17.0	10.5	5.3	9.0	13.9	3.5
Disposable income	83.0	–2.9	2.9	5.1	6.7	8.7
Outlays	67.5	–0.1	3.7	6.0	8.9	7.5
Consumption	67.5	–0.1	3.5	6.1	8.8	7.9
Net remittances abroad	–0.1	–20.0	137.0	–330.6	41.4	–807.2
Saving rate	18.7	16.4	15.7	15.0	13.2	14.2
Balance sheet						
Assets	239.3	6.5	4.7	5.9	11.0	8.5
Fixed	145.2	3.2	1.8	3.5	10.0	6.4
Pensions	48.3	15.3	12.3	12.4	17.0	12.4
Other financial	45.8	7.8	4.8	5.1	6.2	9.5
Debt	65.0	10.4	10.6	8.4	10.1	7.6
Memorandum:						
Debt as a share of total assets (per cent)		28.1	29.7	30.4	30.2	29.9

Source: National Economic Institute, the Central Bank and OECD.

Figure 3. **REAL DISPOSABLE INCOME AND WAGES**

1990 = 100



1. Estimate.

Source: *Statistical Yearbook of Iceland* and OECD.

of physical assets, after a period in which real incomes had fallen. Nonetheless, in contrast to the initial years of the upswing, the growth in consumption was less than the growth in income, and the savings ratio rose. Consistent with the run up in income and consumption, real residential investment also increased in 1997 for the second year in a row after a five-year downward slide (Figure 2).

With the saving rate edging up, household balance sheets have stabilised. Following a four-year period when household debt rose by 46 per cent while disposable income increased by only 12 per cent, debt grew in line with incomes in 1997. Moreover, from 1996 onwards, the growth of total household assets has increased, especially that held by pension funds for retirement benefits, more than offsetting the rise in debt (Table 2). Moreover some of this run up in debt was in the form of mortgages related to increased residential investment, though credit associated with purchases of other fixed assets such as motor vehicles has also risen as well. Overall, debt as a percentage of total assets edged down in 1996 and 1997, but still appears very high when compared with major OECD

countries. In addition, as a good portion of financial wealth is controlled by pension funds and cannot easily be used to repay debt, households may be even more stretched than this comparison suggests.[1]

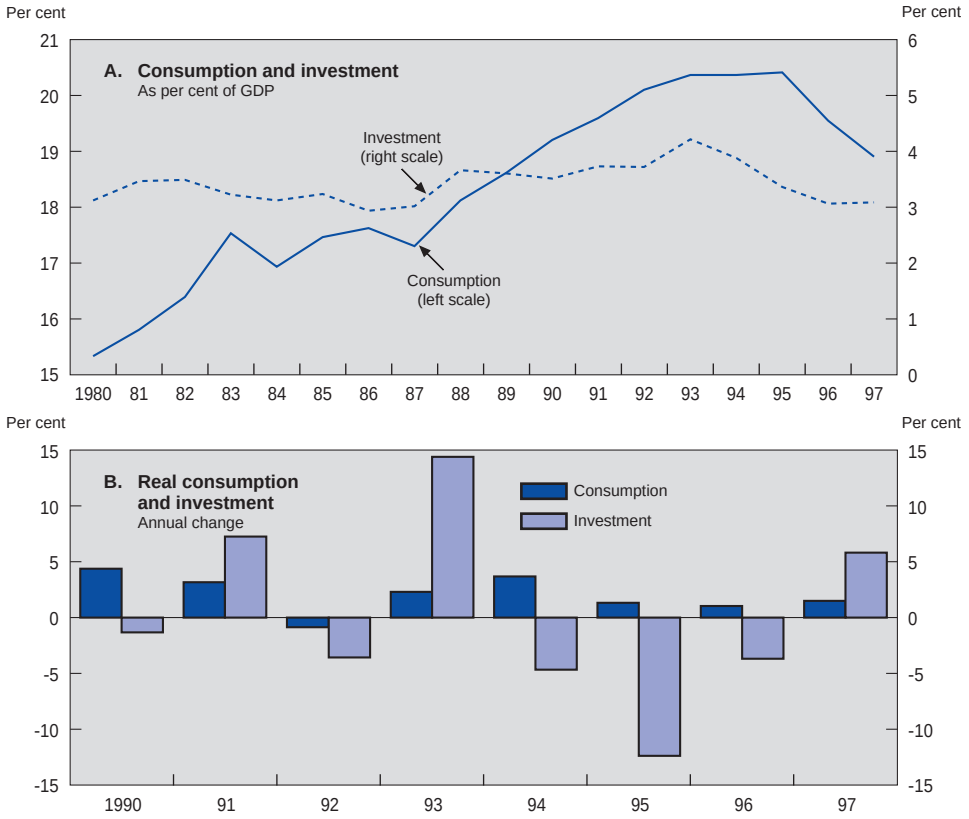
... and public spending increases

A third element boosting demand in 1997 was real government expenditures on goods and services which, after being essentially flat for the previous two years, expanded by about 2 per cent. Investment expenditure picked up (Figure 4), with outlays on roads, bridges and sewers partially reversing two years of declines. Likewise, public consumption accelerated by ½ percentage point, with spending in the areas of education and health care showing the largest increases. In addition, government wages outstripped price inflation by a larger margin in 1997 than in 1996, so providing a further stimulus to domestic demand. Despite these developments, the rapid concurrent growth of GDP has meant that the nominal value of public spending edged down ½ percentage point as a share of output.

As a result, the labour market is tightening, and wages are accelerating...

Such strong increases in demand have helped boost the demand for labour. Overall employment posted a 1.7 per cent increase in 1997. Moreover, with the rise in the participation rate flattening off, after several years of upward movement, the registered unemployment rate fell another ½ percentage point in 1997 to 3.9 per cent; it is now over a percentage point below its peak in 1995 (Figure 5), though still higher than in the years before 1993. Youth unemployment rates have decreased more, while the percentage of the labour force unemployed for six months or more continued its 1996 decline. Indeed, the long-term unemployment rate is low in Iceland when compared with the OECD average.[2] Other indicators also point to tighter labour markets. Employment vacancies have opened up in the past year in most sectors of the economy but especially in the area of the capital city. In addition, net migration was near zero after averaging 0.3 per cent of the population during the three previous years.

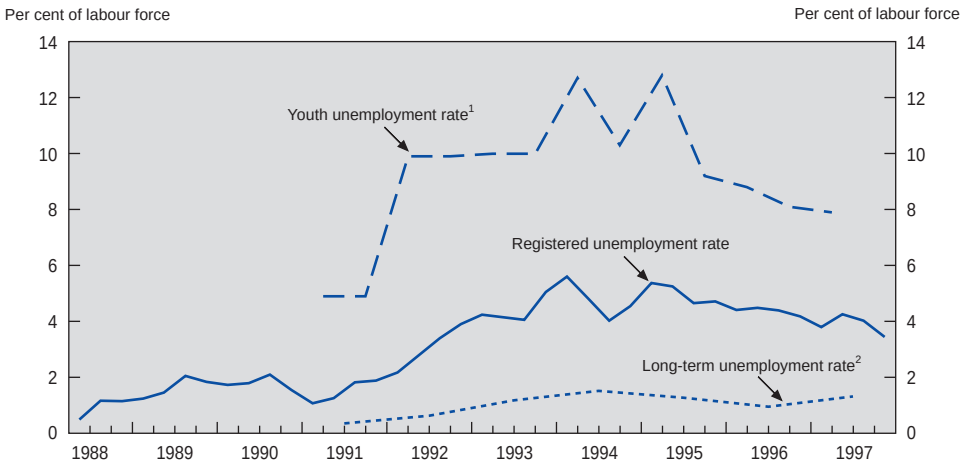
Figure 4. **PUBLIC SECTOR DEMAND**



Source: *Statistical Yearbook of Iceland*, National Economic Institute and OECD.

Under these tight labour market conditions, in March 1997 labour unions and employers reached agreements on new three-year wage contracts that will result in a cumulative wage increase of 12.9 per cent. On signing, wages rose 4.7 per cent, with smaller increases contracted for January 1998 and 1999. To help finalise these pay negotiations, the central government announced that it would lower marginal tax rates over a three-year period, in line with the report of the Marginal Tax Commission. Workers lower down the wage scales received larger pay hikes, with the minimum wage rate in collective agreements set to rise

Figure 5. **UNEMPLOYMENT**

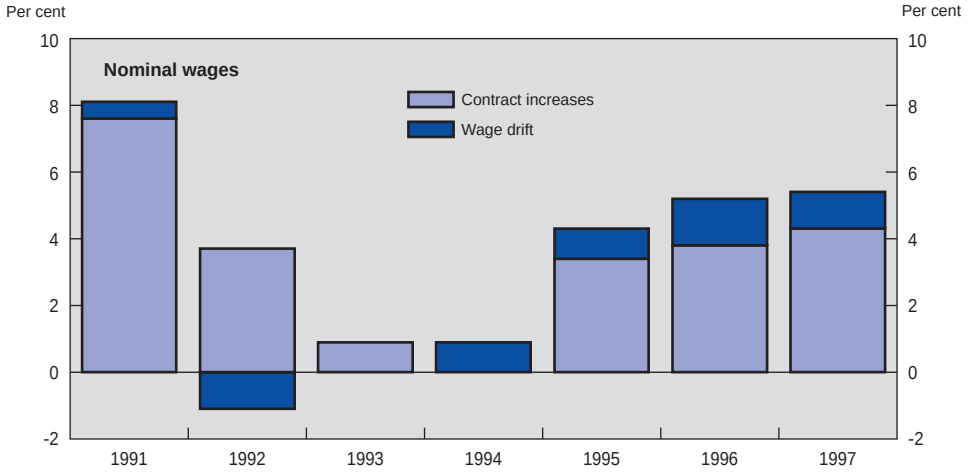


1. 16-24 years.
2. Six months or more over total labour force.
Source: Statistics Iceland, Labour Market Survey and OECD.

by 23 per cent over the period of the contract. Pay for work outside daylight hours received an additional boost when some bonus payments were incorporated into the base wage on which the night hours premium is calculated. Public sector wage increases were similar, in general, to those in the private sector but are spread over a slightly longer period. Primary and junior high school teachers, though, received cumulative pay increases of 33 per cent over three years.

As is usually the case in periods of high labour demand, actual earnings rose faster in 1997 than contractual wage increases would suggest (Figure 6). Over the last three years, such wage drift has averaged a little over 1 per cent and is likely to continue, both because the demand for labour is high and because of special features in the new wage contracts. As a result, the annual growth in wages was 5.4 per cent year-on-year for 1997 and 8.1 per cent over the course of the year. Recent monthly data suggest that wage drift has accelerated somewhat. Increasing levels of employment and average hours worked, as well as the 8 per cent hike announced for social security benefits, should boost personal incomes significantly.

Figure 6. **WAGE DEVELOPMENTS**
Annual change



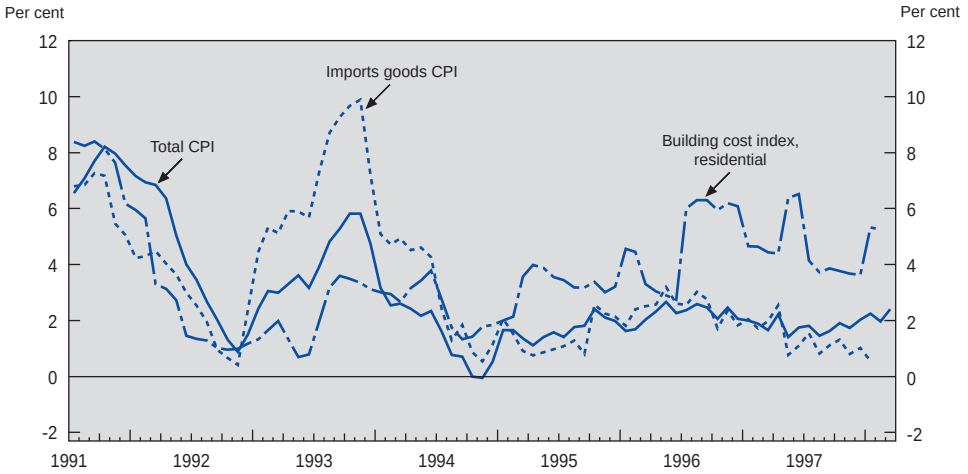
Source: *Statistical Yearbook of Iceland* and OECD.

... but price inflation remains subdued

In spite of tightening resource utilisation and robust wage increases, price inflation remains quiescent (Figure 7). Overall the CPI grew 1.8 per cent year-on-year in 1997, compared with a 2.3 per cent increase in 1996. Some of this relative price stability stems from a marked slowdown in the growth of imported goods' prices as the krona appreciated, oil prices fell and inflation moderated among Iceland's trading partners. This deceleration in import prices also checked domestic inflation, as input prices declined and competitive pressures became stronger. By the end of the year and into the beginning of 1998, there were some signs that inflation was beginning to creep up. In particular, one of the factors behind the good performance in the first half of 1997 – the appreciation of the krona – had been reversed. Since then, however, the krona has recovered, and oil prices fallen further, which may ease the pressures on inflation during the spring of 1998.

Figure 7. **PRICE DEVELOPMENTS**

Percentage change from a year earlier



Source: National Economic Institute and OECD.

The domestic economy has also benefited from some positive sectoral supply shocks. When the domestic air travel industry was deregulated last summer, ticket prices dropped over 10 per cent on average, with airlines reportedly cutting ticket prices on some routes in half. After the government relaxed entry regulations for pharmacies, drug prices tumbled, and healthcare prices overall fell 1.1 per cent since March. Communications prices have remained relatively flat, after falling in the middle of 1996.

A further factor that initially helped to restrain domestic price inflation was the rapid productivity growth that typically occurs in a cyclical upswing. In 1996, economy-wide productivity grew 3.3 per cent, with large increases posted in the fishing, fish processing, manufacturing, construction, transportation, and communication industries (Table 3); among fish processors, productivity surged after two years of declines. Such an increase was well above the average rate posted in the 1990s when output was stagnant. In 1997, productivity growth for the economy as a whole, at 3¼ per cent, remained towards the high end of the range experienced since the 1980s, even when excluding the recession years.

Table 3. **Productivity growth by industry**

Percentage increase

	1992	1993	1994	1995	1996	1991-1996 average
Agriculture	2.3	11.2	7.9	2.9	4.3	5.7
Fishing	8.8	–3.4	1.4	4.9	10.6	4.3
Manufacturing	3.7	5.1	2.5	0.4	6.1	3.5
Fish processing	9.2	3.6	–2.7	–2.1	12.5	3.9
Other	1.4	2.1	5.1	1.4	3.1	2.6
Electricity and water supply	3.0	2.6	–4.1	–2.6	–2.9	–0.8
Construction	–6.3	–1.2	–0.4	–4.9	9.0	–0.9
Wholesale and retail trade, hotels	–4.1	–2.4	1.5	–0.4	2.2	–0.7
Transportation and communication	0.0	4.9	4.9	–0.7	4.8	2.7
Finance, insurance and business services	–2.4	–0.9	4.5	1.1	2.5	0.9
Other private services	–2.9	–0.7	4.2	1.6	3.7	1.2
Private sector	–0.1	0.4	1.9	0.2	4.2	1.3
Public services	–2.8	2.8	2.7	–1.3	–0.8	0.2
Whole economy	–0.7	0.9	2.1	–0.1	3.4	1.1

1. Based on full-time equivalents.
Source: National Economic Institute.

The recent wage increases, so far, have not led to a large decline in profit rates. The National Economic Institute estimates that both the return on equity and profits as a per cent of revenue edged down only 0.1 percentage point in 1997. However profits for corporations traded on the stock exchange fell significantly especially in manufacturing and retailing. Profits in the fisheries were especially poor in the first half of the year and only rebounded when fish prices increased. A surge in competition, as a result of deregulation, cut profits in the air transport industry severely.

The one area of the economy that has seen above-average price inflation is construction, where, as noted, demand for both residential and non-residential structures has been robust. Over the twelve months ended in January (Figure 7), the building cost index for residential buildings grew 5.3 per cent and that for non-residential structures by 4.5 per cent, which represents a recent sharp acceleration. Much of this high rate of increase can be explained by a 10.6 per cent rise in labour costs, although costs other than labour and materials increased

4.1 per cent also.[3] Construction is one sector of the economy that is relatively insulated from foreign competition and requires skilled labour whose supply is relatively inelastic compared with some service industries. Anecdotal evidence suggests that shortages for construction workers have persisted for at least a year, with employers reporting vacancies. Indeed, some contractors have even looked overseas to fill some positions. High building cost increases, however, have yet to show through to housing prices, which have risen in line with overall inflation.

The current account deficit has widened as imports surged...

With aggregate demand accelerating in the past two years, there has been a substantial deterioration of the external account. From a surplus of 1.9 per cent of GDP in 1994 the current account turned into a deficit of 1½ per cent of GDP in 1997 (Table 4). Such a swing was to be expected as the economy recovered. Indeed, Iceland has run a current account surplus in only four of the past fifteen years, and three of these were between 1993 and 1995 when there was a substantial output gap. The average current account deficit over the past fifteen years was

Table 4. **Current account**

Per cent of GDP

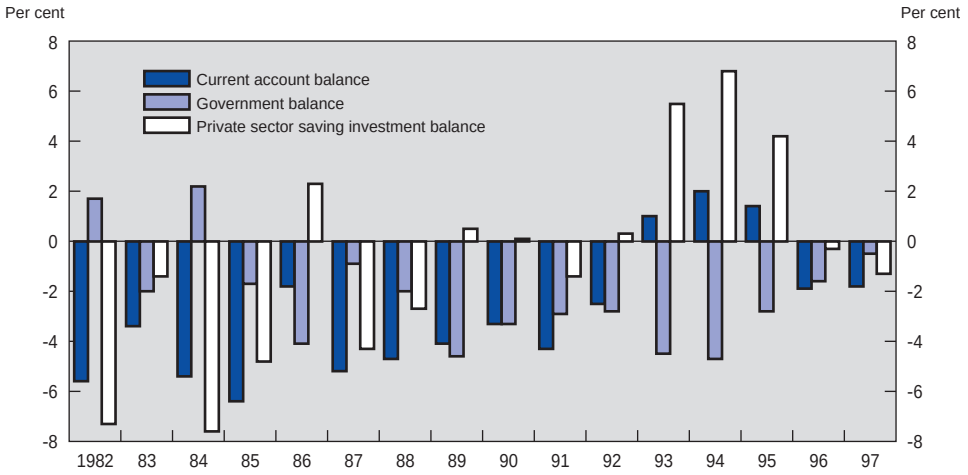
	1992	1993	1994	1995	1996	1997 estimate
Goods and Services	0.0	3.2	5.2	3.7	0.5	0.7
Merchandise	0.0	3.0	4.5	3.0	0.2	0.0
Exports	22.1	23.0	25.9	25.8	25.9	24.9
Marine	17.6	18.1	19.5	18.6	19.1	
Aluminium, ferrosilicon	2.4	2.6	3.1	3.4	3.3	
Other	2.1	2.3	3.3	3.8	3.5	
Imports	22.1	20.0	21.4	22.9	25.7	24.8
Special	3.1	1.7	2.2	2.3	2.6	2.9
General	19.0	18.3	19.2	20.6	23.1	21.9
Non-factor services	0.0	0.2	0.7	0.7	0.3	0.6
Exports	8.5	10.3	11.3	11.2	12.7	14.8
Imports	8.5	9.7	9.6	9.2	10.1	10.6
Factor income	–2.2	–2.4	–3.1	–2.8	–2.2	–2.2
Transfers	–0.1	–0.1	–0.1	–0.1	–0.1	0.0
Current account	–2.4	0.8	2.0	0.8	–1.8	–1.5

Source: Statistical Yearbook of Iceland, National Economic Institute and OECD.

2.9 per cent of GDP, mainly reflecting an average public sector deficit of 2.3 per cent of GDP (Figure 8). The budget deficit has been markedly reduced and is now close to balance (see Chapter II), so the domestic counterpart to the current account deficit is now private sector borrowing.

Figure 8. **EXTERNAL BALANCE AND ITS DOMESTIC COUNTERPARTS**

Per cent of GDP



Source: National Economic Institute.

Import demand was particularly robust in 1997. Led by a surge in demand for capital goods, the volume of imports of goods and services rose 8 per cent. The value of imported investment goods, excluding transportation equipment, increased 16 per cent. Imports of passenger cars jumped 28 per cent, but those of ships fell by a half, leading to an overall decline in other transportation equipment. Imports of consumer products other than automobiles grew roughly in line with overall consumption, while the value of industrial supplies barely edged up.

... and the mix of fish exports became unfavourable...

Exports of goods and services also grew in real terms in 1997 but failed to match the 10 per cent rise of 1996. They benefited from sales of aluminium and other manufactured products as well as a sale of a jet aircraft by Icelandair to a foreign leasing company.[4] In contrast, marine products, the largest element in exports, declined by about 1 per cent, despite the landing of a record tonnage. Nearly all of the increase in the catch came from a surge in the capelin harvest. This species is, however, a less valuable fish, mostly ground up for meal and oil. As a result, such high landings did not offset declines in the catch of more valuable species of fish and crustaceans other than cod. The total cod catch edged up somewhat; a significant decline in the catch in foreign waters was offset by increased landings in Icelandic waters, reflecting a slight rise in the allowable catch in the past two fishing seasons (Table 5). In addition, a drop in the price of shrimp reduced export earnings by 1½ per cent from where they otherwise would have been, offset in past by a rise in the price of fish oil.

... though the stock of cod is recovering

The estimated level of stocks of cod, the principal fish caught in Icelandic waters, has increased in 1997. In 1995 government restricted landings to 25 per cent of the estimated fishable stock subject to a minimum catch of 155 000 tonnes. The Marine Research Institute estimates that such a rule will limit the chances of a collapse in the cod stock to about 1 per cent. With a lag, the spawning mass of fish has also started to climb (Figure 9). Indeed, in 1997, the estimated level of fish hatching was the highest since 1984 and the fourth largest since 1970. Such hatchings will become fishable in four years time, suggesting that the stock should continue to rise to a level that will permit an efficient yield. Indeed, fishermen report finding cod in areas where there had been none in the past several years.

On the other hand, the evolution of other stocks has been poor. Landings in Icelandic waters of species other than cod have remained above the sustainable levels recommended by the Marine Institute in the past. Stocks of Greenland halibut, saithe and plaice have become seriously depleted. In order to remedy this situation, there have been sharp cuts in recommended catch limits for these

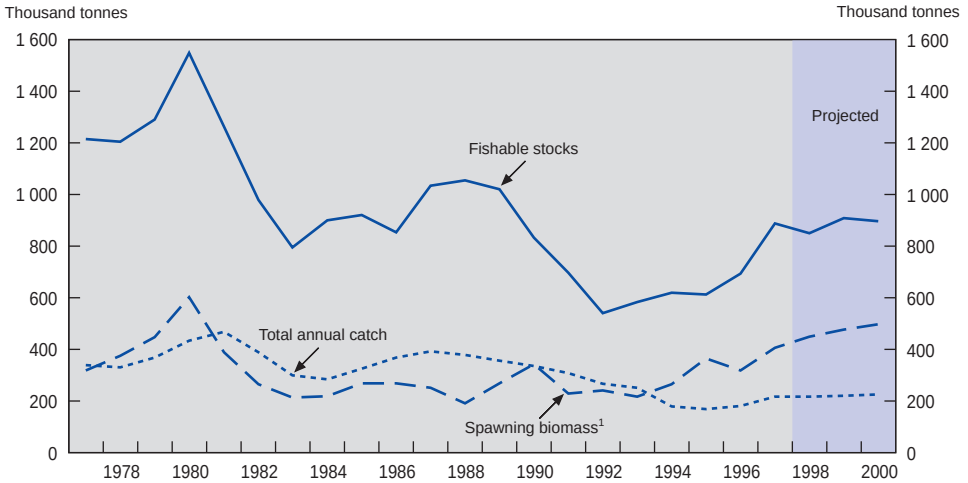
Table 5. **Marine catch**

	Total Allowable Catch (TAC), Icelandic waters						Icelandic waters				Foreign waters		International TAC	
	FY 1997/98		FY 1996/97		FY 1995/96		Icelandic vessels		Foreign vessels		Icelandic vessels		Iceland's quota	
	Recom-mended	Actual	Recom-mended	Actual	Recom-mended	Actual	IKr million	Thousands of tons	IKr million	Thousands of tons	IKr million	Thousands of tons	1997	1996
Cod	218	218	186	186	155	155	12 783	181.0	51	0.7	1 625	23.0		
Haddock	40	45	40	45	55	60	4 178	56.2	47	0.6	0	0.0		
Saithe	30	30	50	50	65	70	1 746	39.3	36	0.8	1	0.0		
Redfish	65	65	65	65	60	65	4 694	67.7	0	0.0	0	0.0		
Oceanic redfish	n.a.	n.a.	n.a.	n.a.	150	n.a.	2 592	46.9	0	0.0	391	6.1	45	45
Herring	100	100	100	110	110	125	2 056	265.4	0	0.0	0	0.0	233	190
Capelin	850	986.7	1 100	1 287	800	1 150	6 491	1 178.5	13	2.4	3	0.5		
Other fish							8 627	82.2	198	1.9	1 755	16.7		
Halibut	10	10	15	15	20	20	4 248	22.1						
Plaice	9	9	10	12	10	13	1 191	11.1						
Wolffish	13	n.a.	13	n.a.	13		900	14.6						
Dab	7	n.a.	7	n.a.	0		456	8.0						
Long rough dab	5	n.a.	5	n.a.	0		346	6.4						
Tusk	n.a.	n.a.	6	n.a.	6		269	5.2						
Witch	1.1	1.1	1.2	1.2	0		164	1.5						
Other[1]							1 053	13.4						
Crustaceans							8 638	86.4	0	0.0	486	91.3		
Shrimp							7 931	68.5						
Inshore	7.15	11.9	7.6	11.9	5.6	11	0	0.0						
Offshore	70	75	55	60	40	63	0	0.0						
Lobster	1.5	1.5	1.5	1.5	1.5	1.5	382	1.6						
Scallops	8	8.9	9.3	8.9	9.45	9.25	274	9.0						
Other							52	7.3						
Miscellaneous							16	0.2	0	0.0	0	0.0		
Total							51 820	2 003.9	345	6.5	4 260	137.7		

1. Includes capelin roe.

Source: *Útvegur*, 1996; the Marine Institute, *State of the Marine Stocks in Icelandic Waters 1996/97*; Prospects for the Quota Year 1997/98; and the NEI.

Figure 9. **ICELANDIC COD**



1. The spawning biomass refers to average number of newly born fish multiplied by their estimated weight at maturity.

Source: The Icelandic Marine Institute.

species. In addition, estimates of the cod stock in the Barents Sea area have fallen, and landings by Icelandic boats from this sector were only 7 600 tonnes in 1997 against the 36 000 tonnes caught in 1994 and 1995.[5]

Short-term outlook

The outlook for output and income continues to look robust for the next couple years, with domestic demand expanding rapidly (Table 6). Real employment and wage gains should keep fuelling private consumption, even if the household saving rate edges up further. Business investment should also increase. There may be a further expansion in capital investment outside the power-intensive industries, now that the recovery is well under way, while construction activity in the electrical power generation and distribution industry should expand in 1998. Moreover, some large-scale projects will continue for several years (Table 7). Icelandair has a programme to purchase four jets, only one of

Table 6. **Short-term projections**

Percentage changes, volume (1990 prices)

	1997 estimate	1998		1999	
		Official forecast March 1998	OECD	Official forecast March 1998	OECD
Private consumption	6.0	5.5	5.8	4.0	3.9
Government consumption	1.5	3.0	2.9	2.3	1.8
Gross fixed capital formation	9.9	11.6	8.7	–4.2	1.5
Final domestic demand	5.8	6.1	5.8	2.0	3.0
Change in stockbuilding[1]	0.0	0.0	0.1	0.0	0.0
Total domestic demand	5.8	6.1	5.9	2.0	3.0
Exports of goods and services	5.6	3.1	5.3	6.1	6.7
Imports of goods and services	8.0	7.2	9.0	3.5	5.6
Change in foreign balance[1]		–1.3	–1.2	0.9	0.5
GDP	5.0	4.6	4.6	2.9	3.4
GDP implicit price deflator	3.5	3.7	3.9	3.6	3.6
Consumer price deflator	1.8	2.7	2.4	3.2	3.2
Unemployment rate (in per cent)	3.9	3.6	3.5	3.4	3.4
Current balance[2]	–1.5	–2.9	–1.8	–2.2	–1.7
3-month Treasury Bill	7.0	n.a.	8.3	n.a.	9.0
5-year indexed Treasury Bill	6.0	n.a.	6.4	n.a.	6.5

1. As a percentage of GDP in the previous period.
2. As a percentage of GDP.

Source: National Economic Institute and OECD.

which has been delivered. In the summer, the enlargement of the ferrosilicon plant will begin, and in the next few years, Columbia Ventures will likely expand the aluminium plant that is currently under construction. Domestic demand could get an additional stimulus from the government sector because the cyclically adjusted budget is expansionary (see Chapter II). At the same time, exports should be boosted by a full year's output from the enlarged aluminium plant, while marine production is expected to grow modestly. With investment in large-scale projects decelerating, the growth of imports is expected to moderate in 1999. Even so, the current account is projected to remain in deficit in the near term. Overall, GDP is projected to grow at about 4½ per cent in 1998 and 3½ per cent in 1999, bringing output above its estimated potential level and unemployment below 3½ per cent in 1999.

Table 7. **Large-scale investment projects**

Billions of 1996-97 IKr

Company	Project	Cost	Temporary imports	Period	Long run		
					Exports	Imports	Investment income
Announced:							
ISAL	Expand aluminium plant capacity by 60 000 tonnes	12	8	1996-97	6.5	3.0	–2.0
Columbia Ventures	Build 60 000 tonne aluminium plant at Grundartangi	12	8	1997-99	6.5	3.0	–1.9
National Power Company and other infrastructure	Build new power plants, expand grid and other improvements	23	16	1997-2000	n.a.	n.a.	–0.2
Iceland Alloys	Expand Grundartangi ferrosilicon plant capacity by 40 000 tonnes	3	2	1998-99	1.9	0.9	–0.6
Icelandair	Buy two Boeing 757-200s and two 757-300s	19	19	1998-2002	5.6	0.5	–1.5
Columbia Ventures	Expand capacity of Grundartangi plant under construction by 120 000 tonnes	24	16	1999-2000	13.0	6.0	–3.9
Speculative:							
Hydro Aluminium	Build 200 000 tonne aluminium plant, including electric power supply	100	67		22	10	–10
Atlantal	Build 180 000 tonne aluminium plant	48	32		19.5	9	–6
Icelandic Magnesium	Build 50 000 tonne magnesium plant	35	24		9	0	–6
Icelandair	Exercise existing options on eight more planes	38	38	2003-13	11	1	3

n.a. = not applicable.

Source: The NEI and OECD.

The assumptions underlying the above projections are as follows:

- the stock of cod in Icelandic waters should continue its upward trend, allowing some increased production for exports; allowable catches of other species are expected to be roughly unchanged;
- export prices for marine products and aluminium are projected to grow no more than the rate of inflation in the OECD area for the next two years;
- output in the OECD Europe area is projected to increase 2¾-3 per cent in 1998 and 1999;
- the exchange rate is assumed to remain unchanged in effective terms, at its level of 18 March 1998;
- short-term interest rates are expected to increase by 130 basis points in 1998 and a further 70 points in 1999, about ½ percentage point more than the pickup in inflation.

Considerable uncertainty surrounds the point at which scarce resources could result in pressure on domestic prices. In the past, Iceland has enjoyed low unemployment rates, and inflation did not clearly accelerate until the unemployment rate dipped below 1 per cent. Unemployment in the 1990s has been much higher, and there is some evidence that the natural rate of unemployment has increased. Indeed, the point at which inflation picks up, the non-accelerating inflation rate of unemployment (NAIRU), may now be as high as 4 per cent.[6] With output rising above estimated potential during the projection period and unemployment continuing to fall, it is expected that there will be a moderate pick up in inflation as wages rise significantly faster than contracted rates. The private consumption deflator is projected to accelerate by 0.6 percentage point in 1998 and another 0.8 percentage point in 1999, despite some further tightening of monetary policy. The impact of continued strength in aggregate demand on consumer prices will be counterbalanced somewhat by falling import prices in 1998 from the renewed appreciation of the krona and the decline in oil prices. Indeed, the terms of trade are expected to improve further in 1998 before levelling off in 1999. Such favourable expected price developments are expected to take some of the pressure off the current account. In spite of a sharp decrease in net export volumes, the current account deficit is thus projected to widen only ¼ percentage point to 1¾ per cent of GDP in 1998 and 1999.

In 1998, however, there is the risk that the strength of domestic demand could more than offset the decline in import prices and provoke a more marked acceleration in consumer prices than expected, especially if wage drift were to

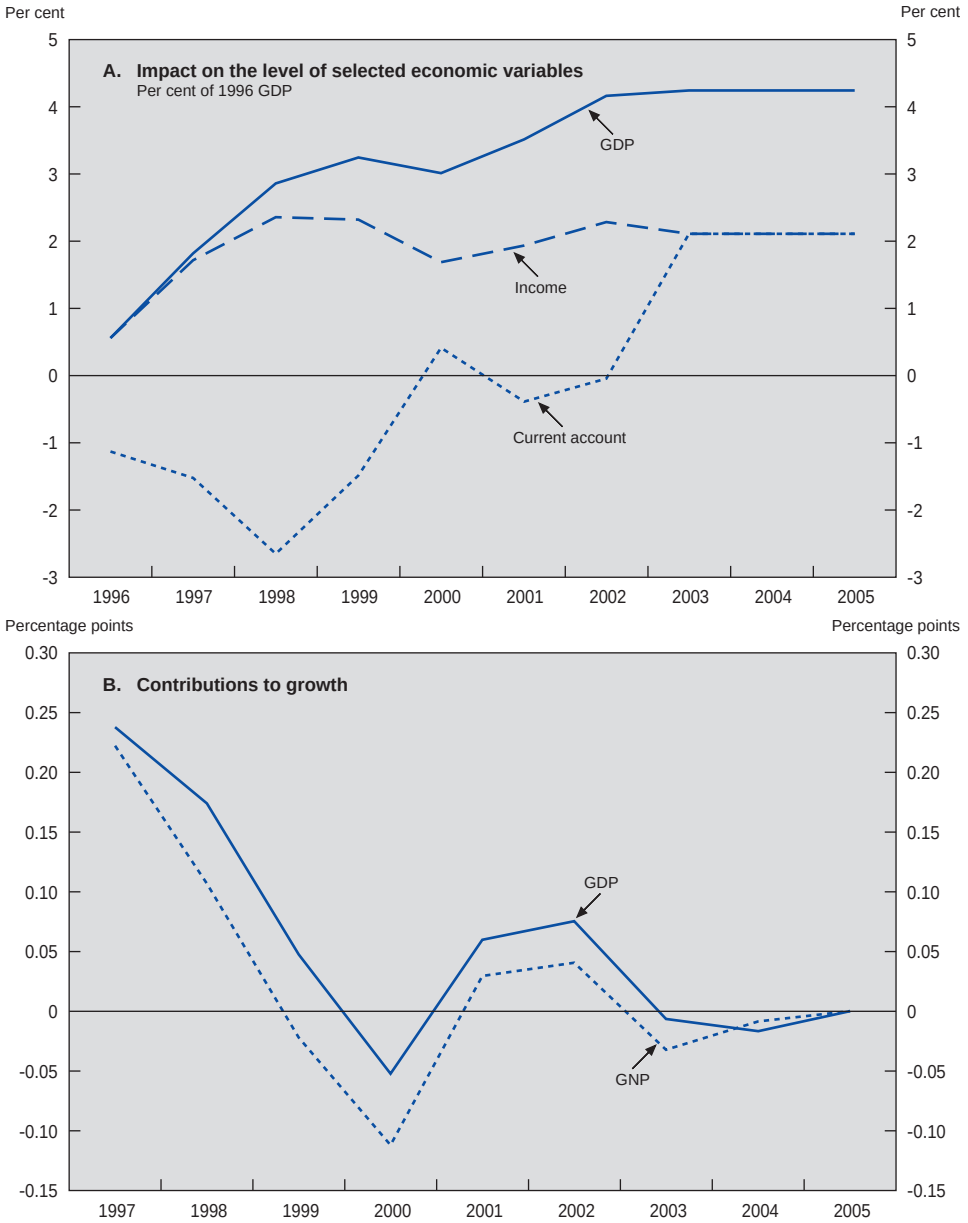
remain elevated. In addition, if new large-scale investment projects were to materialise, there would be further pressure on domestic resources. In such a scenario, monetary policy would have to be significantly tighter to stave off a rapid run up in inflation. In this case the combination of higher interest rates and significant capital inflows from foreign investment would likely counteract any downward pressure on the krona from a pick up in inflation. The fishing industry would be particularly hurt by such a combination of events, and there would be added pressure on the already poor profits in this key sector.

Medium-term prospects

Prospects for the year 2000 and beyond are dominated by the outlook for the fisheries and foreign direct investment. As discussed above, the stock of Icelandic cod appears to be increasing (Figure 8), which should allow some growth in landings in the next five years. The outlook for other species is more uncertain, but there now appears to be a consensus among the fishing industry and the government to protect them. To this end, the government cut the quotas on several species for the 1997-98 fishing year to levels that are in line with the recommendations from the Marine Institute (Table 5). Such a policy should stabilise the level of stocks and eventually lead to some growth in the catch. Because fishing prospects elsewhere in the world are poor, world prices should be well maintained.

On the basis of currently disclosed plans, the level of foreign direct investment may fall after 1999, so reducing demand and output. Indeed the peak impact of these projects on the growth of activity appears to have already passed (Figure 10). However, there is a possibility that additional future projects may take advantage of Iceland's relatively inexpensive electricity.[7] Two additional aluminium projects look promising, especially as several trade reports suggest that demand for aluminium will outstrip supply in the next decade, though events in East Asia could delay the expected run-up in relative prices for a few years. The prospect for a magnesium plant is a little more tentative as several other plants are currently under construction or under consideration elsewhere in the world. Investors are currently investigating other projects, such as an oil refinery, a silica plant and a polyol plant. The realisation of these plans could support investment levels into the next century.

Figure 10. **MACROECONOMIC EFFECTS OF LARGE-SCALE INVESTMENT PROJECTS**



Source: National Economic Institute.

While the ending of construction in existing projects may lower demand, the economy should start to see the benefits of the increased supply coming from the completed plants (Figure 10). Output should be boosted as they start to generate export earnings and their effect on the current account turns positive. National income will rise less than measured output, however, as foreign investors expatriate their profits. Indeed the main long-term gains come through the possibility of indirectly exporting electricity and some labour, while accruing some corporate tax payments.

The current account deficit should also narrow significantly over the medium term. In 1998, it is estimated that these projects will widen the deficit by over 2 per cent of GDP (Figure 9), indeed more than accounting for its 1998 level. Their impact on the current account will gradually diminish over time and then eventually turn positive, pushing it into surplus in the absence of other developments. On the other hand, as investment in other parts of the economy still appears low by historical standards, any pick-up in spending in this area could impact negatively on the current account position in the future.

II. Macroeconomic policies

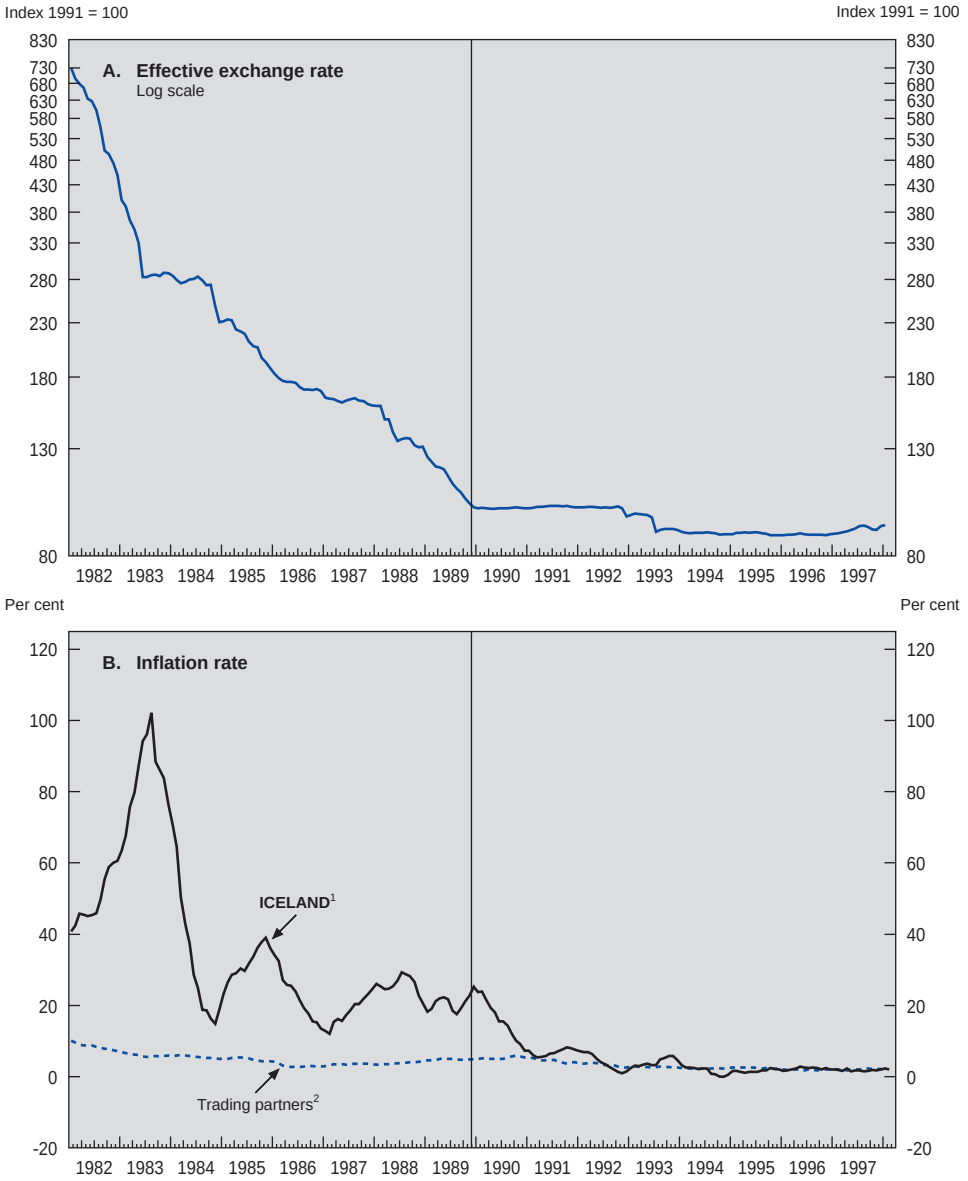
Macroeconomic management in Iceland has undergone profound changes in the past decade. The goal of achieving low inflation has been accorded considerably more weight, while, as a result of substantial fiscal consolidation, public finances have become much sounder. With, as noted above, the economy running close to potential and expected to remain so in the near future, there will be increasing pressure on domestic resources. Such a development poses new challenges to the authorities, which need to be met if the previous gains – in terms of inflation control – are not to be dissipated. After reviewing the changes in policy orientation that have marked the 1990s, this chapter examines recent monetary and fiscal policy developments.

Monetary management

Policy formulation in the 1990s

Since the early 1990s, the conduct of monetary policy in Iceland has been essentially geared towards the maintenance of price stability. This represents a marked shift from the situation in the past. Indeed, in the 1980s, price stability was not given primacy over other policy objectives (such as the maintenance of full employment) contained in the legislation governing the actions of the Central Bank. Monetary authorities attempted to achieve these objectives by allowing devaluation of the currency at certain times. In this way, high employment was preserved despite considerable external economic shocks. Exchange rate adjustments resulted in substantial real wage flexibility, so helping keep unemployment at around 1 per cent of the labour force. However, this policy was inflationary, resulting in annual price increases of over 100 per cent in the early 1980s and still as high as 25 per cent as late as 1989 (Figure 11, Panel A), as currency depreciations were never fully reversed during upswings.

Figure 11. **EXCHANGE RATE AND INFLATION DEVELOPMENTS SINCE THE EARLY 1980's**



1. Measured by the consumer price index.
2. Weighted average of the sixteen primary trading partners.

Source: OECD.

Once a consensus emerged for maintaining low inflation, the role of the exchange rate in the policy framework changed significantly. It became the intermediate target of policy and, in December 1989, the principle of allowing a gradual depreciation of the currency was abandoned.[8] The krona was henceforth to be kept stable against a trade-weighted basket of foreign currencies (Figure 11, Panel A). With the trade unions accepting a low but indexed pay settlement, inflation fell to 5 per cent within eighteen months once the exchange rate stabilised because domestic prices are closely linked to import prices. In January 1992 the currency basket was changed, for the first time, to consist of the European Currency Unit (ECU), the US dollar and the yen,[9] as the authorities expected that there would be little fluctuation in exchange rates between European currencies with the adherence of more countries to the European Exchange Rate Mechanism.

One feature of the policy stance was that some adjustment in the external value of the currency was not entirely ruled out, as long as – in the case of a devaluation – there was clear evidence that the equilibrium real exchange rate had fallen and if there was little danger of such a move provoking a price spiral. Such a situation arose when the pound sterling and several Nordic currencies depreciated against those in continental Europe in the second half of 1992. These exchange rate movements generated a significant terms-of-trade shock for Iceland, given that the countries with depreciating currencies accounted for a higher proportion of Icelandic external trade than their implied weight in the currency basket that was then being used.[10] Another shock occurred in the first half of 1993 when fish prices fell and it proved necessary to restrict the quantity of cod catches in order to allow fish stocks to recover. In response to these events, the exchange rate was devalued first by 6 per cent in November 1992 and again by 7½ per cent in June 1993. Thereafter, the central value of the exchange rate has been held stable, following a second change in the basket of currencies used for the target. Weights are now determined by shares of trade in both goods and services.

Given the depressed state of the domestic economy in 1993, these two devaluations only resulted in a slight, and temporary, increase in the inflation rate above that of trading partner countries (Figure 11, Panel B). Indeed, in the eight years between the introduction of the new policy in December 1989 and the end of 1997, the increase in the domestic *price level* in Iceland was only 2 per cent

more than that in these countries. As a result of such relative stability, at the end of 1997 the real exchange rate was some 7 per cent below the average in the decade prior to December 1989 and 9 per cent below its level in November 1992, prior to the first of the two devaluations that occurred under the post-1989 exchange rate regime.

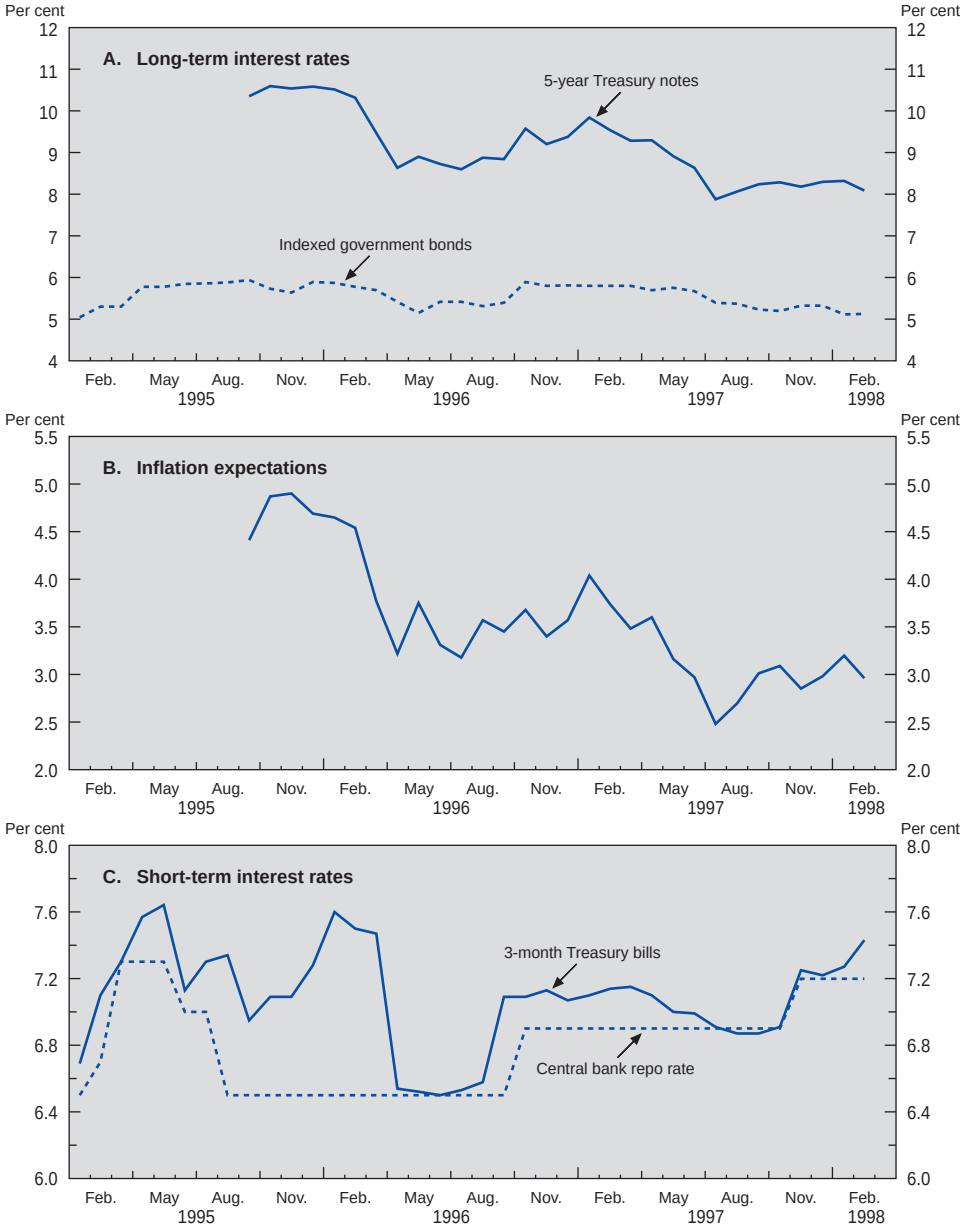
The extent of flexibility around the central exchange rate target was increased as external capital flows became less restricted. Initially, there was a fluctuation margin of ±2¼ per cent. This was never tested, and the exchange rate was kept extremely stable, in the absence of an inter-bank foreign currency market. With the introduction of such a market in 1993, some fluctuations did occur in the value of the currency, but they were slight. However, following the complete liberalisation of capital flows in 1995, wider fluctuation margins (±6 per cent) were implemented in November of that year. The Central Bank nonetheless continued to participate actively in the daily foreign exchange fixing until July 1997. At that point, the nature of the foreign exchange market was changed, with banks expected to maintain continuous two-way quotes for foreign currencies.

Interest and exchange rate developments

During the current economic upswing, the Central Bank has, for the first time, taken advantage of the upward flexibility incorporated in the exchange rate fluctuation margins. Indeed, inflation considerations appear to have been gaining an increasing weight in the determination of interest rates, rather than concerns about the external value of the currency. By the middle of 1996, the strength of domestic demand had ended the drop in government bond yields (Figure 12, Panel A). Moreover, inflationary expectations, as measured by the difference in the yields on ordinary and indexed government bonds, had stabilised and showed some signs of edging up (Figure 12, Panel B). There was also growing uncertainty over the outcome for the wage settlements that were due to be agreed at the end of the year. Thus in September 1996, the official interest rate on three-month repurchases was raised 40 basis points to 6.9 per cent (Figure 12, Panel C), in the absence of any downward pressure on the exchange rate.

In the event, however, such an increase in interest rates gave rise to some conflict with the objective of a stable exchange rate. Indeed, in the second half of 1996, private capital inflows increased (Table 8), in line with the widening of

Figure 12. **INTEREST RATE DEVELOPMENTS**



Source: Central Bank of Iceland.

Table 8. **Capital account of the balance of payments**

IKr billion

	1993	1994	1995	1996	1997[1]
Current account	0.3	8.7	3.7	–8.3	–4.9
Financial flows excluding reserves	–2.1	–20.1	1.3	16.0	0.4
Direct investment, net	0.2	–1.7	–0.8	–0.2	4.3
Portfolio investment, net	–2.8	–7.5	–3.7	–1.6	–8.3
Other capital movements, net	0.5	–10.9	5.9	17.8	4.5
Public sector	4.3	12.9	14.3	0.2	–5.9
Monetary authorities	1.7	3.3	1.3	–6.7	0.0
Public enterprises	–1.7	–1.6	–1.7	–0.2	–0.6
General government	4.3	11.2	14.8	7.1	–5.3
Other sectors	–3.8	–23.8	–8.5	17.6	10.4
Deposit banks	–5.0	–12.3	–1.3	16.3	11.9
Other	1.2	–11.4	–7.2	1.3	–1.5
Net errors and omissions	–2.4	0.9	–4.7	2.5	3.8
Reserves[2]	4.1	10.7	–0.2	–10.2	0.7

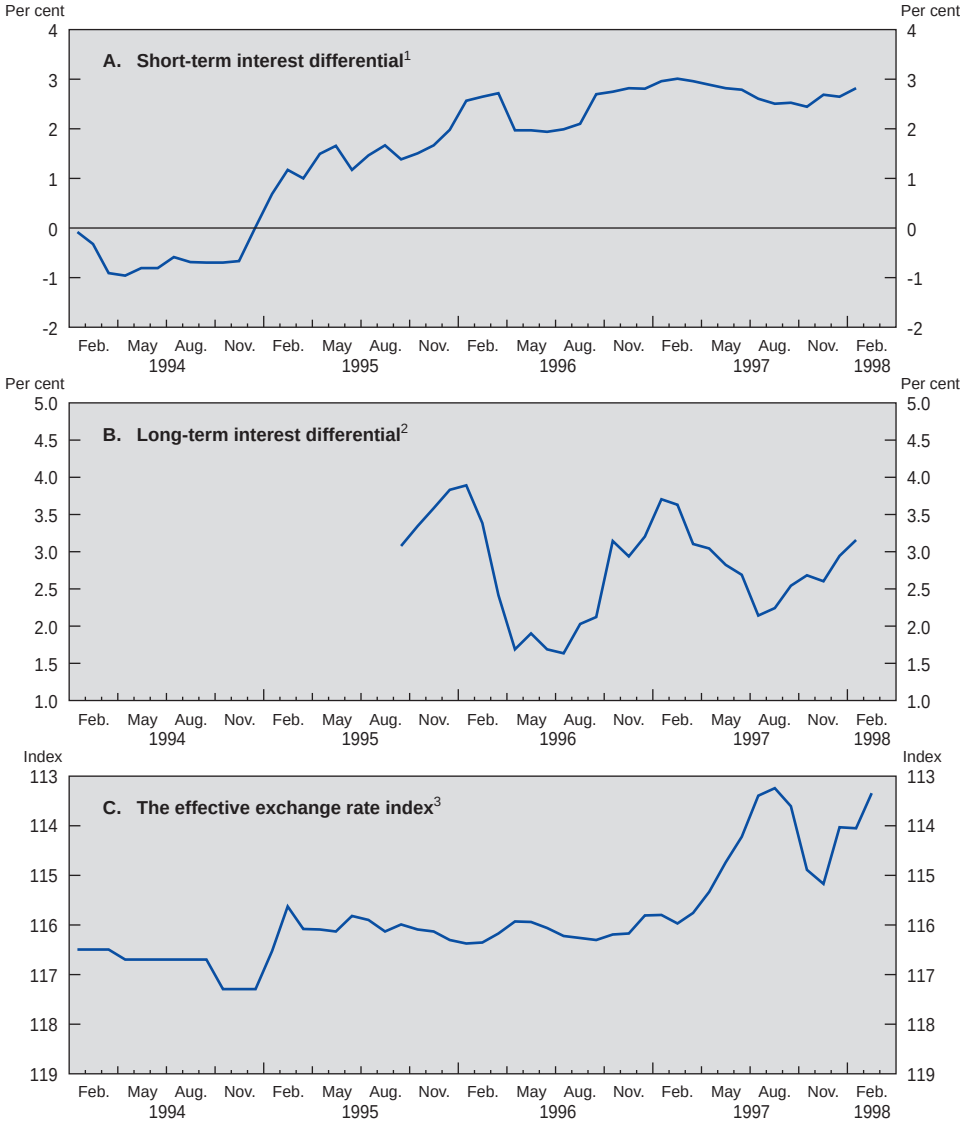
1. First three quarters of the year.
2. A positive figure indicates a decline in reserves.

Source: Central Bank of Iceland.

the margin of Icelandic rates over foreign short-term rates and a widening long-term interest rate differential that was only partly justified by higher domestic inflationary expectations (Figure 13, Panels A and B). Against this, the public sector reduced the extent of its net new overseas borrowing. Nonetheless, upward pressure on the exchange rate occurred at that time, despite the widening current account deficit. The Central Bank initially resisted this movement, so that its net foreign assets more than doubled during the year (Table 9).

After the finalisation of the wage settlements in March 1997, the Central Bank moved to ensure a greater transmission of the monetary tightening to the domestic economy, using the flexibility margin built into the exchange rate objective. A slight appreciation of the exchange rate was allowed in the first half of the year (Figure 13, Panel C) and the extent of official intervention was markedly reduced from the level seen in the second half of 1996. By August 1997, the currency had appreciated some 2½ per cent from September 1996 when interest rates had been increased. Nonetheless, the continued strength of the economy and some relatively poor monthly inflation figures

Figure 13. **INTEREST RATE DIFFERENTIALS AND THE EXCHANGE RATE**



1. The short-term interest rate differential is the difference between the three month rate on Icelandic treasury bills and a trade weighted average of overseas short-term rates.
2. The long-term interest rate differential is the difference between the yield on a 5 year Icelandic government bond and a trade-weighted rate on 10 year overseas government bonds.
3. As defined by the Central Bank.

Source: Central Bank of Iceland and OECD.

Table 9. **Foreign reserves of the Central Bank**

Ikr million, end of period

	1994	1995	1996	1997			
				March	June	September	December
Total foreign assets	24 353	25 365	35 677	31 137	39 482	35 609	33 287
of which:							
Reserves	20 317	20 239	30 468	25 927	34 252	30 338	27 807
Other foreign assets	4 034	5 127	5 210	5 210	5 230	5 271	5 480
Short-term foreign liabilities	9 397	11 378	4 929	6 660	5 184	5 259	5 418
Net foreign assets (reserves)	14 956	13 987	30 748	24 477	34 298	30 350	278 679
Memorandum items:							
Foreign assets (US$ millions)							
Gross	357.0	389.3	534.8	440.3	560.8	501.1	462.4
Net	219.3	214.7	460.9	346.1	487.2	427.1	387.2

Source: Central Bank of Iceland and OECD.

sufficed to raise inflation expectations and to reverse the previous appreciation of the exchange rate despite sales of foreign exchange by the authorities in the market. As a result, in November 1997, the Central Bank raised its lending rates, lifting that for short-term repurchases by a further 30 basis points to 7.2 per cent. The move was sufficient to result in a currency appreciation to its previous high level. However, this did not generate an improvement in inflation expectations, which remained at around 3 per cent. Nonetheless, government bond yields dropped as the rates on indexed securities fell some 50 basis points, reflecting both increased foreign demand for such instruments and lower government borrowing.

The behaviour of money and credit aggregates

Despite the tightening of monetary conditions, there has been an acceleration in monetary expansion. The growth of the broad money stock moved ahead markedly, especially during the middle of 1997 (Table 10). By the end of the year, it was running at more than double the growth of nominal GDP, reflecting, in part, a greater willingness of the banks to expand their balance sheets. Lending by banks increased, with loans to companies and non-bank financial institutions rising particularly rapidly. This development, however, was offset by a decline in

Table 10. **Money growth and credit**

	1994	1995	1996	1997			
				March	June	September	December
	Change as per cent of previous year's money stock						
Money stock (M3)	2.3	2.2	6.2	6.0	6.7	14.8	10.5
Domestic credit expansion	3.8	1.3	12.1	15.4	18.5	23.6	23.9
Government	2.2	0.7	–3.6	–0.4	0.7	2.0	0.8
Companies	–0.2	–1.7	10.7	12.4	14.0	15.6	17.1
Households	1.9	2.3	4.9	3.5	3.9	6.0	6.0
Foreign assets (net)	–0.6	0.6	1.9	–3.9	–7.0	–5.0	–12.6
Other assets (net)	–0.9	0.3	–7.8	–5.5	–4.8	–3.6	–0.7
Memorandum items:							
	Per cent change over previous year						
Components of the money stock							
Base money	1.6	–4.5	23.6	19.3	30.6	27.8	3.3
Narrow money (M1)	10.7	9.6	8.5	11.3	10.7	12.1	16.6
Bank lending	2.8	1.0	8.7	10.9	13.3	16.1	16.9
Companies	–0.3	–2.1	13.7	15.5	17.5	18.8	20.9
Households	5.8	6.9	13.9	9.5	10.5	15.7	15.9
Government	9.5	3.0	–14.4	–1.7	2.9	8.0	3.7
Investment fund lending							
Housing	13.6	11.2	10.2	9.0	7.5	7.5	..
Industry	1.7	2.6	2.3	3.0	5.3	5.4	..

Source: Central Bank of Iceland and OECD.

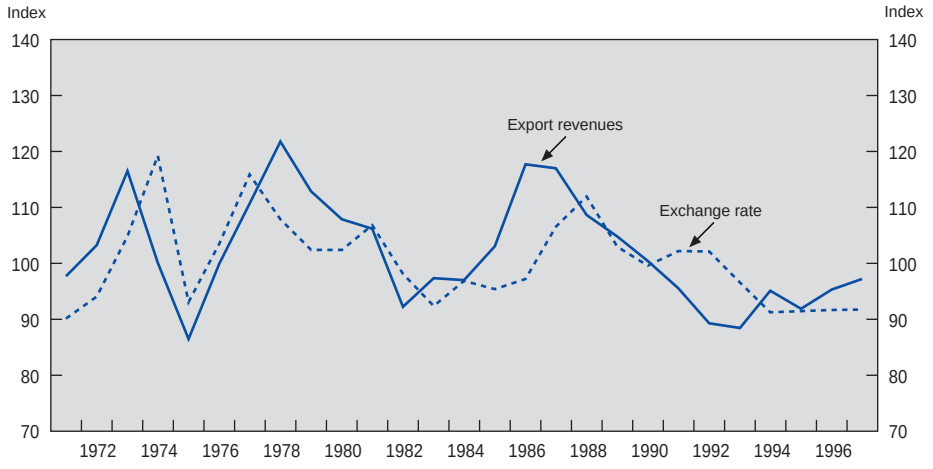
the growth of lending by other non-bank financial institutions such as investment and pension funds (Table 10). Overall in the year to June 1997, the increase in the assets of the credit system (including both banks and non-bank institutions) was stable at around 9 per cent.

The challenges ahead

As noted in the previous chapter, with labour market becoming tight, the Secretariat projects that wage and price inflation may accelerate in 1998 and 1999. In such circumstances, some further increase in short-term interest rates may be necessary to stabilise domestic demand and prevent a rise in inflation expectations. As this may entail upward pressure on the exchange rate, it

is important that such pressure not be resisted, in order to preserve the efficacy of monetary policy. If the pressure was excessive, this may result in the need to either widen the existing ±6 per cent fluctuation margin of the krona's exchange rate or to raise its central rate. Such a course would mean that the current policy framework is symmetric, allowing the devaluations that occurred in the 1992-93 recession to be offset at the top of the cycle. More generally, adhering less strictly to the exchange rate stability target would allow monetary policy to take into account the volatility of the Icelandic economy and its exposure to external shocks. Indeed, past experience shows that movements in the exchange rate (Figure 14) have played an important role in stabilising the economy when it was faced with external fluctuations that were larger than those experienced by other small open countries and which, moreover, were not correlated with economic developments in the rest of the world.

Figure 14. **REAL EXPORT REVENUES AND THE REAL EXCHANGE RATE**[1]



1. Real export revenues are measured as the ratio between actual revenues and Hodrick-Prescott trend. The real exchange rate index is as defined by the Icelandic Central Bank (1990 = 100).

Source: Icelandic Central Bank and OECD.

The fiscal stance

A successful consolidation process

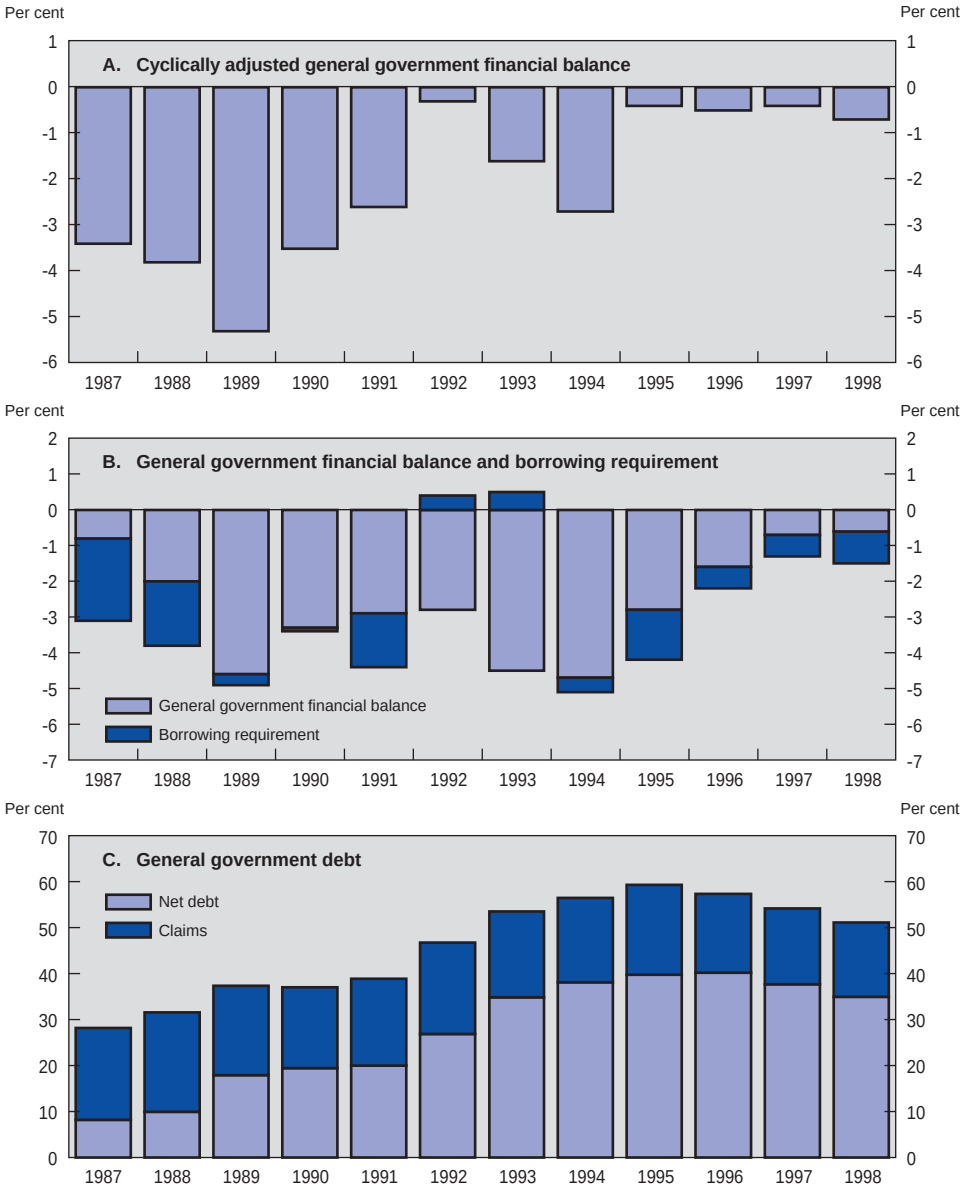
During the 1990s, successive governments have regarded fiscal consolidation as essential to stabilise the economy and, through lower interest rates, to help raise investment and long-term growth. This orientation was reflected in a marked reduction of the budget deficit from 5 per cent of GDP at the end of the 1980s to close to balance by 1997 (Figure 15, Panel A). Given the delay in improving the fiscal position, however, total government debt rose during the process of consolidation (Figure 15, Panel C). Indeed, net debt more than doubled between 1989 and 1997, reaching 36 per cent of GDP. Gross debt, which is higher than net debt due to government financial intermediation, rose from 37 to 52 per cent of GDP, as the government continued with its policy of indirect lending to the private sector.

The consolidation of government finances has essentially been the result of a reduction in public spending. Since 1989, outlays have fallen by almost 5 per cent of GDP, the cuts being concentrated in the field of government support to private sector economic activity, with both subsidies and capital transfers falling. Real spending in these two areas dropped 9 per cent annually in the past eight years, payments to agriculture, housing and health services being most particularly affected. On the other hand, total taxation has been held constant at just over 33 per cent of GDP – one of the lowest rates in the OECD area and almost half below that found in other Nordic countries. Within the total tax yield, there was a move towards income taxation away from indirect taxation, which nonetheless still accounts for over half of total tax revenues.

The initial reduction of the structural deficit in the early 1990s (Figure 15, Panel A) was offset by adverse cyclical developments stemming from both the transition costs of lowering inflation and adverse movements such as the fall in the fish catch. In addition, local authorities added to their borrowing in 1993 and 1994 at the time of the local elections. As a result, the actual general government deficit remained high up to the mid 1990s and only began to fall when activity started recovering subsequently. When local authorities reined back their borrowing after the elections and tax revenue rose for cyclical reasons, the actual deficit was quickly reduced and brought close to balance by the end of 1997 (Figure 15, Panel B).

Figure 15. **FISCAL INDICATORS**

Per cent of GDP



Source: *Statistical Yearbook of Iceland* and OECD.

Moving towards a budget surplus

The 1997 and 1998 central government budgets

The 1997 budget marked the culmination of the consolidation process, with the Treasury budget balance projected to improve markedly, registering a slight surplus (Table 11). The budget was affected, however, by the transfer of primary education from the central government to local authorities, together with compensating revenue. In addition, as in the previous year, the early repayment of certain index-linked loans boosted expenditure.[11] To avoid distorting the year-to-year changes in fiscal positions, the figures for spending, revenues and the fiscal balance have been adjusted to remove the impact of these factors, following the approach adopted by the government. On such an adjusted basis, the outturn of the Treasury budget for the year shows a significantly higher than expected surplus amounting to almost 1 per cent of GDP (Table 11) as the result of a large increase in revenues.

Table 11. **Treasury finances in 1997 and 1998**

Cash basis

	1993	1994	1995	1996	1997 budget	1997 estimate	1998 budget
	Percentage increase from previous year						
Corrected for the transfer of responsibilities and financial transactions[1]							
Revenue	–0.2	6.2	4.4	116	3.5	7.8	6.0
Expenditure	2.0	3.7	5.4	5.2	1.3	2.7	7.7
	Per cent of GDP						
Revenue	25.1	25.2	25.3	26.3	25.0	26.1	25.5
Expenditure	27.4	26.9	27.3	26.7	24.9	25.1	24.9
Revenue balance	–2.3	–1.7	–2.0	–0.4	0.0	0.9	0.5
Uncorrected							
Revenue balance	–2.3	–1.7	–2.0	–2.5	0.0	0.2	0.5

1. The corrections allow for the transfer in 1997 of central government revenue and spending amounting to 4.7 and 4.1 per cent of 1996 revenue and spending levels, representing 1.0 and 1.1 per cent of GDP respectively. In addition, debt repayments of Ikr 10 billion and Ikr 4 billion have been deducted from spending in 1996 and 1997 respectively.

Source: Ministry of Finance.

More rapid-than-expected economic growth indeed resulted in *revenue* expanding almost twice as quickly as allowed for in the budget. Stronger growth in employment and wages boosted both payroll and income taxes above expectations (Table 12). The increase in the latter occurred despite the reduction in the standard rate of income taxation at the beginning of May 1997, the adverse revenue consequences of which were limited by a reduction in the personal income tax credit – so keeping the income tax threshold unchanged.[12] Such changes represented the first part of a three-year tax reform programme (see below). The yield from indirect taxes was also above its budgeted level, helped by the very strong level of auto sales that lifted the receipts of excise taxes and special vehicle taxes. Once again, this buoyancy occurred despite the reduction of a number of excise taxes at the beginning of 1997. The duty on goods subject to a 30 per cent tax was reduced to 25 per cent, that on auto parts was lowered from 20 to 15 per cent, while duties on cosmetics, films, pens and pencils were abolished. The full year effects of the ending of the discrimination between imported and domestic goods in the determination of excise taxes also lowered the yield of indirect taxation.

Table 12. **1997 Budget: initial projections and outcomes**

	Level 1996 per cent of GDP	Overrun 1997 per cent		Level 1996 per cent of GDP	Overrun 1997 per cent
Expenditure			Revenue		
Operating expenditure	10.8	5.0	Direct taxation	8.1	5.8
Wages	8.5	4.8	Individuals	4.4	7.6
Other	3.7	3.2	Companies	0.9	4.3
Service charges	–1.5	–0.3	Social security	2.8	4.1
Transfer payments	10.2	1.9	Indirect taxation	15.8	3.0
Social security	6.0	0.6	Value added tax	9.4	2.4
Unemployment	0.6	0.0	Duties and excises	2.2	8.9
Agricultural subsidies	1.2	3.0	Liquor, tobacco	1.5	–2.2
Student loans	0.3	–0.1	Vehicles	1.9	4.7
Municipal equalisation	0.5	3.6	Other	0.8	0.7
Other transfers	1.6	–9.8			
Interest payments	2.8	–7.9	Wealth taxes	0.8	9.4
Maintenance	0.8		Other revenue	1.7	8.5
Capital expenditures	2.1	1.2			
Total expenditure	26.7	0.9	Total revenue	26.4	4.4

Source: Ministry of Finance.

Government *expenditure* was also above the level projected in the budget, but to lesser extent than for revenues. Three areas accounted for most of the slippage: wages, transfer payments and maintenance outlays. The budget had assumed an increase of 3½ per cent in public sector wages. In the event, the wage agreements that were signed in the middle of the year generated increases of 4¾ per cent, provoking an overrun of IKr 1.8 billion. There was also some slippage in social security spending where the budget had allowed for an increase in benefits in line with inflation. Following the wage agreements, the government raised benefits in March and August – bringing the total increase to 8¾ per cent. In addition, it proved difficult to achieve the planned reduction in health service spending, especially as the Health Insurance scheme is a third party payer. Despite these upward drifts, however, total spending (excluding debt redemption and before allowing for the transfer of functions to local government) rose by only 1 per cent in real terms as there was a significant drop in interest payments and capital expenditure (Table 11).

Adjusted for the special features signalled above, the *1998 central government budget* continues to show a surplus, but somewhat smaller than that registered in the previous year, as revenue is projected to increase somewhat less rapidly than expenditure, (Table 11). Moreover, the *revenue* figures include the profits on the sale of the government share in two companies: the Icelandic Investment Bank and Icelandic Prime Contractor. These gains amount to IKr 2.7 billion (0.5 per cent of GDP) on total proceeds of IKr 7.7 billion. Without such gains, the growth in revenue would only be 4.6 per cent, about half the expected growth of nominal GDP. As for *expenditure*, the full payment of the 1997 increase in pay and social benefits, together with the 1998 hikes, is projected to raise both operating expenditure and social security transfers by over 8 per cent. Such large increases are being partially neutralised by static spending on other types of transfer payments, capital projects and maintenance.

Local authority budgets

In contrast to developments at the central government level, some deterioration appears to have occurred in local government finances in 1997. This turnaround came after a significant improvement in their financial position in 1995 and 1996 (Table 13) when the deficit of the fourteen largest authorities, covering almost 80 per cent of the population, declined to less than IKr 1 billion. The initial budgets of these governments pointed to a further improvement in 1997.

Table 13. **Local government finances**

Accrual basis

	1993	1994	1995	1996	1997	1998
	Percentage increase from previous year					
Corrected for the transfer of responsibilities to local authorities						
Revenue	–1.4	2.4	8.1	14.3	4.6	7.0
Expenditure	7.1	7.6	–6.3	11.1	8.3	7.7
	Per cent of GDP					
Revenue	8.2	7.9	8.2	8.8	9.6	9.6
Expenditure	9.3	9.5	8.5	8.8	9.9	9.9
Revenue balance	–1.2	–1.6	–0.3	–0.1	–0.3	–0.3

1. The corrections allow for the transfer in 1997 from central government of revenue and spending amounting to 4.7 and 4.1 per cent of 1996 of local authorities revenue and spending levels, representing 1.0 and 1.1 per cent of GDP respectively.

Source: National Economic Institute.

However, spending outcomes suggest that the actual deficit will be about ¼ per cent of GDP larger than both the initial budget and the deficit for 1997. Such a deterioration has also occurred in previous election years (1989 and 1993) and has not proved to be long-lasting, so only a small deficit is expected in 1998.

General government financial position

In 1997, there was a reduction in the general government deficit that was mainly accounted for by the move into surplus of the central government budget balance. Despite this surplus, the general government balance remained in deficit for two reasons. *First*, as noted, local authorities ran a small deficit and *second*, the central government budget is calculated on a cash basis, albeit with certain adjustments, while the general government budget is measured on an accruals basis. The move from a cash basis to an accruals basis indeed increases the balance of public spending and taxation by about 1 per cent of GDP, so bringing the general government balance into deficit (Table 14). Starting this year, the central government budget will also be presented on an accruals basis – though with no comparative data for the past (see below). With the central government surplus on an adjusted cash basis projected to decline slightly, the general government deficit should widen somewhat in 1998. Given the projected strength of the economy, such an increase would indicate that the structural budget deficit

Table 14. **General government financial balance**

Accrual basis

	1994	1995	1996	1997	1998
	Per cent of GDP				
Budget balance cash basis	–1.7	–2.0	–2.5	0.2	0.5
Budget balance adjusted for debt redemption	–1.7	–2.0	–0.4	0.9	0.5
Central government balance, SNA basis	–3.2	–2.6	–1.5	–0.2	–0.6
General government balance, SNA basis	–4.7	–3.0	–1.6	–0.5	–0.9
Structural balance	–2.7	–0.4	–0.5	–0.2	–1.0

Source: Ministry of Finance, Central Bank, National Economic Institute and OECD.

was widening, reflecting the delayed impact on spending of the decision to increase wages and social benefits in 1997 and the continued implementation of the three-year tax reform programme (see below).

Implications for public debt

The reduction in the general government deficit implied that the growth of net government debt moderated in 1997. In 1998, the increase should be significantly less than that of nominal GDP, pulling down the ratio of net debt to GDP to 32 per cent by 1998 from 36 per cent in 1997. Nonetheless, net debt is set to remain significantly higher than at the beginning of the 1990s when fiscal consolidation started. In addition, the accumulated pension liability for state employees is estimated by the government to have been IKr 81.4 billion in 1995 (15 per cent of GDP).

The gross debt to GDP ratio is expected to fall slightly more than that for net debt, ending 1998 at 48 per cent of GDP. In Iceland, gross debt is substantially higher than net debt because a substantial part of gross borrowing has funded lending to the private sector, so creating a claim by the government on the rest of the economy. While the government will continue to lend to the Housing Fund, in 1998, this lending can be financed by the planned privatisations without additional borrowing. Indeed, instead of an average increase in financial claims of 0.7 per cent of GDP in the period since 1989, the Treasury will reduce its financial claims on the rest of the economy by 0.3 per cent of GDP in 1998, so creating a negative net borrowing requirement.

Multi-year tax reform

In the 1997 budget, the first stage of a three-year reform of the tax system was implemented, covering the taxation of income from employment, pensions and capital as well as the payroll tax. The standard rate of income tax was reduced from 42 to 40.9 per cent. The reform will continue with a further reduction of 1.9 and 1 percentage points in this tax in 1998 and 1999 respectively. By then, the standard rate will be just under three points higher than when the current income tax system was introduced in 1988. For higher earners the impact of this reduction in the standard rate will be limited by a 2 percentage point increase in the higher income tax rate to 47.9 per cent (a surtax of 7 percentage points). The number of people affected by the surtax, however, will be reduced as the income threshold for payment also will be raised significantly.

The reform will also include a freeze of all tax credits and child benefits, changes in capital income taxation, and the abolition of certain tax incentives for saving. *Child benefits* were held constant in 1997, and the system will be fundamentally changed in 1998. In the past, there was a general flat-rate child benefit and means-tested supplementary benefits for low-income families. The latter were reduced by between 6 and 15 per cent of increases in income above a certain threshold, depending on the number of children. From 1998, the two categories of benefits will be merged and will be means tested. By broadening the base to which the means test is applied, it will be possible to apply a lower marginal rate of reduction that will now vary between 5 and 11 per cent. Housing benefits have also been means-tested through a benefit reduction amounting to 6 per cent of income above a threshold. The government has taken two further initiatives to lower the disincentives caused by means-tested benefits: the repayment rate for student loans has been reduced, as well as the claw-back rate on supplementary pensions. But despite all these changes, some taxpayers will still face marginal tax rates of up to 51 per cent, once the means-tested benefits are taken into account.[13]

A new system of *capital income taxation* was also introduced in 1997. However, as it is only paid once annual income is known, the full revenue from this tax will not be received by the government until 1998. Under the new scheme all capital income and gains are taxed at 10 per cent. This uniform rate replaces those applied on different forms of capital income, thereby reducing the distortion between investments in different financial instruments. While the

resulting taxation on capital income might appear low in relation to that on employment income, it comes in addition to a tax on net wealth. Individuals' net worth above IKr 3.5 million ($50 000) is indeed subject to a tax of 1.2 per cent, which rises to 1.45 per cent when net wealth is above $75 000. Thus an investor holding securities that produce a real return of 5 per cent, faces a total maximum tax of just over 38 per cent on income received – in line with the tax on employment income.[14]

Finally, the *payroll tax*, levied to help finance social security benefits, is being made neutral across industries. This change will be implemented over a three-year period starting in 1997. Prior to the change, the tax was levied at two rates. Employers in the agriculture, fisheries, manufacturing, hotels, restaurants, car rentals and computer services industries pay a low rate of 3.55 per cent, while other service industries pay a rate of 6.85 per cent. As from 1999, there will be one uniform rate of 5.5 per cent, which corresponds to the average rate levied in 1996.

As such, the three-year tax reform programme is expected to result in an income tax reduction of IKr 5.2 billion (about 1 per cent of GDP). This will be partially financed by the new capital income tax, the ending of tax concessions on share purchases and the freezing of tax credits and benefits in 1997. These three measures will still leave a margin of IKr 2 billion, which the government intends to finance by expenditure reductions.

Achieving increased budget transparency

As noted, the government presented its 1998 budget more transparently, using accrual accounting techniques, with the objective of promoting more efficient and effective financial management practices in central government. As far as measuring government income and expenditure is concerned, the reform has two aspects: adopting accrual accounting and reducing the extent of netting revenues against expenditures and conversely. With respect to *accrual accounting*, the principal differences occur in the area of public employees pension payments, debt interest payments and tax write-offs. For the former, it is the change in the present value of liabilities that now enters into expenditure rather than the actual pensions paid in a given year. For debt, the increase in value of indexed and foreign currency liabilities now enters into expenditure directly as the value of the bond changes.[15] In addition, as taxation will be accounted for on

an accruals basis, it will be necessary to add a write-off line to expenditure to the extent that accrued taxes are not collected. As to *netting procedures*, the income tax credits for child benefits, housing and churches are now counted as expenditure. On the other hand, various user charges are counted as revenue rather than as negative spending.

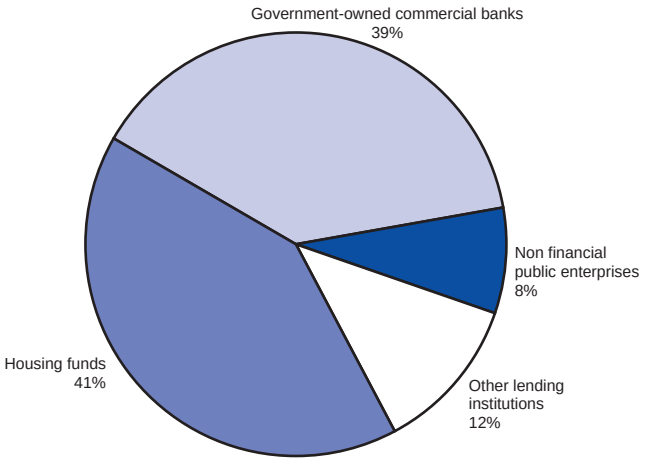
Overall, these changes in accounting procedures will have the impact of increasing the stated total of government spending by 21 per cent but raising revenues by only 19 per cent. As a result, the 1998 budget balance, measured on an accruals basis, should fall by 0.5 per cent of GDP to register only a slight surplus. However, Central Bank estimates show that the adjustment to an accruals basis may have a larger impact on the budget position.[16]

As another example of its move towards more transparent accounting practices, the government has decided to publish accounts for all its lending and borrowing guarantees at the same time as the budget. To that end, the government has started to publish four consolidated statements, replacing the previous credit budget. Two of these cover the activities of wholly publicly owned non-financial and financial enterprises, while the others concern the activities of partially publicly owned enterprises. They show that public institutions continue to make large calls on financial markets, amounting to a projected 2.3 per cent of GDP in 1998, a large part of which will be covered by state guarantees. Overall, the outstanding stock of such guarantees amounted to IKr 330 billion in 1995, some 73 per cent of GDP (Figure 16). Such a stock should decline in the future as the financial sector is progressively privatised (see Chapter III). The combined total of gross government debt, the unfunded liability for state employees' pensions, and publicly guaranteed borrowings amounts to some 140 per cent of GDP.

Prospects and policy implications

The overall current position of public finances in Iceland is good. The general government deficit has been almost eliminated, and public debt is low relative to most European countries, the more so when account is taken of the assets of the central government. This performance has been achieved at the same time as a welcome reform of income and payroll taxation has been introduced, lowering marginal tax rates and reducing discrimination between industries. In addition, public finances will be relatively little affected by ageing as the government has contained its liability to paying just a basic pension. Supplementary

Figure 16. **THE DISTRIBUTION OF GOVERNMENT GUARANTEES BY SECTOR**
1995



Source: Ministry of Finance.

pensions are made through fully funded private-sector pension plans that are mandatory for all economically active people. Moreover, the budget has become more transparent, being presented on an accruals basis.

Over the medium term the government envisages a continuing slight surplus in Treasury finances (on an accruals basis). This reflects a scenario in which output growth is expected to continue at a rapid pace, lifting the economy further above the level of potential output estimated by the Secretariat, and in which inflation is expected to accelerate. In its projections, the growth of public expenditure slackens progressively from the rapid pace expected in 1998, reflecting slower growth in wages and a stabilisation in debt interest payments. This will allow taxes to grow less rapidly than GDP while maintaining a slight surplus. If local authorities also continue to balance their budgets, the net debt of the general government sector could fall to 29 per cent of GDP by 2001, as much as for gross debt.

Despite such a favourable past performance and promising outlook, there is a case to be made for a tighter fiscal policy in the current conjuncture. As noted, the domestic economy is growing rapidly and yet there is little or no further

improvement expected in the general government deficit, indicating an increase in the cyclically adjusted deficit. If the recent improvement in public finances is not to be significantly reversed in any future downturn, the increase in the cyclically adjusted deficit needs to be reversed. Moreover, at the moment, a more restrictive fiscal policy would help ease the burden on monetary policy in keeping inflation under control. To this end, the government needs to identify areas where expenditure can be reduced if it wishes to maintain its tax cuts and move the budget well into surplus. Such a move would also contribute to raising national savings, thereby allowing a reduction of the current account deficit over the medium term.

III. Deregulation of the financial sector

In the 1990s there has been a considerable change in structural policies in Iceland, characterised by a move away from government intervention in markets towards creating a more competitive environment. In part this came through the need to align Icelandic practices in many areas of the economy with those of its European partners in the context of the European Economic Area Agreement. Government policy, though, has also moved to ensure that state-ownership of enterprises and regulation does not distort competition nor result in unnecessary burdens on businesses. This chapter reviews the extent of the changes that have occurred, or are planned, in the financial sector and assesses the scope for further action in that field. The next chapter deals, more generally, with structural policy initiatives in the rest of the economy.

Reforming financial markets

The medium-term objective of the government in the financial area has been to achieve a market-based framework open to external influences. Some steps were taken in the late 1980s, such as the deregulation of bank interest rates. However, the bulk of the change was made in the 1990s. Until 1996, the principal measures introduced by the government were intended to create and enhance the operation of financial markets. Since then, the focus has moved to reducing the extensive public-sector ownership of the financial institutions operating in these markets.

Foreign exchange market

At the turn of the decade, the government heavily regulated foreign exchange transactions. Residents were required to sell foreign currency to licensed banks, and approval was needed to hold assets or borrow abroad.

Opening financial markets to external influences required major changes in the foreign exchange legislation. A step-wise procedure for liberalising capital movements was introduced in 1990, but the process gained momentum with the Treaty instituting the European Economic Area (EEA) at the beginning of 1993, which Iceland ratified. In order to conform with existing European Union legislation, adopted by the EEA, Iceland started liberalising long-term capital flows in 1992. The process was completed two years later (Table 15), followed by the removal of restrictions on short-term flows by the beginning of 1995.

As noted in Chapter II, once capital movements became completely free, the margins around the government's exchange rate target were widened to allow for the possibility of greater fluctuations in the exchange rate. However, the Central Bank retained a close grip on the foreign exchange market, as it was the counterparty to most transactions. It was not until the middle of 1997 that the Bank withdrew from its *de facto* market-making operation in the currency

Table 15. **Chronology of liberalisation measures in the foreign exchange and money markets**

Foreign exchange market.		
July	1990	Liberalisation of capital movements is begun.
November	1992	New foreign exchange legislation is adopted.
January	1993	Iceland becomes a member of the European Economic Area.
May	1993	Inter-bank market for foreign exchange created.
January	1994	Long-term capital movements are fully liberalised.
January	1995	Short-term capital movements are fully liberalised.
May	1996	New legislation on foreign direct investment is passed.
July	1997	Central Bank withdraws from market-making in the foreign exchange market.
Money market		
June	1992	Ceiling placed on the automatic overdraft facility of the Treasury with the Central Bank.
January	1993	Treasury overdraft facility is closed, but the Central Bank agrees to buy bills if necessary.
January	1993	All direct lending of Central Bank to Treasury is ended.
	1997	Central Bank considers withdrawing from market making in Treasury Bills.

Source: Guomundsson and Kristinsson (1997).

market, leaving this role to commercial banks. Daily fixing sessions with the Central Bank were abolished, and the commercial banks assumed the obligation to post continuous buy and sell quotes. With, in addition, a new credit institution entering the market, foreign exchange turnover in the last four months of 1997 almost quadrupled compared with the first four months of the year. Nonetheless, the Bank continues to intervene in the exchange markets, being the counterpart to over a fifth of all transactions in the second half of 1997.

Money market

Up to 1992, there had been no market in short-term government paper as the Treasury had the right to borrow from the Central Bank through an automatic overdraft facility. Between 1992 and 1994 this facility was gradually phased out, with the Treasury relying on the issue of short-term bills to cover its financing needs. While commercial banks traded credits in an informal market that the Bank facilitated, there was no general market for interbank deposits. However, at the same time as Treasury bills were introduced, a market in overnight deposits started (Table 16) and, in 1993, an inter-bank market for deposits at the Central Bank was established. Finally, a market for bank bills was introduced at the end of 1996.

With the development of a money market, the Central Bank conducts monetary policy in a manner similar to other central banks. To adjust short-term interest rates, it favours the use of ten-day repurchase agreements or reverse repos on treasury bills that are available on a tap basis. The Bank will also substitute certificates of deposit that it occasionally issues on demand if the supply of bills is low. Other open market operations are also available. It sets the interest rate at the discount window for bank overdrafts on their current accounts, but access to this facility has been limited. Since November 1993, reserve requirements remained unchanged until recently, although the liquidity requirement[17] was raised in 1996 from 10 to 12 per cent of domestic assets to absorb liquidity from the system generated by the early call of some government bonds.

The Central Bank introduced a further set of changes, between March and May 1998, designed to improve the implementation of monetary policy. First, the liquidity ratio imposed on commercial and savings banks was abolished. In addition, the number of institutions subject to reserve requirements was extended, and the reserve ratio was reduced in proportion to the increase in the base used

Table 16. **Chronology of measures concerning the banking sector and bond markets**

Indexation and banking		
March	1993	New legislation governing commercial and savings banks is enacted.
May	1993	Government-owned National Bank is recapitalised through a series of loans amounting to Ikr 4.25 bn (1.1 per cent of GDP).
July	1993	Central Bank limits the banks imbalance between indexed assets and liabilities to 20 per cent of their capital base, to be achieved by end-1995.
September	1993	Central Bank offers to swap indexed assets for non-indexed liabilities based on a projection of future inflation.
January	1995	Indexation of sight deposits is no longer allowed. Indexation is still permitted on deposits with an initial maturity of one year.
April	1995	New legislation mandates the use of the CPI for indexation, rather than the credit conditions index.
October	1995	Treasury issues 5-year indexed notes.
December	1995	Central Bank ends its swap facility for indexed assets.
January	1996	Indexed loans must have an initial maturity of 3 years, up from 2 years.
January	1998	Indexed deposits must be for an initial maturity of at least 3 years. Indexed loans must be for an initial maturity of at least 5 years. Banks can count part of their equity in calculating their indexed assets.
January	2000	Target date for the abolition of indexed deposits and lengthening of indexed loans to an initial maturity of 7 years.
Bond market		
October	1993	Government attempts to lower long-term real interest rates by adopting a long-term intervention rate on indexed securities.
	1994	Central Bank intervenes in bond markets including housing bonds.
May	1995	Introduction of first non-indexed government bond with three-year maturity.
October	1995	A second non-indexed bond with five year maturity is introduced.
February	1996	Securities firms take over role of market-maker from central Bank.
February	1997	The Government Debt Management Agency consolidates government bonds into nine issues.
May	1997	Real-time electronic quotes is provided internationally.
June	1997	Market making is introduced for Housing Bonds.
June	1997	Preparation for electronic registration of bonds starts is to be completed in 1999.

Source: Guomundsson and Kristinsson (1997) and Central Bank *Annual Report*, (various issues).

for calculating reserves. Short-term bank liabilities will continue to face a higher reserve ratio than long-term liabilities. Reserve deposits at the Central Bank will no longer be indexed but remunerated. Once the new reserve requirements came into effect, the limited overdraft facility of commercial and saving banks at the Central Bank was abolished, though a new overnight collateralised overdraft facility will be available. Banks that require reserves will have to participate in a weekly auction for repurchases; for this purpose, the Bank will now accept all government guaranteed stock registered on the Icelandic Stock Exchange. These changes are designed to limit the automatic access of credit institutions to Central Bank facilities and so stimulate transactions in the money market.

The exercise of monetary policy, however, is hampered by the fact that the Central Bank is the only market maker in short-term paper; in 1996 it was the counter-party to 93 per cent of all transactions. Such an active role could slow down the operation of monetary policy as the banking sector can, in practice, generate significant liquidity by taking advantage of the Bank's obligation to post two-way quotes before the Bank reacts. The Bank also wants to improve the credibility of the interest rate quotes in this market. For this reason, the Bank is considering transferring its market making activity to commercial banks, who have gained greater experience in recent years in operating in these short-term markets.

Indexation

Throughout the 1970s inflation was high and variable in Iceland, while nominal interest rates were controlled. As a result, real rates were significantly negative, leading to a sharp fall in saving and a process of disintermediation. Starting in the late 1970s the indexation of government bonds was introduced to support real interest rates, and by 1979 the practice was extended to all maturities, as well as to bank deposits and loans. As inflation was brought under control in the early 1990s, however, the government started a policy to unwind some of the indexation (Table 16) for four reasons. *First,* the banking sector had markedly more indexed assets than liabilities, which, in the view of the authorities, exposed these institutions to uncovered inflation risk in the event that inflation declined and short-term nominal interest remained unchanged. *Second,* the government was concerned that the preponderance of indexed lending made fixed interest rate credits expensive, due to the high level of real rates. Moreover, the Central Bank

was concerned that movements in short-term nominal rates were dominated by changes in indexed yields at the longer end of the spectrum. *Third*, the extensive use of indexation at the short end made the development of a money market in nominal treasury bills that more difficult. And *fourth*, because indexation was not widespread outside of Iceland, it was thought that it discouraged foreign participation in Icelandic markets.

To reduce the amount of indexation, the Central Bank progressively issued several regulations. In 1993 the Bank instructed commercial banks to limit their exposure by keeping the imbalance between indexed assets and liabilities to 20 per cent of their capital base. To assist in these measures, for a time the Bank agreed to swap indexed securities for non-indexed assets based on its projection of inflation. In 1994 it raised the minimum maturity for indexed bank deposits from six to twelve months, and a year later, it increased the minimum to three years, effective at the end of 1998, while the minimum maturity of indexed loans and securities will be five years. Indexation of all bank deposits will be forbidden in the year 2000, and the minimum maturity of indexed loans will be set at seven years. To facilitate the reduction in indexation, the Treasury issued three-year non-indexed notes in May 1995 and five-year non-indexed notes in October, and a further issue was made in January 1998. The reforms have had some success. Currently, only about one-third of bank deposits and one-half of bank loans are indexed, but almost all securities denominated in krona are indexed.[18]

Bond market

One early move in the financial liberalisation process was the creation of the Icelandic Stock Market in 1985. Initially, the market dealt only in government and government-guaranteed bonds. By the beginning of the 1990s, however, still only one-third of all marketable securities were quoted. In order to reduce the role of the monetary authorities in the bond market, the responsibility for marketing and managing government debt was transferred from the Central Bank to a newly created National Debt Management Agency (Table 15). Trading remained sufficiently thin that the government-set rates on new issues differed from those on the secondary market. Significant differentials meant that new placements were sometimes difficult – as happened in 1991 in the savings bond market. Official intervention in bond markets peaked in 1993 and 1994 when the government decided that the Central Bank should intervene in long-term bond markets

again, in order to push real interest rates down from an average of 7¾ per cent to around 5 per cent. Such real interest rate targets had adverse consequences for monetary policy and were abandoned in early 1995.

Since then, government action has concentrated on broadening and deepening bond markets. In 1995, the first three and five-year non-indexed government securities were introduced, while in 1996 the Central Bank ended its market making role on bond markets by contracting out the management of its portfolio to three market makers. In February 1997, the Government Debt Management Agency announced a programme to consolidate government bonds from 46 to nine issues. The proportion of outstanding Treasury bonds in the nine largest classes subsequently rose from 65 to 75 per cent in the next nine months. The government is also actively renovating the market's infrastructure. The stock exchange provides real-time quotes internationally, and a central clearing house is scheduled to begin operations at the beginning of 1999. With the development of the market, real interest rates have fallen to between 4½ and 5 per cent, while turnover has increased from IKr 5.8 billion in 1992 to IKr 27.2 billion in 1996. Turnover probably doubled again in 1997.

Equity market

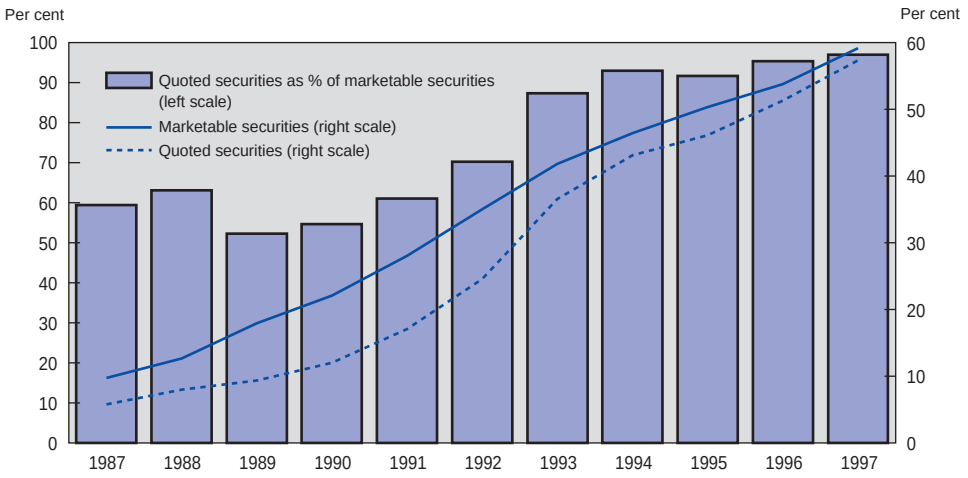
The modernisation of fixed-income markets has also encouraged the development of an equity market. An informal over-the-counter (OTC) market had existed since 1985, but equity trading on the Stock Exchange did not begin until 1992. In 1993 the parliament passed a new act modernising the exchange, which it subsequently amended in 1996 in accordance with EC directives. Between 1992 and 1997, the number of quoted companies has grown from only two companies to 51. Turnover has roughly doubled each year. Another 60 smaller firms are registered in the OTC market. Nonetheless, the markets are immature. New equity issue is the same order of magnitude as secondary trading, as only 11 per cent of shares changed hands in 1997, and overall capitalisation of the two markets is still only one-third of GDP, somewhat below the EU average. Only half of the registered stocks on the OTC were traded in 1997. In early 1998, pension funds held only 8½ per cent of their assets in equity, and foreign investors have so far participated little in the market. The equity markets could get a boost, however, from the privatisation programme of the government (see below), as well as the removal at the end of 1997 of the Stock Exchange's legal monopoly.

Impact of the reforms

The above reforms generated considerable changes in Icelandic financial markets. Transactions in fixed income securities have become markedly more transparent, with the proportion of those instruments traded on the stock exchange rising from 30 per cent at the beginning of the reform process to almost 95 per cent in 1996 (Figure 17). Perhaps partly in response to this, the stock of such quoted instruments has grown more than six fold relative to GDP between 1990 and 1997. Turnover has also increased, rising from 6 per cent of the outstanding stock annually to 42 per cent by 1997, though this partly reflects the introduction of quotations for money market instruments in this period. As was already noted, the modernisation of Iceland's credit markets improved the ability of the Central Bank to implement monetary policy and aided the development of an equity market. With most of the policy reforms concerning financial markets completed, the attention of the government has turned to reforming its financial institutions.

Figure 17. **GROWTH OF QUOTED FIXED INTEREST SECURITIES**

Per cent of GDP



Source: Central Bank of Iceland.

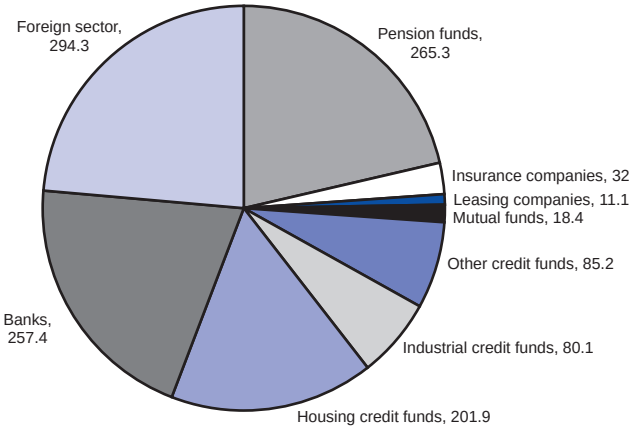
Reforming financial institutions

While the reforms outlined above have contributed greatly to expand domestic financial markets, the major credit institutions remain publicly owned, with a structure that essentially reflects a period when these markets were poorly developed. Short-term credit for households and business is provided by commercial banks, while longer-term financing both for business and housing is largely provided by various government investment funds that draw their resources from private sector pension schemes and foreign borrowing. A series of measures was introduced recently to adjust that structure to the needs of a market-oriented economy, but progress so far is still limited.

The basic structure

There are three main groups of financial institutions in Iceland: banks, pension funds and government credit funds, with the latter being divided into three specialised categories (Figure 18). These institutions have supplied most of

Figure 18. **SOURCES OF CREDIT**[1]
1996 Year-end assets, IKr billion



1. Does not net out inter-institution lending of IKr 353.0 billion.
Source: Central Bank of Iceland.

the private-sector demand for credit with the government credit funds taking, in aggregate, the largest part of the market. Insurance companies, mutual funds and leasing companies provide only a small share of credit to the economy. The foreign sector has also supplied a significant amount of outstanding credit, mainly to the government and its credit funds.

Banking sector

The banking sector is currently dominated by the three commercial banks dealing with the public. Between 1930 and 1990, all the major banks were government-owned, as the only private bank closed in 1930 and was taken over by a new state-owned bank. At the end of the 1980s, most of the public-sector banks were making large losses. To deal with this problem, one insolvent public-sector bank was sold to a number of small banks and became the nucleus of what is now the only private-sector commercial bank. In 1993, another was rescued through a series of loans and equity injections amounting to over 1 per cent of GDP. As a result, there remains only two banks in the public sector, accounting for 68 per cent of total bank assets. Until 1997, they were not constituted as joint-stock companies subject to commercial law and benefited from government guarantees on their non-deposit liabilities.

Savings banks provide some competition to the commercial banks. However, the average assets of each of them are low – on average less than 3 per cent of those of the smallest commercial bank. Indeed, the asset base of all 29 savings banks together amounts to only one-quarter of that of the commercial banks. In order to improve their competitiveness, the savings bank association created a consortium bank to provide them with banking services and to enable them to enter areas such as fund management and brokerage where the individual savings banks would not be of a competitive size. While this consortium is effectively a fourth commercial bank, it does not deal directly with the public. One of the reasons for the formalities of the savings banks is that they generally serve one locality but there has nevertheless been some reduction in their numbers, even though such moves are complicated by their ownership structure.

Savings banks are owned by the holders of the bank's guaranteed capital provided by the original depositors. These owners, however, have the right to elect only 60 per cent of the board of directors, the remaining 40 per cent coming from local government. Moreover, they do not have the right to distribute the

profits of the bank, nor can they transfer the ownership of the capital without the permission of its board. In addition, if the articles of association of a savings bank limit it to operation in a certain geographic area, the owners must live in that area. The shareholding of each owner must be equal, and no one holder can own more than 5 per cent of the capital. In practice, these provisions mean that the savings banks are self-owned and, while profit-making, distribute no dividends.

As currently structured, the banking sector in Iceland appears to suffer from high costs. In 1995, total annual operating costs represent close to 5 per cent of assets under management.[19] Indeed, within the OECD area, total costs are higher only in Poland. Although considerable progress has been made in reducing staff numbers in the past decade,[20] both labour and other costs are around twice as high as the OECD average, and there are almost twice as many bank branches per person. In part, this reflects the dispersed nature of the population and the small size of the market, though modern technology has lessened this disadvantage. Such a situation generates a wide spread between borrowing and lending rates, with the difference between interest received and interest paid representing 4.6 per cent of total assets in 1995. Again this is one of the widest spreads in the OECD area. By 1997 this differential had fallen somewhat to 3.9 per cent.

Banks' costs are also high in relation to their domestic competitors. Icelandic mutual funds, for example, have average administration and sales expenses of around 1 per cent of assets under management, while pension funds (see below) have expense ratios of only 0.3 per cent. Consequently such financial institutions represent significant competitors for banks, notably for longer-term time deposits. The demand for money transmission services, as evidenced by the size of sight deposits, has remained buoyant. Nonetheless, this competition may have constrained the banks' ability to lend as total bank credit fell by 2½ per cent of GDP between 1993 and 1996, at a time when the total value of outstanding securities (other than shares) rose by 12 per cent of GDP.

The profitability of the banks has also been depressed by the need to make a high level of provisions against bad loans. However, in 1996, with new provisions for losses on bad loans falling to 1.3 per cent of their lending portfolio – close to the average prior to the stabilisation of inflation at the beginning of the 1990s – the profitability of the banking sector rose some 30 per cent. In the first half of 1997, profits rose 54 per cent from the same period a year ago, helped

by a marked increase in fee income and a further decline in provisions to 0.7 per cent of outstanding loans. The stock of non-performing loans that have not yet been written off remains high, at about 8.5 per cent of total assets – down from a peak of over 12 per cent in 1992.

Industrial credit funds

In addition to commercial banking activity, the Icelandic government has been heavily involved in credit markets through a series of financial institutions that were known as the industrial investment credit funds. The largest of these institutions (the Fisheries Fund and the Industrial Loan Fund) were founded in 1905 and 1935 respectively. Another was created in the early 1970s, when Iceland joined EFTA. While the relative importance of each of these funds has varied considerably over time (Table 17), the total lending of these institutions

Table 17. **Lending by housing funds and industrial investment credit funds**

End year outstanding balances, per cent of GDP

	1990	1994	1995	1996
Total housing funds	23.5	38.4	41.0	41.7
Total industrial investment credit funds	15.8	17.5	17.3	16.5
By fund				
Fisheries Investment Fund	3.9	5.8	5.2	4.9
Industrial Loan Fund	2.6	3.7	3.6	3.5
Industrial Development Fund	1.2	1.3	1.2	0.9
Total of above	7.8	10.8	10.0	9.3
Agricultural Loan Fund	2.1	2.1	2.2	2.1
Commercial Loan Fund	0.6	1.0	1.8	1.9
Municipal Loan Fund	0.9	1.1	1.1	1.1
Regional Development Fund	2.4	1.7	1.7	1.6
Development Fund	1.4	0.4	0.3	0.3
Tourism Fund	0.2	0.3	0.3	0.2
Other Funds (4)	0.4	0.1	0.0	0.0
Total of above	8.0	6.6	7.3	7.2
By sector				
Agriculture	2.8	2.3	2.5	2.2
Fisheries	5.4	7.1	6.6	6.6
Commerce	0.7	0.4	0.5	0.5
Manufacturing	4.1	4.8	4.7	4.2
Unclassified	1.4	1.6	1.7	1.8
All industries	14.4	16.2	16.0	15.2

Source: Central Bank of Iceland.

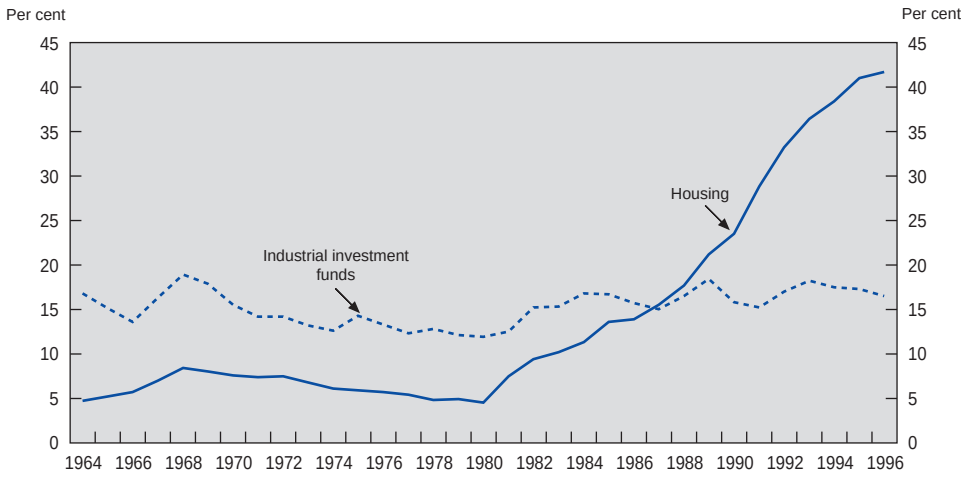
has remained relatively stable – fluctuating between 12 and 18 per cent of GDP over the past twenty years (Figure 19). About half of their borrowing is in foreign currencies, the remainder coming from pension funds and the government. Overall, the volume of their loans to the business sector is three-quarters smaller than that of commercial and saving banks.

Housing funds

The government-owned housing funds are currently more than double the size of the industrial investment credit funds, with assets amounting to 42 per cent of GDP. Three institutions have almost equal shares of the market: the State Housing Fund, the Housing Bond Department and the Workers Building Fund. These funds represented a relatively small source of lending until 1980, but they experienced a rapid expansion of their balance sheets following the indexation of nearly all long-term debt instruments and the progressive increase in the supply of funds from pension plans (Figure 19).

Figure 19. **THE EVOLUTION OF LENDING BY THE GOVERNMENT INVESTMENT FUNDS**

End period balance outstanding, per cent of GDP



Source: Central Bank of Iceland.

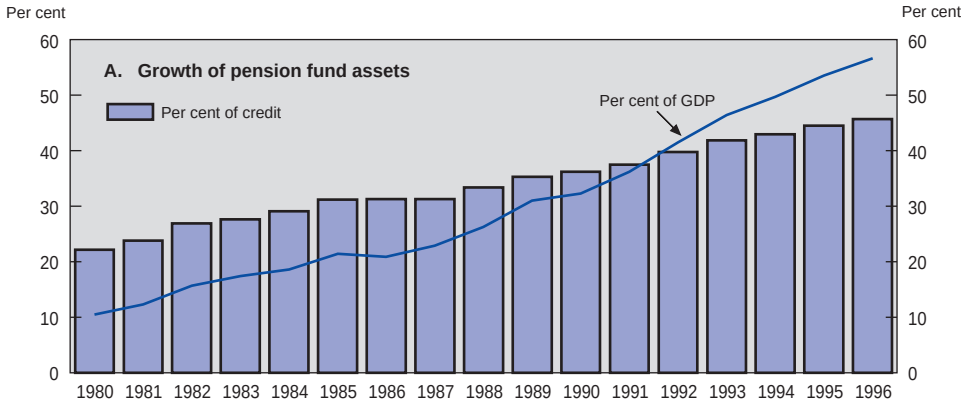
Pension funds

Pension funds are now amongst the largest financial institutions in Iceland. While having a long history, their importance dates essentially from 1969 when private-sector employers and trade unions agreed to establish a second-tier, earnings-related, pension scheme designed to supplement the flat-rate state pension benefit. In 1974, the government made membership of such a scheme compulsory for employees, and in 1980 the self-employed were also obliged to join it. At the end of 1996, 41 open private sector funds and 16 for public-sector employees were in operation.[21] In 1996, the assets of the three largest pension funds were equivalent to just over 40 per cent of those held by commercial banks, while total pension fund assets were slightly larger than those of commercial and saving banks taken together. As a result, they represent close to half of the assets of the credit system (Figure 20, Panel A).

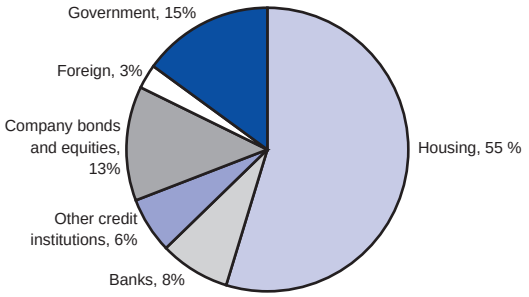
After a poor start in the 1970s, the investment performance of these funds has been good. They have earned a real rate of return over 7 per cent since 1980, mainly reflecting the high yields from indexed government securities. In addition, their expense ratio has been kept low, recently averaging 0.3 per cent of assets under management or 4 per cent of premium income. The future liabilities of the funds are valued assuming a 3 to 3½ per cent real discount rate, lower than both current market rates and past returns. Consequently, most funds have become solvent – in contrast to their position at the end of the 1970s when many had deficits. Currently, three open pension funds are classed as having an insufficient or uncertain financial position, as well as four that are closed to new entrants. The assets of these seven funds, however, are quite small. In addition, total benefits payments have been low as the ratio of contributors to beneficiaries is still higher than the eventual long-run equilibrium. With yields high, payments low and membership increasing, the assets of the funds have steadily risen from 10 per cent of GDP in 1980 to 55 per cent in 1996 (Figure 20, Panel C). They should continue to increase until 2025, peaking at close to 100 per cent of GDP.

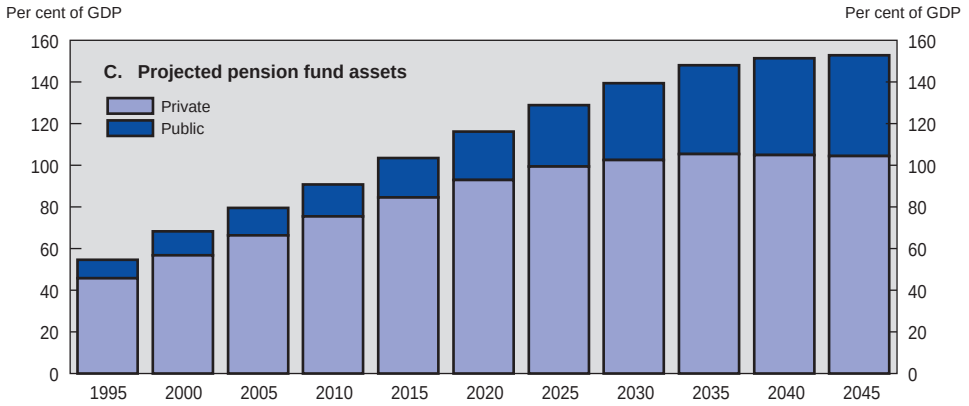
Most of the assets of the pension funds are invested in the housing sector of the economy (Figure 20, Panel B). Three-quarters of their lending to this sector is either as private placements with the State Housing Fund or holdings of Housing Bonds, both of which are state guaranteed (as is their lending to the industrial investment credit funds). In total, only about one-third of their assets are invested

Figure 20. **EVOLUTION OF PENSION FUNDS**



B. Distribution of pension fund assets by sector, 1996





Source: Central Bank of Iceland and National Economic Institute.

in risky assets, mainly in the form of direct housing loans to members of the pension fund or loans to enterprises. Equity shareholdings represent less than 8½ per cent of their total asset portfolio.

Recent changes

In May 1997, the Parliament passed three laws designed to restructure government-owned financial institutions. The principal provisions of these laws, which deal with both commercial banks and the industrial investment credit institutions, came into force in January 1998. In addition, a new regulatory regime for pension schemes has been introduced. The objective of this new legislation is to enhance competition in the financial sector, increase its efficiency and reduce the extent of state guarantees on financial instruments.

In the *commercial banking sector*, this objective will be achieved in stages. First, at the beginning of 1998, the legislation transforms the two state banks into corporations subject to commercial law. Newly issued liabilities of these corporations will not carry a government guarantee, though deposits will continue to be insured. Secondly, the government has the intention to open the capital of these two companies to the public. The implementing legislation gives the government the authority to sell new shares to the public up to a limit equivalent to 53 per cent of the existing equity, effectively limiting the eventual total private sector participation in the new banks to 35 per cent. The timetable for such an equity sale has not been decided. If, in a third stage, the government decided to sell the existing shares of the banks, it would have to return to Parliament to amend the legislation.

With respect to the *industrial investment credit funds*, the new legislation partially rationalises this sector by creating a new institution, the Icelandic Investment Bank (IIB), which consolidates most of the activities of four existing funds: the Fisheries Fund, the Industrial Loan Fund, its subsidiary the Export Credit Fund, and the Industrial Development Fund. The first two of these funds have been profitable, earning respectively a real rate of return of around 6 and 13 per cent on their equity in 1996. As such, they will form the core of the new institution. By contrast, the profitability of the Industrial Development Fund has been lower – at only 4 per cent – due to its portfolio of loans made on soft terms. The new bank will regroup about half of total lending by the industrial investment credit funds. A minority stake in the bank (of 49 per cent) will be sold to

the public during 1998, with the objective of encouraging an active market in the shares and obtaining a stock-market quotation. The government has not yet decided on the precise modalities of the sale: in particular a cession of part of the 49 per cent to a financial institution has not been ruled out.

Another institution – the New Business Venture Fund – was also created by removing parts of their existing activities from the four above-mentioned funds before they were merged into the IIB. This institution is publicly owned, with its activities circumscribed to the provision of start-up and seed capital for new business and the promotion of new businesses in rural areas. The activities of the new fund will be limited as it will not be allowed to issue bonds, loans or other redeemable securities to the public and is authorised to borrow only to balance its short-term cash flow. Its initial assets, with a total value of IKr 4 billion, are composed of a 7 per cent stake in the only private commercial bank plus a ten-year claim on the Icelandic Investment Bank that will be redeemed gradually; the annual cash flow from these sources is likely to be less than IKr 200 million. In addition, the government will provide further capital of IKr 1 billion. This money will be used to fund rural development projects, but such loans will be managed by private institutions on a competitive basis and will be repaid to the Treasury after seven to ten years. The Fund will be run by a board with five members: two from the government, two from the trade associations and one from the trade unions.

The government has made a proposal to reform the *Housing Funds.* The three housing funds will be merged into one unit that will remain in the public sector. In addition, it will end the system of low-interest housing loans for poor households. Instead, the loans will carry market rates of interest but will be eligible for interest rebates based on family income and wealth through the tax system. In addition, local authorities will be able to guarantee that part of a home loan obtained by low income families that lies between 70 and 90 per cent of the collateral value.

In the area of *pension funds,* parliament passed legislation introducing a new regulatory framework at the end of 1997. Securing a stable environment for these funds is particularly important in that they depend entirely on their assets and future contributions to pay pension benefits. Private-sector employers have, in general, no liability to meet deficiencies in a pension fund,[22] and there is no other guarantee for the benefits.[23] Moreover, the beneficiaries of the scheme are unable to withdraw their assets.

The objective of the new law is to ensure that pension schemes remain solvent and to specify minimum benefit levels. Funds will be licensed by the Ministry of Finance, which will approve their benefits, as well as the rules and regulations governing their investment policy. A minimum size will be established for a pension scheme, so increasing the speed of consolidation in this sector. Moreover, the legislation foresees that funds whose estimated liabilities are 10 per cent greater than their assets in a given year, or 5 per cent greater for five consecutive years, will have to submit a plan to remedy the situation either by reducing benefits or increasing contributions. The new legislation also specifies the asset classes in which funds can invest, but the list is so wide as to permit most investments. Moreover, there are no limits on the proportion of total assets that can be invested in any given security. The major limitation on the funds is that any residential and commercial mortgages must be limited to 65 and 35 per cent, respectively, of the value of the collateral asset.

In addition, the new legislation will introduce more competition between pension funds. In the future, minimum contribution and benefit rates will be established for all economically active persons, but it will be possible to negotiate higher contribution rates collectively, and individuals will be able to make voluntary contributions. Contributions over the minimum will accumulate in defined contribution plans, with individuals free to choose their own fund manager. These excess contributions will be tax deductible in the same way as normal pension contributions.

In 1997, the government also reformed the public employees pension scheme in order to contain the growing deficit that had been revealed by accrual accounting. This scheme, which is closed to new employees, has a large deficit. New entrants to the public service have to join a funded scheme, with different benefits and contribution rates to those for existing employees. Overall, the long-run cost of the new scheme should be 15 per cent of the payroll against an estimated cost of 20 per cent for the existing one.[24] The build up of assets in the fund will be slow – in the next twenty years assets should grow by 10 per cent of GDP – but by the time the fund is mature, in 2050, its assets could represent around 50 per cent of GDP.

Scope for further action

Financial markets have been thoroughly modernised in the past five years. As a result, Icelandic markets are now fully open to external influences and function competitively. The Central Bank is able to conduct monetary policy in a manner similar to practices abroad, while capital markets have become deeper and more liquid. There are three areas where some improvement may be possible. The Central Bank remains the only market-maker in Treasury Bills, which can complicate its management of the money and foreign exchange markets from time to time. This function should be quickly moved to the private sector, as envisaged by the bank. In addition, there should be a study of how to increase the turnover of the equity market and whether derivative markets could be introduced.

In the field of commercial banking, the current reforms should only be considered as a first stage in reducing the excessive costs of banking services in Iceland. A major programme of cost reduction and branch closure would indeed seem to be required to achieve this objective – a move that would be all the more feasible when increased competition in telecommunications lowers prices in that area. In such a perspective it will be important, at the same time, to achieve a corporate governance system that emphasises profit maximisation. In this regard, the government should consider a much more rapid complete privatisation of the two commercial banks it still owns in full. This could result in further consolidation of the system, and hence, increase the danger of monopolistic pricing. The rationalisation of the banking sector would have to be monitored by the competition authorities.

The process of consolidation in the savings bank industry would also need to be speeded up. This is made more difficult given the ownership structure, defined in the 1993 legislation governing these banks. A number of reforms should be considered in this regard. Local government representatives should no longer sit on boards. There should be no limit on the shares held by an individual, and they should be freely transferable, with the right to a dividend as determined by the board. Legislation might be required to determine the ownership of the accumulated reserves of the banks. One possibility would be to create new shares, in line with the reserves of the bank that could be distributed to depositors. Such moves would help reform the governance of the savings banks, placing greater emphasis on profitability and so enhancing efficiency. They would, moreover, remove barriers to a demutualisation of the savings banks.

The reform of the government-owned investment credit funds is not yet complete. The move to sell a minority stake in the newly created Icelandic Investment Bank has been welcomed by the capital markets. Indeed, the government should be able to obtain a price higher than the book value of assets, given the returns earned by the constituent parts of the bank in recent years. An even better price might, however, be obtained by reducing the state share in the company to well below that of a controlling shareholder. The government should aim for the complete privatisation of this Bank as quickly as possible. In addition, there are still a considerable number of investment credit funds for which no privatisation plan is in prospect yet. The development of private capital markets and the moves towards transparency in the government accounts argue for the privatisation or closure of these funds. If capital is to be provided for selected activities at below market rates, then it is appropriate that, at the least, the government counts the ensuing subsidies in its expenditure.

Similar arguments apply to the government-owned housing finance sector. With the existence of a thriving financial market, the need for continued state involvement in this area can be questioned. The government is already reviewing the precise form which its future participation in this area should take. One option would be to create a Housing Bank and then sell it to the private sector. At the same time, the legal infrastructure for asset-backed securities should be created so that the bank could sell its portfolio of mortgages to the market.

In the area of pensions, the move to accrual accounting has revealed the extent to which the pension scheme for public employees' pensions is in deficit. In order to deal with this problem, new recruits to the civil service will have to join a fully funded scheme. The government has appointed a task force to consider the future role of private sector pension funds in the provision of retirement income. A large number of issues will need to be discussed, most importantly, how to separate the redistributive element of the basic pension from the element that represents the savings for retirement. As with the recent change in the presentation of the budget, the aim should be to increase the transparency of the system, clearly identifying its different components. The study group will also need to evaluate the balance between the current and prospective levels of real interest rates, population growth and real income growth before coming to a decision as to whether to move some of the financing of the state pension to the private sector. At the moment, such a study is likely to show, as in the Canadian case, that the quicker a move to a fully funded scheme is made the better.

IV. Other structural policy developments

The past five years have also seen considerable changes outside the financial sector of the economy. With increased emphasis on price and exchange rate stability, a number of reforms have been introduced in the labour market to improve its flexibility and help reduce the level of unemployment. At the same time, competition policy has been strengthened, with deregulation proceeding in some areas. Moreover, privatisation of the extensive state sector has been pursued, while the functioning of the remaining public sector is being modernised. Other structural initiatives have been taken in the fields of agriculture, fishing and the environment. The paragraphs below review these developments in more detail, assessing the areas where further progress may be needed.

Labour market

The labour market in Iceland has consistently delivered one of the lowest unemployment rates in the OECD area. Overall unemployment is under 4 per cent of the labour force, and long-term unemployment amongst the lowest in OECD countries. This has been achieved through considerable labour market flexibility, with the fluctuation of real wages being more than twice that experienced in other small open economies (Table 18) despite economic activity being dependent on external forces that, in the past, proved to be quite volatile. At the moment, it appears that this market continues to be relatively flexible, current wage pressures being essentially the reflection of tight labour market conditions. Nonetheless, the *OECD Job Study* identified a number of areas where the functioning of the labour market could be improved (OECD 1997). In the previous year or so, some important reforms were implemented in the fields of industrial relations, public employment services, unemployment insurance and education, which were regarded as key areas for improvement. Some of these changes have been in the direction of the recommendations made in the study (Table 19).

Table 18. **Volatility of economic variables in selected small countries**

Standard deviation (percentage points), 1960 to 1995

	Iceland	Finland	New Zealand	Sweden	Portugal	Norway	Denmark	Ireland
GDP	5.20	4.05	4.40	2.50	2.45	2.35	2.60	3.20
Employment	3.45	4.75	2.40	2.85	2.95	2.85	1.75	2.35
Productivity	3.75	3.50	3.90	3.85	1.95	1.90	2.50	2.15
Real wages	8.85	4.10	3.90	4.55	3.50	3.40	4.75	2.75
Real labour costs	6.35	3.25	3.85	3.90	4.90	4.75	2.75	2.90
Price inflation	19.20	4.35	5.75	3.40	3.05	3.05	3.55	5.80
Wage inflation	18.00	5.50	6.70	3.15	3.60	3.60	3.90	6.35
Average volatility of above real variables	5.5	3.9	3.7	3.5	3.2	3.1	2.9	2.7

Source: Palle S. Andersen and Guomundsson M. (1997).

In 1996, the Parliament adopted a new Industrial Relations Act that was intended to speed negotiations between employers' federations and labour unions. The act stipulates that employers and labour must agree to a schedule for negotiations at least ten weeks before the end of a contract, and unions are required to submit formal demands during the period covered by this schedule. A conciliation and mediation officer is supposed to ensure that such a schedule is followed, and if none is negotiated, the officer can impose one. During negotiations, the officer can submit compromise proposals, especially when negotiations are undertaken with multiple unions or federations. The act also clarifies the mandate that negotiating committees have when representing several unions and requires a secret ballot before a work stoppage can be started by unions or federations.

The 1997 wage negotiations were the first under this new Act. They proved to be longer and more decentralised than expected. Although the previous contracts were due to be terminated at the end of 1996, the first new agreements were not signed until the beginning of March, while negotiations with some unions and federations dragged on into the summer. These agreements were generally concluded with individual unions, and even though the Federation of General Workers Unions represented its members in talks, many of them rejected the settlements. Several, generally small, work stoppages occurred, especially in the fisheries sector. In retrospect, because the talks took place soon after the act

Table 19. **Implementing the OECD Jobs Strategy – an overview of progress**

Proposal	Action	Assessment/recommendation
I. Decentralise industrial relations system		
• Move away from two-stage wage bargaining process.	Industrial Relations Act required schedule for negotiations but no mandate for who negotiates.	1997 negotiations were very decentralised. New law was expected to streamline negotiations, but this did not happen, though expectations are for an improvement in the next round.
II. Reform the unemployment benefit system		
• Tighten benefits eligibility.	New law clarifies and tightens eligibility slightly.	
• Introduce time limits.	New law introduces five year limit. Worker and counsellor sign agreement for search and training.	Five-year limit still long compared with most other OECD countries, but if stringent conditions on search and training between the unemployed and counsellor becomes standard policy this disincentive could be offset. It should also be noted that long-term unemployment in Iceland is low.
• Lower replacement rates and make uniform across skill groups.	Currently flat rate with extra children benefits.	Flat rate with extra benefits for children near 100 per cent for minimum wage earners means rates are very high for the low skilled.
• Introduce experience rating.	Currently no experience rating.	Seasonal employers in fishing and industry take advantage of the unemployment system.
III. Improve education system		
• Lengthen school year, and increase hours in foreign languages, natural sciences and mathematics.	Transfer of responsibility of primary schools to local authorities with increased funding. Primary school teachers' pay increased. 1996 legislation makes secondary schools more autonomous. Comprehensive review of primary and secondary curriculum underway.	Transfer of primary schools to local authorities is expected to increase schooling hours per week in the next four years, but secondary school hours still especially short compared with the European mean.
• Sharpen performance incentives.	An agreement on ''performance management'' is being prepared between Ministry of Education and secondary schools.	Transfer of primary schools may increase monitoring.

Table 19. **Implementing the OECD Jobs Strategy – an overview of progress** *(cont.)*

Proposal	Action	Assessment/recommendation
• Increase graduation rates for upper secondary schools.		Curriculum review and ''performance management'' agreement may alleviate dropout rate that is high compared with the rest of the OECD.
• Shorten time to complete post-secondary schooling.	1996 law lengthens school year by 15 per cent.	Time to complete post-secondary schooling well above OECD norms, perhaps due to short school year.
• Improve vocational training with more money and improved relationships with employers.	1996 law creates advisory committees for curriculum.	Law has had little practical effect except in traditional skilled trades.
IV. Boost product market competition		
• Increase reform in agriculture and financial sector, and resume privatisation programme.	Various reforms in financial sector and additional privatisation programmes.	Progress has been outstanding in many areas, but privatisation in finance means the Investment Bank and the two commercial banks must be fully privatised, not just minority stakes sold.
V. Enhance working time flexibility		
• Reduce non-daylight hours premium in public and private sectors.	No change in negotiated premium. EU working-time directive adapted that extends minimum rest period from 10 to 11 hours.	Premium for an eight-hour shift during non-daylight hours is very high compared with OECD countries.

Source: OECD.

was passed, hopes for a quick outcome may have been overly optimistic. Now that both sides are more familiar with the new environment, many analysts expect future negotiations to conclude more quickly.

In 1997, the Parliament passed two acts concerning unemployment insurance. The first clarified and tightened eligibility requirements somewhat and introduced a five-year maximum duration for benefits where previously there was none. After one year of work, a person now becomes re-eligible for benefits, while time spent on training and courses does not count to the five-year limit. The system for determining unemployment benefits was unchanged and remains a flat rate almost equal to the negotiated minimum wage of workers in fisheries. Compared to other OECD countries, replacement rates for low-income workers are high, but they are fairly low for higher-income workers. No experience ratings were introduced either; firms with large swings in seasonal hiring such as fish processors are able to exploit the unemployment insurance system to compensate laid-off workers during the off-season. While the reforms are modest in scope and will save the government only a little money in the near term, it should be noted that long-term unemployment does not appear to be a problem in Iceland (Chapter I).

The second act moved funding and responsibility for unemployment services to the central government. The quality of services provided by the job placement offices run by the local governments varied greatly across regions, and there was little help in finding job opportunities outside of the locale. The new system replaces the 56 local offices with eight regional ones that are electronically linked, creating a national job-bank. The act ties together the job placement agency with the unemployment benefits administration and so is able, for the first time, to condition the receipt of unemployment benefits. The new legislation requires that, in order to continue receiving payments, an unemployed person must have a meeting with a counsellor within ten weeks of applying for benefits and sign an agreement concerning job search and training. After a further four weeks, the contract is re-evaluated, and the person can lose benefits if the agreement is not met. For the longer-term unemployed (four to six months), counsellors provide additional guidance, including courses, re-training and even psychological help if needed. Such a system with its emphasis on effort by the unemployed may be able to offset the disincentive effects from an otherwise generous compensation system.

In the past year there have also been some important reforms to the educational system, designed to improve its performance.[25] The responsibility for primary schools was transferred to local authorities with increased funding. It is expected that over the next four years, hours taught per day, which currently are among the lowest in the OECD area, will be raised. After intense negotiations and a one-day walk-out, a new contract with teachers was signed with pay increases above those signed elsewhere. Teacher pay had been very low by OECD standards. In 1996 a law was passed that makes secondary schools more autonomous, and the school year was lengthened by 15 per cent. Currently, an agreement on ''performance management'' is being prepared between the Ministry of Education and secondary schools, and a comprehensive review of the curriculum of primary and secondary schools is underway.

Competition policy and deregulation

In the 1980s, the central government was still deeply involved in the economy. Intervention in price setting was substantial. In that spirit, competition law was geared more to regulating pricing behaviour than to promoting free markets and, moreover, it largely ignored public enterprises, which accounted for a large proportion of the economy. Since the turn of the decade, the situation has changed progressively as the government pursued a policy that favours free market competition over active management. As a result, government interference in the price mechanism is substantially more limited today.[26] In this respect, the development of policy in Iceland has followed the emphasis on improving the performance of product markets through competition policy and deregulation that has been a common theme in the *OECD Jobs Study*.

There has been considerable progress in the area of competition policy in recent years. In part to comply with European Economic Area (EEA) directives, Iceland implemented a new competition law in 1993, replacing one introduced in 1982 that aimed to limit the maximum margins producers could apply.[27] In contrast, the new act was designed ''to promote effective competition'' by covering price collusion, unfair trade practices, abuse of monopoly position, mergers that affect competition and, importantly, public enterprises with market power (Section 14 of the law). In order to promote competition, the law created three administrations: a competition authority with broad powers to follow up

complaints and initiate investigations; a competition council to rule on disputes with the power to fine, deny mergers, and issue cease and desist orders; and a competition tribunal to hear appeals. The law also provides for further appeals through the court system.

Recently, the Icelandic courts have judged several cases that have clarified the competence of the competition authority, allowing it to move into areas where government regulation had created monopolies. Generally, plaintiffs have argued that other legislation supersedes the competition law, but the courts have usually sided with the council using Section 14 of the Act to extend the reach of the competition authority.[28] The courts sided with young doctors in a suit against the Ministry of Health who allowed older doctors to dictate who performed surgery, successfully arguing such an arrangement constituted an unfair trade practice. The authorities have brought several agricultural cases where the defendants lost after arguing that the law was not supposed to cover agriculture. Farmer co-operatives are now pressuring the parliament for specific exemptions. Currently, important investigations are underway in finance where it appears the banks may be denying others access to the electronics payments system and where banks and savings institutions allegedly are co-operating unfairly in the insurance business. The Ministry of Finance is expected to issue a policy paper soon, requiring government institutions in competition with private enterprises to act in a manner consistent with the aims of the competition law and refrain from abusing privileges and advantages.

In spite of the determination with which the competition authority has fulfilled its mandate, it has shown some flexibility in the application of the law, considering that Iceland's small size often leads to efficient scales where one firm is dominant. Thus, for example, with the deregulation of the domestic market, the competition authority allowed a merger between Icelandair and a domestic airline after some provisions for arms-length transactions were included. The authority also investigated apparent market sharing between two shipping companies for cargo between Iceland and the United States where one company stopped sailing and contracted cargo space on the other's ships; the practice was allowed on efficiency grounds as the market supported only one carrier.

Iceland has also encouraged more competition through lowering the barriers to the entry of foreign companies. A 1991 act liberalised most direct investment, but it kept ownership limitations of 49 per cent in domestic airlines, 25 per cent

in banks, and denied foreign ownership entirely in the fisheries and energy businesses. In 1996 the limits on the banking and the energy sectors were dropped. In that year, Iceland ratified a treaty that partially freed the ownership of domestic airlines by exempting citizens of the EEA from existing restrictions. The same treaty also allowed some indirect ownership in fisheries, while a 1997 act permitted foreign investors to own 25 per cent stakes in firms that own shares in the fisheries.

Competition has also been improved through deregulation. In the summer of 1997 the domestic airline industry was liberalised. This led to a precipitous drop in fares and a 12 per cent increase in traffic. It appears, however, that prices may move back up, as profits have tumbled. Deregulation has forced the closure of some unprofitable routes, especially to remote areas, and the government will underwrite Air Icelandic flights to Raufarhöfn in north-east Iceland.

The government has also begun to review regulations that set standards in the area of consumption and production. In part this reflects the need for Iceland, as a member of the EEA, to align domestic laws with EU legislation. Some specific measures passed so far include simplified electrical safety controls and new consumer product safety legislation, so easing non-tariff barriers to international trade. In some instances the government has stopped inspection and monitoring activities, allowing the private sector to perform some of these functions. The government now sets standards for fish and uses licensed private companies to carry out the inspections. In 1997 it privatised the government agency that carries out motor vehicle inspections.

Privatisation

Iceland has had extensive direct government ownership of businesses in several sectors of the economy. Often because of scale economies, no firm had found it profitable to enter the market until the central or local governments demanded their services (OECD, 1995). As such, when the need arose, the authorities found it necessary to create their own enterprises, and they became involved in such diverse activities as data processing, pipe manufacturing and drilling. The National Power Company, jointly owned by the central government and the cities of Reykjavik and Akureyri, generates almost all of the country's electricity. The government also owns other utilities, two commercial banks, a

newly formed investment bank, a TV channel and a couple of radio channels. The Post and Telecommunications firms are government owned, and the school, housing loan and health systems are largely government controlled. Larger municipalities operate local bus systems, as well as several other miscellaneous service providers. Overall in 1994, the public enterprise sector of the economy accounted for 6¾ per cent of total employment in Iceland (Table 20). Such a figure was above the European average of 5 per cent in 1991.[29] In addition to the public enterprises, the government undertakes a large amount of construction work using its own employees.

Table 20. **Employment in public sector and public enterprises**

1994 figures

	Person years	Per cent of employment
Utilities	3 666	3.0
Electricity	1 056	
Water	384	
Communications	2 224	
Transport	1 152	0.9
Finance	2 612	2.1
Recreation and culture	748	0.6
Total of public enterprises	8 178	6.7
Government account construction	2 858	2.3
Health and non-profit welfare institutions	6 466	5.3
Government services	24 352	19.9
Total of above	33 676	27.5
Total public sector	41 852	34.1

Source: National Economic Institute.

In 1991 the government began a policy of privatisation, and since then, it has sold fifteen companies, the total value of which in 1997 prices was roughly ½ per cent of GDP (Table 21). In 1998 the government plans to offer for sale several small companies, as well as its remaining stake in Icelandic Prime Contractors and a minority participation in the Icelandic Investment Bank (Table 22). In addition, there are some tentative plans to sell additional equity in the two publicly owned commercial banks. Future sales may include the remaining 38.5 per cent stake in Icelandic Alloys, worth IKr 3.9 billion.

Table 21. **Iceland's privatisation programme**

1991-1997

	Firm	Contents	Proceeds IKr million
1991	Skipaútgerd rikisins (coastal shipping line)	Company disbanded. Assets sold to private lines. Services now in competitive market	223
1992	ÁTVR (alcohol and tobacco monopoly)	Production facilities, trademarks and patents sold	15
1992	Prentsmidjan Gutenberg (printing plant)	Complete privatisation	86
1992	Jardboranir (geothermal drillers)	Sale of entire 50 per cent Treasury holding	120
1992	Ferdaskrifstofa Íslands (travel agency)	Sale of entire 1/3 Treasury holding	19
1992	Menningarsjódur (arts fund)	Publishing unit sold	57
1992	Próunarfélag Íslands (venture capital)	Sale of entire 29 per cent Treasury holding	130
1992	Islensk endurtrygging (insurance)	Sale of entire 1/3 Treasury holding	162
1993	Rýni (fisheries quality control)	Complete privatisation	5
1993	SR-mjöl (herring processing)	Complete privatisation	725
1994	Pormódur rammi (trawlers and fish processing)	Sale of entire 1/6 Treasury holding	88
1994	Lyfjaverslun Íslands (pharmaceutical manufacturing and distribution)	Sale of 50 per cent Treasury holding	201
1995	Lyfjaverslun Íslands	Sale of remaining 50 per cent Treasury holding	201
1997	Skýrr (data processing)	Sale of 28 per cent treasury holding[1]	81
1997	Bifreiðaskoðun (automobile safety inspection)	Sale of 44 per cent Treasury holding	91

1. The city of Reykjavík also sold its 28 per cent holding.

Source: Ministry of Finance.

Table 22. **Privatisations envisaged in 1998**

Firm	Description
Icelandic Investment Bank	Sell 49 per cent of shares of newly corporatised bank.
Búnaðarbanki (Agricultural bank)	Sell new equity up to 35 per cent.
Íslandsbanki (National Bank of Iceland)	Sell new equity up to 35 per cent.
Icelandic Prime Contractors (construction)	Sell all holdings, currently owns 52 per cent stake.
Sementsverksmiðjan (Iceland Cement)	Sell ¼ of stake.
Skýrr (data processing)	Sell remaining stakes (22 per cent).
Áburðarverksmiðjan hf. (fertilizer manufacturer)	
Hólalax (Fish breeding plant)	
Stofnfiskur (Fish breeding plant)	
Endurvinnslan (recycling plant)	

Source: Ministry of Finance.

The government has also started a programme to reorganise publicly owned utilities. As part of the Telecommunications Agreement made in March 1997 under the auspices of the WTO, Iceland agreed to liberalise essentially all basic telecom services. First, in summer 1997, the government granted a licence to a private company to operate a mobile phone network, breaking the telecommunications monopoly.[30] Then, at the beginning of 1998, the Post and Telecommunications firm was corporatised as two separate companies. The shares of these enterprises may be offered on the stock exchange in the future.

Late in 1997 the Prime Minister's office announced the government's intention to reorganise the electricity market as well. Driven in part by an EU directive on common rules in that field, the planned reforms aim to introduce some competition and generate additional financing sources for new power stations. Such a move in a market that currently operates under a fifty-year law, however, represents a significant task. Opening the market to direct foreign trade is not economically feasible at the moment, and Iceland's small size makes it unlikely that other firms could find it profitable to enter the market immediately. Generation is dominated by the National Power Company, which has a 93 per cent market share.

Given these obstacles, the government has drawn up a tentative three-phase schedule that allows for a reorganisation over several years:

Phase 1: *Preparation* (1997-98). This puts in place some of the reforms necessary for privatisation. The generation and transmission accounts of the utilities will be separated and technical and financial options will be studied. At this time, competition for end-users with large consumption needs (60 gWh or more) may be instituted. Currently, three companies qualify, but each has signed long-term contracts. New projects in the power-intensive industries, however, may be started soon, allowing such competition in the near future (see Chapter I).

Phase 2: *Reorganisation* (1999-2003). This second phase adjusts the legal framework to allow more competition. A new act will formalise the separation of the generation, transmission and retail accounts. After this step, each of the constituent parts of the industry will be turned into separate limited-liability corporations. The reorganisation of the transmission system will receive special emphasis because it is a natural monopoly. During this period, the government will need to pay particular attention to designing a pricing system to promote competition (Hope, 1996).

Phase 3: *Implementation* (2004-09). This last phase introduces competition and ultimately an electricity market. Depending on technological developments and price developments, at this time it may be economically feasible to connect to the European grid via an underwater cable providing additional market involvement.

Overall, such a plan is similar in structure to New Zealand's restructuring, another relatively small OECD country that has an isolated, hydro generation system and would proceed at the same pace if the government were to push the reforms as quickly as the plan allows.[31]

The reorganisation plan currently does not include the district heating companies that the local governments own. These enterprises mainly provide hot water for space heating, but they also have the potential to produce electricity through co-generation. Indeed, even though large-scale geothermal electricity plants are not economically attractive compared with hydro sources, small-scale

plants that produce electricity as well as distribute hot water are economically competitive. Several such plants are currently in operation around Iceland.[32] The regulatory framework for the electricity market will have to account for these publicly owned dual-use facilities. Otherwise, these enterprises could subsidise electricity generation from their monopoly position in hot water distribution. In this regard, there has been some discussion to draft legislation that would require the heating companies to separate electricity generation in their accounts.

In addition to the efforts of central government, the City of Reykjavik has also started a privatisation programme. It has already sold a firm that manufactures sewer pipes, and in 1997 it sold its share of a data processor. In the future it hopes to privatise companies that mine stones for construction and produce asphalt for roads. It would also like to sell its drilling firm and an insurance company. The city reorganised 1 100 public housing units, creating a corporation to run them. In order to introduce other private sector initiatives, it has contracted out for services for youth recreation, pre-school education and the operation of the sports stadium.

Modernisation of the public sector

The government is in the process of implementing a multi-year programme to improve its operations by making the public sector a purchaser of services on behalf of taxpayers rather than a provider of services and employment. The objective is to replace the system in which the Ministry of Finance was involved in the detail of budgetary items with one where agencies have more responsibility. The first part of the plan is to introduce performance targets and a surveillance system to strengthen the productivity of government agencies and to establish better control over their expenditures. As from November 1996, Ministries and agencies have to establish framework agreements that spell out the expectations of the government for the performance of the agencies and the responsibilities of ministries to the agencies. These agreements include a general statement of the budgetary appropriations that the agencies can expect. Agencies then formulate their own three-to-five-year operational plans and provide criteria for performance measurement. Each year they review their performance and publish financial accounts of their operations. All agencies and ministers are expected to have adopted the system by the end of 1998.

Along with the decentralisation of decision-making authority to public agencies, management processes within ministries are being reformed. Line managers have been given more responsibility, including control over their staff. In July 1996 a revised Act on the Rights and Obligations of State Employees took effect, allowing a decentralised pay structure that would gradually be based on performance rather than seniority. In 1996 a new Civil Service Act replaced the previous act of 1954, which guaranteed lifetime employment to a large number of government employees. Most public-sector employees no longer enjoy such guarantees,[33] allowing management additional flexibility.

The government also moved to increase its use of contracted services from the private sector. Early efforts had been concentrated on outsourcing ancillary services such as catering and maintenance. The government then reformed its procurement methods in 1994. It expanded the use of tendering for legal, accounting, engineering and architectural services, following its practice in the area of construction. Under EEA rules, these tenders are now open to foreign companies and should become more common in the future. The Ministry of Finance estimates that in some cases tendered offers have generated 15-20 per cent cost reductions.

In addition, the role of central government has been reduced by the devolution of some services to local governments, reversing a trend towards centralisation that started in 1989.[34] This has been achieved either through service contracts or legislated transfers along with the tax resources. In a pilot programme, municipalities can choose to take over some functions of the central government under a contract for a trial period, and so far, the programme has been judged a success. The main areas covered have been health services, services to the old and disabled and various cultural programmes. Recently, responsibility for the management of rent subsidies was moved to municipalities. More significantly, primary education is now controlled by local governments.

Local governments have also started a reform plan. Most importantly, in the past three years, several municipalities have consolidated. In November 1993 a vote was taken, leading to a decrease in the number of municipalities from 200 to around 170. In 1995 the process continued, and by the first quarter of 1998 there were only 129 jurisdictions. With the exception of one town that merged with Reykjavik, all of the recent mergers have involved small jurisdictions that created no municipality bigger than 5 000 persons. Several more consolidations are possible in the future.

Other structural initiatives

Agriculture

Even though agriculture represents less than 2½ per cent of GDP, it is one sector that the government regulates heavily.[35] In the past few years there have been small incremental reforms to agricultural policy. Following the 1995 sheepmeat agreement from the Uruguay Round, the government stopped administering wholesale sheepmeat prices and will cease doing so for producer prices in 1998. Sheepmeat production is no longer subject to quotas, and administered producer prices for eggs and poultry were abolished at the start of 1997. Overall, market price supports are being replaced with payments based on land area or heads of livestock. Due to both policy reform and marked growth in world prices, the share of market price support to total support fell from 65 per cent in 1986-88 to 59 per cent in 1996. The decline in price support can also be seen in producer subsidy equivalents (PSEs) as a percentage of farm production, which have fallen from 82 per cent in 1986-88 to 68 per cent in 1996, and in consumer subsidy equivalents, which measure the implicit tax on consumers and have fallen from 50 to 34 per cent over the same period. The shift away from price support to direct subsidies does reduce market distortions somewhat. In addition, the government has also cut direct payments to agriculture, and the 1998 Treasury budget calls for an additional 4.3 per cent reduction in agricultural subsidies.[36]

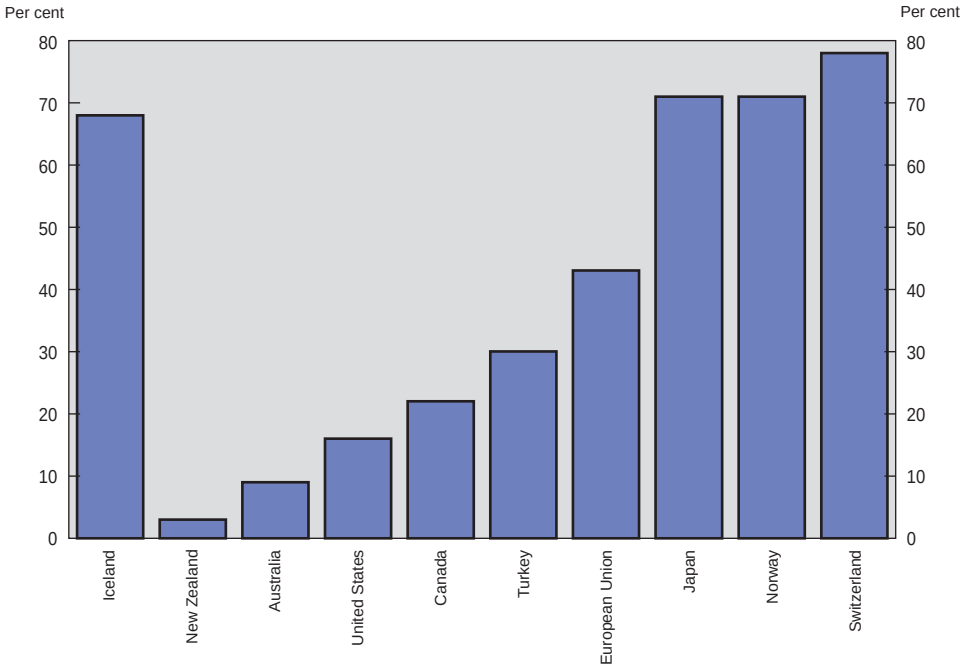
Nonetheless, agricultural protection, as measured by the PSEs, was still one of the highest in the OECD in 1996, the latest year available (Figure 21). The government still sets many producer prices and some wholesale prices, and milk production is subject to quotas. Imports of agricultural products are restricted through licences and are granted to competing importers only when local producers cannot satisfy demand.[37] Overall, agricultural trade policy is geared to ensuring domestic sources of supply are utilised first, in contrast with other sectors where Iceland is relatively more open.

Fishing

In February 1998, fishermen staged a one-week strike, charging that the method by which fish prices were determined at the docks unfairly reduced their payments. (Their compensation is largely based on the fish prices received.) They returned to work in time for the important capelin season but resumed the strike

Figure 21. **PRODUCER SUBSIDY EQUIVALENTS**

1996, per cent of agricultural production



Source: OECD.

in March before the parliament passed legislation implementing the recommendations of an appointed committee. This new legislation creates a commission that will oversee fish prices and fishermen's salaries based on their share of the catch. Another law establishes an open market for trading quotas, while a third law limits the transfer of each vessel's quota to 50 per cent annually. Otherwise, there has been little change to the fish quota system in the past couple of years. A few years ago, a small tax on quotas was imposed, the revenue from which was dedicated to the Development Fund. This fund has since been merged with three other funds (Chapter III). There has been some thought to taxing proceeds from the sales of quotas, which have been allowed for some time, but no decision has been taken. There is a bill currently in parliament that would limit ownership of quotas to about 8 per cent of the total, as it is feared some firms could gain a dominant position, though no firm currently has a share close to the ceiling.[38]

Several outstanding issues in regards to fishing concerns were resolved over the past year. In July, Iceland reached an agreement with Denmark over a dispute over the boundaries between Iceland and Greenland. At issue was whether a tiny island called Kolbeinsey could be used as a base point for the Icelandic Exclusive Economic Zone (EEZ); a compromise was reached where Iceland's EEZ extends much less than 200 miles from the island. Representatives resolved a similar dispute in November over boundaries between Iceland, Greenland and the Norwegian island Jan Mayen. In August Iceland signed a co-operation deal with Russia, which is a prerequisite for buying and renting Russian fishing quotas and will also help solidify relations between Russian fisherman and Icelandic processors. In November 1997 the North East Atlantic Fisheries Committee (NEAFC) set a new ocean perch quota and agreed to a June 1998 meeting to discuss a new surveillance regime. A year earlier the committee set a total allowable catch for atlanto-scandian herring that straddle international waters between Norway and Iceland in the so-called herring loop.

In a number of areas no agreements have yet been reached. Concerning the Barents gap, where Iceland had landed 23 000 tonnes of cod in 1996, Iceland and Norway agreed to consultations, but they still need Russia to join in the talks. At one point Iceland considered imposing a quota on itself, as it does with the shrimp catch in the Flemish Cap, but did not. The whole issue may be moot, however, with the decline in cod stocks in the Barents Sea. No agreement has been reached with the Faeroe Islands over the precise delineation of their respective EEZs. Preliminary talks with Greenland and Norway began in December to renew the capelin treaty after Iceland announced it would not renew the old Capelin Fisheries Agreement after April 1998.

Another development that will have a significant impact on the fisheries and trade in general is a trade and economic co-operation agreement that Iceland signed with Canada in March 1998. The agreement aims to lower trade barriers and strengthen the relations between the two countries. It is also seen as an important first step in negotiations for a free trade agreement between Canada and the EFTA that have recently started. Iceland holds the EFTA chair for the first six months of 1998.

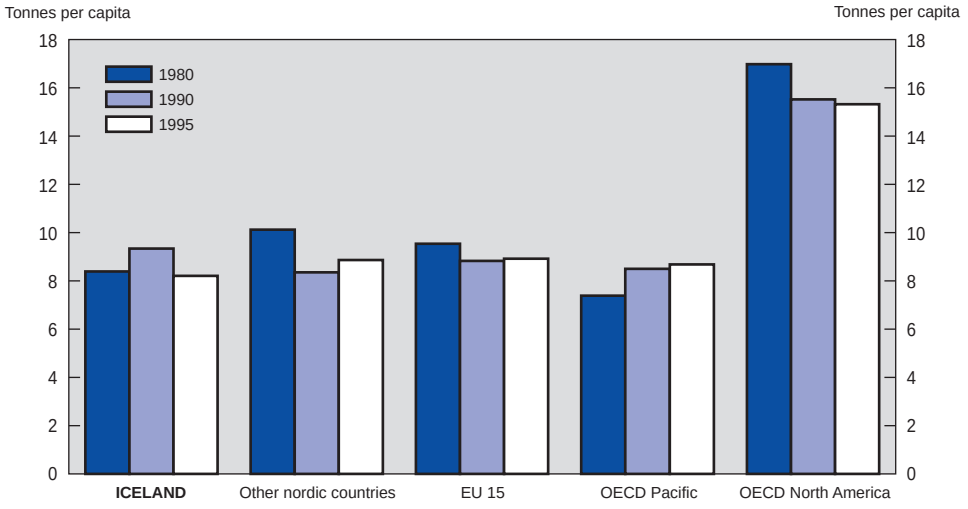
After some consideration the government decided that at present it would not allow the resumption of whaling, even though Iceland does not intend to rejoin the International Whaling Commission. The Marine Research Institute has argued that some limited whaling should occur to conserve stocks of other fish

species. There is also some suggestive evidence that the cod stocks could be reduced if the whale population around Iceland is left unchecked. The Institute has recommended that 100 fin whales and 200 minke whales could be caught each year without decreasing their stocks. The hotel and travel industry, however, fears that the resumption of whaling would cut into tourism receipts that are ten times the level of revenue in today's prices that was generated by whaling in the first half of the 1980s.[39]

Environment

Iceland successfully negotiated most of its positions at the Kyoto Conference on greenhouse gas emissions. At the meetings, Iceland argued for a lenient exemption from required emissions cuts based on three points. First, the common use of hydro and geothermal energy means Iceland produces low levels of emissions per capita compared to other countries. Indeed, their carbon dioxide emissions are a bit lower than Europe and the developed nations of the Pacific and far below levels in the United States (Figure 22). Although data are unavaila-

Figure 22. **PER CAPITA CARBON DIOXIDE EMISSIONS**



Source: IEA, OECD.

ble to do a comparison with other greenhouse gases across countries, it is possible that Iceland produces even a smaller share of other gases measured in carbon equivalents as their emissions of hydrofluorocarbons and polyfluorocarbons is much lower than in the United States.[40] Second, a larger proportion of pollution in Iceland comes from mobile sources, which should prove harder to reduce (see Box 1). And third, building more plants in the energy-intensive sector will raise emission levels from Iceland but could reduce world-wide emissions, provided they all use hydroelectricity. Indeed, about 90 per cent of the greenhouse gas emissions from an aluminium plant comes from electricity generation, while only 10 per cent comes from the chemical processes involved in the plant operations. Building a plant outside of Iceland could then lead to larger emission levels because the electricity used by that plant would probably be generated by burning fossil fuels.

At the conference Iceland was allowed the largest increase in emissions of any industrialised country; emissions in 2008-12 can be 10 per cent above 1990 levels. In contrast, the European Union agreed to an 8 per cent reduction, the United States 7 per cent and Japan 6 per cent. In addition, the treaty recognises soil reclamation and not just reforestation as CO_2 sinks, which is important because 58 per cent of the carbon dioxide sequestered in Iceland in the first half of the decade came from soil reclamation. Delegates also agreed to Iceland's proposal to give special consideration to small countries at the Buenos Aries meeting in November 1998. A final decision on whether Iceland will sign the treaty has not been made; regardless, the government has announced a policy of reducing greenhouse gas emissions, excluding the power-intensive industries, to 1990 levels by the year 2000.

Concluding remarks

Overall, deregulation outside the financial sector appears to have been significant in Iceland in recent years. With respect to the *labour* market, it is too early yet to judge the efficacy of the reforms introduced. The new Industrial Relations Act may encourage smooth negotiations and flexible wages, but talks in 1997 were still protracted. The reform of the job placement service, with its emphasis on job search effort by the unemployed, may compensate for the disincentive effects from an otherwise generous compensation system as long as officials are given sufficient authority to withhold benefits to those who try to

Box 1. **Reaching emission goals**

In 1995 the government adopted a National Climate Change Action Programme to reduce greenhouse gas emissions for several sectors:

- Industry: reduce fuel use in industry by switching to electricity and geothermal sources by improving the infrastructure and lowering electricity prices. Reduce allowable hydrofluorocarbon emissions. For instance, polyfluorocarbon emission from the aluminium plant were cut by 80 per cent through concentrated efforts, explaining a good deal of the decline in carbon equivalent emissions between 1990 and 1995.
- Mobile sources: review all taxation and levies and exempt pollution-free vehicles from excise taxes. Oblige vendors to provide standardised data on fuel use of new vehicles. Increase public transport, including electric vehicles on some routes, and make available more routes for pedestrians and cyclists. Increase the availability of electricity in ports to reduce fuel consumption by vessels.
- Refuse: expand the current project to collect methane at the major landfill for the Reykjavik area. Increase recycling, and terminate the open incineration of waste.
- Carbon Sequestration: accelerate carbon sequestration with a goal of 100 000 tonnes of CO_2 per year by the year 2000. From 1990 to 1996, 16 300 ha of land were set aside for sequestration through reforestation and soil reclamation, reducing CO_2 by 46 700 tonnes. The government has allocated IKr 450 million to reach its goal. Besides decreasing net emissions, the project would reduce soil erosion, which has been a big problem in Iceland for quite awhile.*
- Other: limit the use of solvents in road construction. Begin specific conservation campaigns and increase overall education and information. Increase research and monitoring. Establish working groups to investigate additional measures in specific sectors and appoint a special co-ordinating committee at the ministerial level to implement the programme.

The action programme will be backed by increased government spending with the aim of allocating about 0.3-0.4 per cent of GNP in this area in the year 2000.

As part of the Kyoto agreement, the limit on the emission levels of greenhouse gases in Iceland was set as 110 per cent of 1990 levels in the period 2008-12. In addition the government has announced its own goal of lowering emissions excluding those from the power-intensive industries to 1990 levels by the year 2000.

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* It has been estimated that since the time of the settlement of the island, the share of barren land has increased from 20 per cent of the total land mass to 58 per cent, and the share of woodlands has fallen from 24 per cent to 1 per cent. Satellite imagery, however, suggests the amount of barren land today may be overstated by as much as 18 percentage points, so the exact figures are controversial. Nonetheless, it is clear that Iceland's climate and land can support more vegetation than currently exists.

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Current projections, however, show that reaching the goals set at the Kyoto Conference will be difficult even with the above action programme. Iceland generates almost all of its electricity and heating from hydro and geothermal sources, so reduction techniques available to other countries such as productivity improvements in fossil-burning plants and a switch to cleaner-burning coal are unavailable. Moreover, Iceland generates

Carbon equivalent emissions, estimates and projections[1]

Per cent change since 1990

	1995	2000	2010
Total	–3.1	14.8	31.9
Excluding power-intensive industries	1.4	4.9	14.0

1. Carbon equivalent levels are calculated as (12/44) times the Greenhouse Warming Potential (GWP). GWPs are: Carbon dioxide = 1; Methane = 21; Nitrous oxide = 310; HFCs = 9 255.4; PFC = 7109. The GWP for HFCs are calculated as a weighted average of GWPs of specific HFCs using United States weights from the Energy Information Agency. Following convention, the so-called criteria gasses (carbon monoxide, other nitrogen oxides and non-methane volatile organic compounds) are excluded as their is no agreed upon analysis. Projections are based only on policies already in place, except it assumes 100 000 tonnes of CO_2 are sequestered each year by the year 2000.

Source: *Second Status Report for Iceland, 1997*, and OECD.

a higher proportion of its gases from mobile sources (Figure 23), which may be harder to reduce than emissions from other sources without an advancement in technology. For instance the use of fuel cells in automobiles and boats is unfeasible now, and large improvements in fuel efficiency appears to be several years away. Even if there were a significant advance in fuel economy or a new technology, converting the existing stock of vehicles will take time. The government is especially wary of hurting the fishing sector, which generates most of its foreign exchange earnings. Some rationalisation of the fleet may be available now, but this improvement may be offset by increased processing at sea and more deep-water fishing. Projected increases in total emissions from the expansion of the power-intensive industries by itself are 21 per cent of total 1990 emissions, requiring deep cuts elsewhere or less additional investment in these industries, an option the government has turned down.

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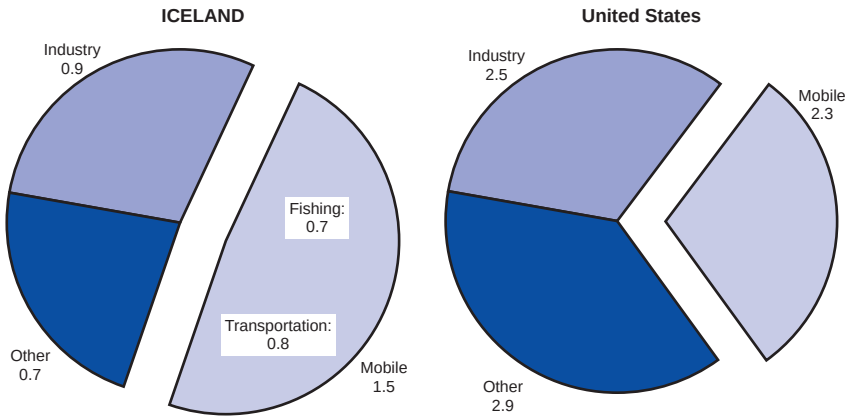
Figure 23. **CARBON EMISSIONS BY END USE**

1990, tonnes of carbon equivalent per capita



Source: OECD calculations based on data from the Ministry for the Environment, Iceland and the Energy Information Agency, United States.

exploit it. Seasonal and temporary layoffs, however, are still effectively subsidised as the system does not vary contributions by industry according to experience ratings, and this should be reconsidered. In addition, social partners should revisit the issue of compensation during non-daylight hours, which is significantly higher than in other OECD countries. Elsewhere, such jobs traditionally have provided work to those entering the labour force.

The changes that have been made to the *education* system will also have their effect only in the longer term. Moving the responsibility of primary education to local authorities should increase the responsiveness of the system to changing demands. Teachers negotiated a significant pay raise last summer; it was generally recognised that their compensation was low. The plans to boost teaching hours need to be implemented as they are currently low compared to the average in other OECD countries. In this regard, the government needs to examine why the average graduation ages at upper-secondary schools, as well as

universities, are so high. As for the latter, fees could be boosted to encourage the quicker completion of studies, using the revenue to increase investment in new facilities as overcrowding is a problem. In addition, capital spending should be boosted for vocational training centres, which need significant upgrading.

Since the turn of the decade, Iceland has pursued a vigorous policy that favours market *competition* instead of active management of the economy. The coherent programme that includes privatisations, antitrust enforcement of government-owned enterprises, deregulation and reform of government procurement practices has been impressive. Some of the initiatives have been introduced to satisfy membership requirements of the EEA, but given the scope of the reforms implemented, the underlying relationship between the government and the economy appears to have changed. The emphasis on the private market should serve Iceland well in the future.

Some of the reform programmes could be pushed further. Given the development of the financial markets and the stock exchange, there is little reason why *privatisation* of more enterprises cannot be faster. In particular, with increasing international competition now possible following the new world trade agreements, the national telecommunications company should be privatised quickly in order to meet future challenges, rather than just being transformed into a firm subject to corporate law. As the experiences of other countries show, it may be preferable to restructure the electricity industry at the earlier end of the timetable laid out by the government. Elsewhere in the utility sector of the economy, local authorities should give some consideration to including the district heating and water distribution in the reform programme. Deregulation of the heating companies could give a boost to the expansion of Iceland's ample geothermal energy supply and could provide competition to hydro generation of electricity.

Two other areas in need of reform is agriculture and the environment. Government involvement in agriculture is still substantial, in spite of recent developments. In particular, the policy that unduly stresses domestic suppliers needs to be rethought in favour of one that emphasises greater efficiencies. Reduction of price management should continue, and import barriers should be relaxed. Moreover, while some improvements have been made, overall agricultural support – one of the highest in the OECD – should be cut further. With respect to the environment, some additional policy measures will likely be needed if Iceland is to achieve its own stated goals of greenhouse gas reductions as well as the emission targets agreed to at the Kyoto Conference.

Notes

1. In 1995 (the latest year of available data) household debt as a percentage of total household assets in the G-7 countries were: United States 16.4, Japan 14.5, Germany 8.0, France 14.3, the United Kingdom 16.3 and Canada 14.9. In Iceland, debt represented 30.4 per cent of household assets in 1995 (Table 2). The figure for Germany excludes real assets in the denominator and its associated debt in the numerator, so it is not directly comparable with the other countries. Data since 1991 are unavailable for Italy; in that year debt was only 5 per cent of assets.
2. Five OECD countries (the United States, Canada, Mexico, Korea and Norway) had smaller shares of long-term unemployment (six months or more) to total unemployment in 1996, and only three of them (the United States, Mexico and Korea) had a smaller ratio of long-term unemployment to the total labour force.
3. Material costs, however, edged up only 1.3 per cent.
4. Icelandair continues to use the aeroplane as the sale was part of a leaseback arrangement.
5. Icelandic boats can fish in the ''gap'' between the exclusive economic zones of Russia, Norway and the Svalbard Islands.
6. See Gudmundsson and Zoega, 1997.
7. Electricity for industrial use in Iceland cost 1.3 US cents per kWh in 1994 against an average cost of 7.7 cents elsewhere in the OECD area.
8. See 1990 and 1991 *OECD Economic Survey* for a fuller description of these events.
9. The ECU was given a weight of 76 per cent, the US dollar 18 per cent and the yen 6 per cent.
10. The currency basket used by the Icelandic authorities did not have explicit country-by-country weightings for European currencies; rather their weight was that used within the ECU.
11. The cash-based accounts do not treat the increase in the value of the debt as an expense during its life, waiting to book the outlay as expenditure when the debt is redeemed.
12. The personal income tax credit (equivalent to a tax allowance) was reduced in proportion to decreases in the marginal tax rate.
13. Some low-income pensioners may face even higher marginal tax rates.
14. Some assets, such as government bonds, are exempted from the wealth tax.
15. Under the cash system, the increase in the value of such debt was not recognised until the bond was redeemed, and this could give rise to large swings in interest payments, especially if there were early redemptions of indexed-linked bonds – as in 1996 and 1997.

16. According to these estimates, the increase in pension liabilities amounted to 0.6 per cent of GDP annually over the past four years while booking interest payments on an accruals basis raised public expenditure by a further 1.2 per cent of GDP. The corrections may prove less, however, as inflation is low – so reducing the increase in value of indexed linked debt – and given a reform to the pension system for state employees (see below). Nonetheless, the Central Bank puts the accrual adjustment at more than double that by the government, so pushing the budget balance back into deficit.

17. For this purpose, eligible assets have three elements: central bank liabilities (cash, demand deposits and certificates, less short-term borrowing), central government liabilities (Treasury bills, bonds and housing bonds) and finally, deposits with other banks both in Iceland and abroad.

18. Less than 4 per cent of government debt is in the form of non-indexed domestic currency liabilities. However in 1995 and 1996, non-indexed notes accounted for 35 per cent of sales of government paper, other than Treasury bills and by end 1997, 13 per cent of central government long-term debt was in the form of non-indexed securities.

19. Operating costs in some savings banks are even higher.

20. The ratio of labour costs to total assets has fallen by almost 1 percentage point between 1986 and 1996.

21. A further nine funds are closed to new entrants. Only nine of the open schemes are defined contribution plans, the remainder are defined benefit schemes. Payment of pensions generally starts at age 70. It is possible to draw a reduced pension from age 67, the age at which the state pension is paid.

22. Only three of the 40 private-sector funds are guaranteed by the employer. All public-sector funds are guaranteed either by the central or local governments.

23. However, most of the funds' assets are invested in government-guaranteed securities.

24. The Central Bank, though, estimated the cost of the existing state pension at 27 per cent of the payroll and noted that the employees' contribution rate had not been increased for 50 years, despite increases in benefits.

25. The *1997 OECD Economic Survey* discusses Iceland's educational system in more detail.

26. The government continues to be involved in price setting of agricultural products, taxi and delivery vans, drugs and medicines and public utilities. Until September 1994, it equalised prices for a number of products across the country; since then only transportation costs have been equalised, with prices otherwise freed from regulation.

27. The 1995 *OECD Economic Survey* extensively reviews developments in competition policy. It also presents an analysis of the effects of Iceland's small size and its remote location on its ability to promote effective competition.

28. For example, in a case where the competition council ordered a firewall created between a municipally owned funeral home and a cemetery, the Supreme Court upheld the order. At issue was whether the link between the city-owned cemetery and the public funeral home gave the latter an unfair advantage when competing with private funeral homes that also had to use the cemetery.

29. Indeed such a figure is at the high end of the proportion found in other European countries. In 1991, the total employment in public enterprises was 8 and 7 per cent of total employment in France and Italy respectively. In contrast in the United Kingdom, Luxembourg and the Netherlands, the proportion was only 2¾ per cent.

30. Iceland has more than 20 mobile phones per 100 inhabitants, significantly more than Britain, Germany and France, though less than other Nordic countries.

31. From 1986 to 1991 New Zealand corporatised and restructured its generation, transmission and distribution systems. A 1992 act led to competition for small customers in 1993 and large users in 1994.

32. Hydro electric plants generated 94 per cent of Iceland's electricity in 1994; geothermal plants produced 5 per cent with the rest generated by conventional thermal facilities. These conventional plants have under-utilised capacity that is used to meet peak needs.

33. Senior civil servants now have five-year fixed-term contracts. Other employees are hired on indefinite contracts that can be terminated provided severance payments are made. Judges, policemen and ministers of the established church continue to benefit from lifetime contracts.

34. In 1989 the division of responsibilities between municipalities and the central government was reviewed. At that time, local governments were under significant budgetary pressures, and so more functions were moved to the central government.

35. The *1994 OECD Economic Survey* reviews Iceland's agricultural sector.

36. Because price support has been cut more than direct subsidies, the share of these to total payments has risen from 18 per cent in 1986-88 to 23 per cent in 1996 in spite of the absolute reductions.

37. Iceland also regulates imports of live animals and meat products for zoo-sanitary reasons. Past experience has shown that the herds are susceptible to foreign diseases because immunities have not developed.

38. New Zealand also has an extensive tradeable quota system. They restrict ownership to 35 per cent of the total deep-sea quota and 20 per cent of the total inshore quota. See Runolffson (1997) for further discussion.

39. Over half of the respondents to a poll conducted by the Tourist Council last summer said that the resumption of whaling would adversely affect their decision to visit Iceland. According to the Icelandic daily newspaper *Morgunbladid*, however, tourism in Norway has not been seriously affected by the decision to resume whaling, perhaps in part due to the ban on whalemeat exports.

40. For instance, because of its climate, leakage from air conditioners is not a significant source of greenhouse gases, and Iceland does not have large chemical plants or oil refineries.

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Annex

Calendar of main economic events

1997

March

The Central Bank renews its custodian and market making agreement with three securities firms.

The government introduces tax reductions planned for the next four years and also announces that it will raise social security benefits in line with the wage contracts.

After several small strikes, unions and employers sign the first wage contracts. The contracts run for three years and provide a cumulative wage increase of 12.9 per cent.

Icelandic alloys announces that it will add a third furnace to the ferrosilicon plant. Work does not begin until 1998.

April

The Government Debt Management Agency implements the second phase of its plan to reorganise outstanding Treasury bonds, swapping 22 bond issues for benchmark issues in a special auction.

The Minister of Finance proposes a comprehensive bill on pension funds.

About 800 workers mainly in fish processing and cleaners go on strike, which ends in June.

May

Parliament passes separate legislation converting the two state-owned banks into limited liability corporations and restructuring four investment credit funds.

The National Power Company secures a $50 million loan from the Nordic Investment Bank, the largest loan ever granted to an Icelandic borrower.

The government announces a 4 per cent across-the-board increase in government benefits retroactive to the beginning of March. The increase comes on top of a 2 per cent rise at the beginning of the year.

June

Íslandsbanki hf. and Landsbréf hf. assume the roles of market makers for housing bonds after a public tender offer from the State Housing Fund.

The Ministry of Fisheries issues the total allowable catch for species during the 1997-98 fishing year, raising the cod quota 17 per cent from last year's level.

July

Banks assume the responsibilities for market making in the interbank market for foreign exchange. The daily fixing meetings are abolished. The Central Bank issues new rules on foreign exchange exposure, reserve requirements and liquidity.

Moody's Investors Service upgrades Iceland's credit rating on foreign long-tern debt from A1 to Aa3. It rates short-term debt Prime-1.

Icelandic and foreign vessels send almost 325 000 tonnes of capelin to Icelandic processing plants, easily surpassing the previous record high.

Icelandair agrees to purchase four Boeing 757 jets with options to buy up to eight more.

The government announces a further 2½ per cent increase in social payments after a row starts when the State Salaries Arbitration Court recommends a 13 per cent pay hike for senior government officials.

Domestic air travel prices drop with deregulation of the industry.

August

Iceland and Russia sign a fisheries co-operation pact.

Columbia Ventures and the Icelandic government sign an agreement allowing the firm to build an 180 000 tonne aluminium smelter in Grundartangi.

September

Norway permits Icelandic vessels to catch 10 000 tonnes of herring in its exclusive economic zone.

October

After a one-day strike, teachers agree to a new contract that provides substantial wage increases over the next three years.

The expansion of the aluminium smelter in Straumsvík is completed three months ahead of schedule and 15 per cent under budget.

The central government issues its 1998 budget, projecting a small surplus on an accruals basis. The 1998 budget is the first one using the new accounting standards.

November

Iceland successfully negotiates most of its positions at the United Nations Conference on Climate Change held in Kyoto, Japan. They receive the highest exemption of any developed country from greenhouse gas emissions reductions.

The Central Bank raises short-term interest rates 30 basis points.

December

Representatives from Iceland, Greenland and Norway meet to discuss the renewal of the capelin treaty. A second meeting takes place in February in Reykjavik.

Hospital doctors sign a new wage contract. Many young doctors, however, are unhappy with the provisions and quit, leading to under-staffing at some hospitals.

The Post and Telecommunications companies are formally split.

Parliament passes a pension reform law.

1998

January

Stock in the new Investment Bank begins to be traded on the Icelandic Stock Exchange.

February

About 5 000 capelin fishermen stage a one-week strike.

March

Capelin fishermen renew their strike for two weeks before parliament passes legislation creating a commission to rule on fish prices and their compensation share.

The Central Bank abolishes the liquidity requirements on banks and modifies the maturity and instruments it uses to conduct monetary policy. It also announces that over the next two months it will extend coverage of reserve requirements to most credit institutions, lower the ratio and modify other features of the system.

Iceland and Canada sign a trade co-operation agreement.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Supply and use of resources**

IKr. million, current prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996 [1]	1997 [2]
Private consumption	161 068	190 254	223 176	248 366	248 339	248 182	256 949	272 708	296 840	320 314
Public consumption	50 537	60 341	69 989	78 157	80 375	84 818	89 424	94 080	100 358	106 540
Gross fixed asset formation	50 498	58 698	70 103	76 173	69 589	64 177	65 876	65 950	86 752	97 694
Expenditure on final domestic use	262 103	309 293	363 268	402 696	398 303	397 177	412 249	432 738	483 950	524 548
Change in stocks of export products	–3 080	–8 143	–4 247	–891	–486	1 016	–26	2 285	–1 202	–840
National expenditure	259 023	301 150	359 021	401 805	397 817	398 193	412 223	435 023	482 748	523 708
Exports of goods and services	81 721	106 282	124 936	125 671	121 597	135 694	157 436	161 250	176 147	190 223
Imports of goods and services	84 100	99 240	119 556	130 491	121 782	122 466	134 631	144 725	173 727	186 732
Gross domestic product (market prices)	256 645	308 192	364 401	396 985	397 632	411 421	435 028	451 548	485 168	527 199
Net income from abroad	–6 506	–11 164	–12 849	–10 644	–8 911	–9 821	–13 445	–12 502	–10 848	–11 779
Gross national product	250 139	297 028	351 552	386 341	388 721	401 600	421 583	439 046	474 320	515 420
Depreciation	28 905	37 308	43 692	47 933	50 881	54 144	56 453	57 702	59 911	63 320
Net national product (market prices)	221 234	259 720	307 860	338 408	337 840	347 456	365 130	381 344	414 409	452 100
Indirect taxes	61 324	72 152	79 167	83 943	83 714	77 009	79 004	82 373	89 811	95 509
Subsidies	9 107	12 835	12 953	12 319	13 225	10 504	9 607	9 492	10 054	10 363
Net national income	169 017	200 403	241 646	266 784	267 351	280 951	295 733	308 463	334 652	366 954

1. Provisional.
2. Forecast.

Source: National Economic Institute and Central Bank of Iceland.

Table B. **Supply and use of resources**

Ikr. million, constant 1990 prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996 [1]	1997 [2]
Private consumption	231 752	222 092	223 176	232 402	222 060	212 017	215 959	225 185	239 468	253 751
Public consumption	65 108	67 071	69 989	72 159	71 579	73 240	75 936	76 923	77 688	78 878
Gross fixed asset formation	73 863	68 055	70 103	71 534	63 513	56 230	55 638	54 061	68 421	75 150
Expenditure on final domestic use	370 724	357 217	363 268	376 095	357 152	341 487	347 533	356 169	385 576	407 779
Change in stocks of export products	–945	–3 613	–4 247	1 121	–86	1 016	169	2 343	93	–38
National expenditure	369 779	353 604	359 021	377 215	357 066	342 503	347 702	358 512	385 669	407 741
Exports of goods and services	121 396	124 968	124 936	117 565	115 316	123 437	135 805	132 807	145 550	153 796
Imports of goods and services	131 914	118 383	119 556	126 371	116 362	106 303	110 725	114 893	134 022	144 663
Gross domestic product (market prices)	359 261	360 189	364 401	368 409	356 020	359 636	372 782	376 426	397 197	416 875
Gross national product (market prices)	347 484	347 230	351 552	357 177	345 576	348 388	358 888	363 856	385 301	404 285
Effect of changes in terms of trade	7 653	2 225	0	5 625	1 988	–3 952	–5 152	–3 400	–8 984	–6 142
Gross national income	355 137	349 455	351 552	362 802	347 564	344 436	353 736	360 457	376 317	398 143

Note: Estimates of real income coincide with output in real terms on the assumption of unchanged terms of trade. Due to particularly strong fluctuations in Icelandic terms of trade national expenditure in real terms may deviate substantially from real gross national product without adverse effects on the balance of payments. This is explicitly introduced in the Icelandic national accounts, as shown above. The item ''Effect of changes in the terms of trade'' equals the external purchasing power of export earnings (nominal exports deflated by a price index for imports) minus the volume of exports of goods and services.

1. Preliminary.
2. Forecast.

Source: National Economic Institute.

Table C. **Production and employment**

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996 [1]
Fisheries and fish processing										
Output (volume change over previous year)	12.0	–0.5	–3.9	1.7	–5.9	–0.2	5.4	–1.9	–1.8	8.7
Export production, value (IKr million)	40 189	45 946	55 662	67 692	74 333	70 209	76 752	85 567	86 694	92 056
Fishing fleet: [2]										
Trawlers (GRT)	53 113	55 200	56 241	55 612	55 952	56 756	59 189	62 427	67 165	73 100
Motor boats (GRT)	65 477	64 490	64 505	64 544	65 678	63 640	62 040	59 360	55 819	57 843
Total (GRT)	118 590	119 690	120 746	120 156	121 630	120 396	121 229	121 787	122 984	130 943
Employment (man-years)	16 788	15 145	14 893	14 539	14 303	13 023	13 744	13 739	13 735	12 870
Agriculture										
Output (volume change over previous year)	6.0	–10.2	–3.6	–2.0	–0.2	–2.4	6.1	2.3	–4.1	8.8
Export production, value (IKr million)	1 015	997	1 288	1 765	1 657	1 449	1 795	2 127	2 060	2 548
Employment (man-years)	7 147	6 470	6 399	6 873	6 709	6 514	5 859	5 621	5 300	5 054
Electricity generation										
Installed capacity (MW)	922	923	929	913	937	1 039	1 044	1 050	1 049	1 049
Output (GWh)	4 152	4 416	4 475	4 447	4 427	4 541	4 721	4 774	4 977	5 112
Manufacturing (excluding fish processing)										
Output (volume change over previous year)	9.3	–5.0	–3.4	–3.7	2.5	–4.0	–4.3	3.7	3.8	8.5
Export production, value (IKr million)	10 059	13 677	19 460	18 742	15 927	15 131	16 999	21 089	25 006	25 526
of which:										
Aluminium	4 761	6 705	10 146	9 629	8 222	7 996	8 428	10 741	12 198	12 332
Diatomite	296	348	416	522	406	422	379	532	667	576
Ferro-silicon	1 195	2 203	2 899	2 180	1 637	1 691	2 413	2 839	3 296	4 069
Employment (man-years)	18 439	17 057	16 195	15 903	15 586	14 776	13 850	13 611	14 024	14 873

1. Provisional.
2. Including whale catchers, excluding open boat; stock at end of year.

Source: National Economic Institute and Central Bank of Iceland.

Table D. **Gross fixed asset formation and national wealth**

IKr million, current prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997 [1]
Gross fixed asset formation, total	50 498	58 698	70 103	76 173	69 589	64 177	65 876	65 950	86 752	97 694
Classification by end-use:										
Industrial asset formation	29 402	32 470	38 612	42 294	36 109	28 932	30 578	33 576	52 927	60 614
Agriculture	1 593	1 584	1 537	1 943	1 618	1 884	1 804	1 606	2 542	2 550
Fishing	6 788	4 216	2 711	2 911	5 950	2 146	3 471	1 546	5 174	2 000
Fish processing	1 509	1 265	1 730	1 413	1 408	1 679	2 429	3 166	4 659	3 800
Manufacturing other than fish processing	4 671	5 865	5 387	6 642	5 432	5 419	5 347	6 114	13 857	17 614
Electricity, heating and water supply	3 532	5 443	6 526	7 101	3 857	3 513	3 044	3 497	5 137	10 750
Various machinery and equipment for construction	1 305	1 377	1 744	2 473	1 562	1 528	1 274	1 749	2 624	2 800
Trade, restaurants and hotels	4 190	4 700	4 721	4 330	4 397	3 884	3 548	3 850	4 698	5 800
Transport and storage	2 559	5 280	10 941	11 348	7 387	3 952	2 825	4 642	5 465	6 000
Communications	752	905	1 003	1 466	1 854	2 036	2 909	2 659	2 718	3 300
Computers and office equipement	2 502	1 835	2 312	2 668	2 646	2 892	3 925	4 747	6 053	6 000
Residential construction	12 586	15 936	18 666	19 104	18 912	18 212	18 667	17 338	18 935	20 780
Public works and buildings	8 510	10 293	12 825	14 775	14 568	17 033	16 630	15 036	14 890	16 300
Public buildings	5 026	6 344	7 970	8 800	8 770	9 300	9 100	8 800	8 735	9 100
National wealth	738 511	934 774	1 114 927	1 220 985	1 271 101	1 326 663	1 381 081	1 428 717	1 506 983	1 590 171
Industrial sector	348 571	441 675	504 474	547 700	566 357	589 688	609 967	621 718	653 883	687 884
Private sector [2]	265 465	334 786	403 167	442 068	460 342	476 819	495 127	514 883	546 099	576 690
Public works and buildings	124 475	158 313	207 286	231 217	244 402	260 156	275 987	292 116	307 001	325 597

1. Provisional data.
2. Residential housing.

Source: National Economic Institute.

Table E. **Gross fixed asset formation and national wealth**

Volume indices, 1990 = 100

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997 [1]
Gross fixed asset formation, total	105.4	97.1	100.0	102.1	90.6	80.2	79.4	77.1	97.6	107.2
Classification by end-use:										
Industrial asset formation	109.5	94.3	100.0	103.7	85.6	64.4	64.8	69.5	106.3	120.0
Agriculture	150.2	113.5	100.0	121.0	97.2	106.6	97.1	81.6	133.1	127.4
Fishing	370.0	182.1	100.0	101.5	201.8	66.5	100.3	44.9	142.1	55.4
Fish processing	123.5	82.8	100.0	77.7	74.5	83.5	115.3	147.2	211.3	169.2
Manufacturing other than fish processing	126.2	125.3	100.0	116.8	92.4	85.7	80.1	89.4	198.3	249.0
Electricity and heating	75.0	96.4	100.0	101.1	50.8	44.0	37.6	41.7	62.5	132.4
Water supply	82.3	77.4	100.0	110.6	82.3	82.8	66.2	76.0	68.1	78.6
Various machinery and equipment for construction	97.8	80.2	100.0	135.9	82.5	72.5	56.2	76.4	111.2	118.7
Trade, restaurants and hotels	117.2	106.7	100.0	85.2	84.5	73.1	65.1	68.5	81.0	97.4
Transport and storage	36.1	56.3	100.0	98.8	61.8	31.3	22.0	34.3	39.8	42.1
Postal and telecommunications	106.6	101.3	100.0	135.8	167.7	173.9	236.3	211.7	209.8	251.3
Computers and office equipement	144.0	80.6	100.0	110.6	105.5	103.5	130.6	156.4	193.5	191.8
Residential construction	97.8	100.6	100.0	95.1	91.9	86.7	86.7	78.1	81.2	85.4
Public works and buildings	102.6	101.4	100.0	107.3	103.5	118.3	112.7	98.7	95.1	100.5
Public buildings	98.2	100.6	100.0	102.6	99.9	103.6	98.9	92.7	89.3	90.5
National wealth	95.5	97.7	100.0	102.4	103.9	104.7	105.6	106.3	108.2	110.6
Industrial sector	96.4	98.0	100.0	102.1	102.7	101.6	100.7	100.3	102.6	105.7
Private sector [2]	95.6	97.8	100.0	101.9	103.6	105.0	106.4	107.4	108.5	109.7
Public works and buildings	92.7	96.4	100.0	103.9	107.5	111.8	115.8	118.7	121.3	124.2

1. Preliminary data.
2. Residential housing.

Source: National Economic Institute.

Table F. **Central government and social security income and expenditure**

Kr million, accruals basis

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Current revenue	57 907	78 166	93 984	106 143	119 375	121 734	119 048	124 974	130 530	143 643
Direct taxes	10 184	18 395	22 387	29 267	38 199	39 345	39 763	42 778	46 871	53 237
Indirect taxes	42 468	53 459	62 666	69 562	72 601	72 143	70 105	71 816	74 115	81 149
Other	5 255	6 312	8 931	7 314	8 575	10 246	9 180	10 380	9 544	9 257
Current expenditure	51 855	72 191	88 913	103 880	113 545	116 743	119 301	123 304	131 188	136 479
Public consumption	30 838	39 603	46 843	54 864	60 537	60 696	63 445	65 925	70 005	72 304
Interest expenditure	4 340	7174	9 685	11 370	12 962	12 875	13 678	14 795	16 499	15 812
Current transfers and subsidies	16 677	25 414	32 385	37 646	40 046	43 172	42 178	42 584	44 684	48 363
Current balance	6 052	5 975	5 072	2 263	5 830	4 991	–253	1 670	–658	7 164
Capital revenue	1 234	1 492	1 714	1 976	2 346	2 330	2 448	2 669	2 825	3 070
Capital transfers	371	474	469	468	610	517	531	688	729	817
Consumption of fixed capital	863	1 018	1 245	1 508	1 736	1 813	1 917	1 981	2 096	2 254
Capital expenditure	8 847	12 481	19 040	16 540	19 186	16 828	15 892	18 144	14 117	17 538
Gross fixed investment	3 078	3 972	4 444	6 338	7 100	6 434	8 025	8 290	7 316	8 486
Capital transfers	5 769	8 509	14 596	10 202	12 086	10 394	7 866	9 854	6 801	9 052
Capital balance	–7 613	–10 989	–17 326	–14 564	–16 840	–14 498	–13 444	–15 475	–11 292	–14 468
Financial balance	–1 561	–5 014	–12 254	–12 301	–11 010	–9 507	–13 697	–13 805	–11 950	–7 304
Net increase in claims	3 523	3 839	707	–1 053	5 827	–1 934	–780	1 603	3 582	2 935
Borrowing requirement	5 087	8 852	12 961	11 249	16 836	7 574	12 917	15 408	15 533	10 239

Source: National Economic Institute.

Table G. **Fish catch, wages and prices**

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Fish catch (thousand of metric tons)										
Total	1 757	1 495	1 505	1 049	1 575	1 716	1 555	1 609	2 059	2 190
Demersal species	698	692	673	655	585	585	558	511	502	477
Herring	93	97	90	79	123	117	130	110	100	72
Capelin	911	653	694	260	798	941	754	716	1 179	1 320
Crustaceans	42	40	44	50	61	70	85	93	101	93
Wages and prices (indices 1990 = 100)										
Hourly wages	82.0	92.9	100.0	108.7	113.3	115.2	115.8	121.1	128.9	. .
Consumer price index	71.9	87.1	100.0	106.8	110.8	115.3	117.0	119.0	121.7	123.9
Credit term index	73.1	86.6	100.0	107.5	111.5	114.3	116.4	118.2	120.9	123.0
Building cost	68.6	83.9	100.0	108.0	111.0	113.2	116.2	119.8	125.2	131.1
Export price of fish products (1991 = 100)[1]										
Groundfish:										
frozen on shore			88.0	100.0	94.5	83.3	79.6	79.3	78.3	82.2
frozen at sea			86.8	100.0	99.6	89.0	84.1	88.4	93.4	91.1
fresh			95.0	100.0	94.5	81.0	78.3	79.1	78.2	. .
salted			89.9	100.0	95.3	74.8	76.7	79.7	82.9	81.3
Fish mean and oil			92.1	100.0	98.1	84.8	84.1	99.3	111.5	132.6

1. The index shows the development of export prices (fob) in terms of SDR's.

Source: National Economic Institute and Central Bank of Iceland, Economic Statistics.

Table H. **Foreign trade, total and by area**

US$ million, monthly rates

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Total imports, cif	133.1	116.7	138.7	143.0	140.4	112.4	122.7	146.3	169.5	173.6
OECD countries	123.8	107.1	126.6	132.0	130.0	103.4	113.0	133.4	155.9	158.7
EEC (15)	84.7	72.6	83.4	89.5	81.3	65.2	71.6	87.4	95.9	100.7
Other Europe	15.5	11.4	10.4	10.7	22.8	16.9	20.2	19.4	29.2	26.2
United States	10.1	13.0	19.7	14.8	11.6	10.5	10.9	12.3	16.1	17.1
Eastern Europe	4.8	5.4	7.4	5.3	2.7	0.2	0.4	1.2	0.9	1.2
Non-oil developing countries	4.5	4.1	4.7	5.6	7.5	8.7	9.1	11.4	12.4	13.4
Total exports, fob	118.6	116.7	132.6	129.1	127.3	116.4	135.3	151.3	158.0	154.7
OECD countries	109.6	106.2	123.7	123.7	121.6	110.7	130.5	143.4	145.4	142.5
EEC	73.8	69.6	93.8	89.0	90.8	72.0	83.3	94.9	98.1	94.5
Other Europe	10.3	11.2	8.1	7.5	6.1	8.3	6.4	8.5	9.6	13.3
United States	16.0	16.4	13.1	16.1	14.4	18.5	19.6	18.7	19.5	23.1
Eastern Europe	4.5	4.2	3.4	0.8	0.5	0.1	0.1	0.3	0.4	1.2
Non-oil developing countries	3.2	5.8	4.7	3.6	4.2	4.6	3.8	7.0	11.5	9.9

Source: Central Bank of Iceland and OECD, *Foreign Trade Statistics*, Series A.

Table I. **Foreign trade by commodity group**

US$ million

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Imports										
Total	1 590.5	1 395.2	1 654.8	1 739.0	1 682.0	1 407.0	1 472.4	1 757.6	2 041.5	2 012.5
Transport equipement, SITC 78-79	266.8	183.8	260.8	220.7	246.3	124.1	151.1	187.1	260.8	238.5
Other imports	1 323.6	1 211.4	1 394.0	1 518.3	1 435.7	1 282.9	1 321.3	1 570.5	1 780.7	1 773.9
Food and live animals, SITC 0	106.6	110.7	122.5	126.7	127.4	132.3	137.7	168.8	178.0	165.9
Manufactured goods, SITC 6	286.4	256.6	278.4	309.0	286.5	253.4	256.3	304.3	342.0	331.9
Machinery and apparatus, SITC 71-77	318.2	269.1	318.9	370.7	331.2	283.9	312.1	380.0	471.1	514.5
Other goods	612.4	575.0	674.2	711.9	690.6	613.3	615.1	717.4	789.5	761.6
Exports										
Total	1 431.1	1 400.9	1 590.7	1 571.4	1 527.8	1 456.2	1 613.8	1 802.0	1 887.9	1 854.0
Fish products, total	1 016.9	994.3	1 197.7	1 239.5	1 213.0	1 105.2	1 213.1	1 296.2	1 390.6	1 323.2
Frozen fish fillets	367.1	398.2	522.5	589.9	569.8	521.6	508.4	504.8	464.8	438.8
Herrings salted	24.6	21.8	26.0	14.8	11.1	8.8	11.9	12.0	15.8	10.7
Herring and capelin meal	75.1	63.1	53.9	20.4	72.8	72.9	63.3	72.6	120.7	121.2
Agricultural products	24.8	23.9	30.5	27.5	28.3	24.0	30.5	31.8	38.3	29.7
Aluminium products	153.8	180.1	164.3	136.7	139.8	122.4	154.9	190.1	181.8	214.7
Other manufactured products	167.0	164.8	159.8	133.3	122.7	122.5	145.9	196.2	194.3	191.6

Source: Central Bank of Iceland and OECD, *Foreign Trade Statistics*, Series C.

Table J. **Money and credit**

IKr million, end of period

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Central Bank										
Penalty rates (annual rate), per cent [1]	43.9	35.9	26.0	25.0	19.5	17.7	14.2	14.6	15.8	16.3
Net position of government	9 117	8 237	3 594	8 748	1 551	4 444	13 849	12 300	3 680	6 422
Money supply										
M1 [2]	14 853	19 725	24 644	29 553	29 942	31 564	34 955	38 316	41 557	49 518
M2 [3]	73 271	92 548	105 731	116 697	124 391	129 905	113 725	111 109	113 416	118 442
M3 [4]	88 802	112 998	129 802	148 436	154 101	164 127	167 918	171 656	183 407	200 665
Deposit money banks										
Required reserves and other funds with Central Bank	12 162	14 711	11 223	10 418	9 255	7 185	7 070	6 852	7 728	8 642
Demand deposits	12 302	16 750	21 586	26 314	26 350	27 658	30 315	33 807	36 085	43 773
Net liquid foreign assets	1 700	2 943	2 379	3 506	3 413	5 851	6 182	2 217	2 893	–1 242
Lending (exclusing foreign funds relent) of deposit money banks										
Total	95 504	118 130	136 539	152 927	159 640	172 020	181 542	193 578	211 453	245 043
of which to:										
Agriculture	7 681	9 254	10 589	10 768	8 891	8 252	7 805	7 491	6 830	7 253
Fishery and fish processing	17 161	19 261	17 938	19 306	19 545	20 254	19 981	26 279	31 446	33 815
Manufacturing and commerce	28 737	34 518	36 038	40 854	41 929	41 371	41 628	41 642	44 864	46 893
Dwellings	8 064	11 325	12 915	14 051	14 771	14 409	14 036	13 869	16 419	17 606
Foreign exchange										
Central Bank net liquid foreign assets	11 919	20 005	23 412	24 066	31 053	28 639	18 786	19 259	36 025	33 246
Deposit money banks' net short-term foreign position	–15 399	–11 512	–8 596	–10 008	–12 898	–9 288	–4 294	–8 972	–19 623	–23 060

1. Annual average.
2. Notes and coins plus demand deposits.
3. M1 plus general savings deposits.
4. M2 plus time deposits.

Source: Central Bank of Iceland.

Table K. **Public sector**

	1960	1970	1980	1990	1992	1993	1994	1995	1996
General government accounts (as a per cent of GDP)									
Current revenue	28.2	30.2	33.1	36.0	37.8	35.8	35.3	36.0	36.8
Tax revenue[1]	27.2	29.6	30.9	33.5	34.9	33.2	32.6	33.3	34.3
Interest income	..	..	1.9	1.6	1.7	1.7	1.6	1.5	1.4
Capital revenue	..	..	0.2	0.7	0.8	0.8	0.8	0.8	0.8
Total expenses	25.8	29.9	32.4	40.0	41.4	41.1	40.9	39.8	39.2
of which:									
Current expenditure	..	..	26.0	33.3	34.8	34.9	34.4	35.1	34.2
Current transfers	..	..	4.3	7.0	7.6	8.0	7.8	8.0	7.7
Subsidies	..	..	3.1	3.6	3.3	2.6	2.2	2.1	2.1
Capital expenditure	..	..	6.4	6.7	6.6	6.3	6.5	4.7	5.0
Gross fixed investment	..	..	3.7	3.9	4.0	4.3	4.1	3.0	3.2
Capital transfers	..	..	2.7	2.7	2.6	2.0	2.4	1.7	1.8
Tax receipts as a per cent of general government total taxes									
General government									
Direct taxes	31.5	30.9	26.9	35.2	39.7	43.6	44.2	45.2	45.9
Indirect taxes	68.5	69.1	73.1	64.8	60.3	56.4	55.8	54.8	54.1
Central government and Social security									
Total taxes	77.5	77.2	79.8	80.9	80.3	80.4	80.9	80.4	80.9
Direct taxes	13.3	16.8	15.9	24.0	28.3	29.1	30.2	31.2	32.0
Indirect taxes	64.2	60.4	63.9	57.0	52.0	51.3	50.7	49.3	48.8
Local government									
Total taxes	22.5	22.8	20.2	19.1	19.7	19.6	19.1	19.6	19.1
Direct taxes	18.2	14.1	11.0	11.2	11.4	14.5	14.1	14.1	13.9
Indirect taxes	4.3	8.7	9.2	7.9	8.3	5.1	5.1	5.5	5.2

1. Direct and indirect taxes.

Source: National Economic Institute and Sögulegt Yfirlit Hagtalna, 1945-1988, National Economic Institute.

Table L. **Labour market**[1]

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Working age population										
15 to 64	160.7	162.8	164.1	166.3	168.1	169.4	170.9	172.5	173.5	175.4
16 to 74				173.4	175.0	177.8	179.0	179.8	180.7	182.4
Labour force										
Man-years	128.8	128.2	127.2	126.7	126.9	127.7	128.9	131.0	133.3	134.9
ILO definition				140.5	143.0	144.2	145.4	149.0	147.5	147.8
Unemployment rates										
Total Man-years	0.6	1.7	1.8	1.5	3.0	4.4	4.8	5.0	4.3	3.9
Total ILO definition				2.5	4.3	5.3	5.3	4.9	3.7	3.9
Males ILO definition				2.3	3.8	5.0	5.1	4.8	3.4	3.3
Females ILO definition				2.9	4.9	5.6	5.5	4.9	4.1	4.5
Employment										
Man-years	127.9	126.1	124.9	124.8	123.0	122.1	122.7	124.5	127.5	129.7
ILO definition				136.9	136.9	136.6	137.7	141.8	142.0	
Participation rates										
Total Man-years	79.3	78.4	77.5	76.2	75.5	75.3	75.4	75.9	76.8	76.9
ILO definition				81.0	81.8	81.1	81.3	82.9	81.6	81.0
Males Man-years	91.2	90.0	89.1	87.4	86.4	85.8	85.0	86.0	86.3	86.7
ILO definition				87.4	87.6	85.9	85.8	87.7	86.4	..
Females Man-years	67.1	66.4	65.5	64.7	64.2	64.6	65.5	65.7	66.6	66.1
ILO definition				74.6	75.8	76.1	76.7	77.9	76.8	..
Work stoppages										
Man-days lost	100 773	79 970	231	3 413	385	120	97 343	217 186	..	..
% of total working time	0.30	0.24	0.00	0.01	0.00	0.00	0.30	0.67	..	..

1. Aged 15 to 65 except for ILO definitions where 16 to 74.

Source: National Economic Institute.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

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